

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
SAN FRANCISCO BAY REGION**

1515 Clay Street, Suite 1400, Oakland, California 94612  
waterboards.ca.gov/sanfranciscobay

**ORDER R2-2021-0026  
NPDES PERMIT CA0037770**

The following Discharger is subject to the waste discharge requirements (WDRs) set forth in this Order:

Discharger **Mt. View Sanitary District**  
Name of Facility **Mt. View Sanitary District Wastewater Treatment Plant  
and its collection system**  
Facility Address **3800 Arthur Road  
Martinez, California 94553  
Contra Costa County**

**Table 1. Discharge Location**

| <b>Discharge Points</b> | <b>Effluent Description</b>                     | <b>Discharge Point Latitude (North-South)</b> | <b>Discharge Point Longitude (East-West)</b> | <b>Receiving Water</b>                         |
|-------------------------|-------------------------------------------------|-----------------------------------------------|----------------------------------------------|------------------------------------------------|
| 001                     | Advanced Secondary-Treated Municipal Wastewater | 38.021111°                                    | -122.103611°                                 | Peyton Slough, a tributary to Carquinez Strait |

This Order was adopted on: **December 15, 2021**  
This Order shall become effective on: **February 1, 2022**  
This Order shall expire on: **January 31, 2027**  
CIWQS regulatory measure number: **446204**

The Discharger shall file a Report of Waste Discharge as an application for updated WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than **May 6, 2026**. The U.S. Environmental Protection Agency (U.S. EPA) and the California Regional Water Quality Control Board, San Francisco Bay Region (Regional Water Board) have classified this discharge as “**major**.”

I hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the Regional Water Board on the date indicated above.

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Michael Montgomery, Executive Officer

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## 1. FACILITY INFORMATION

Information describing the Mt. View Sanitary District Wastewater Treatment Plant and its wastewater collection system (collectively, Facility) is summarized on the cover page and in Fact Sheet (Attachment F) sections 1 and 2. Fact Sheet section 1 also includes information regarding the permit application.

## 2. FINDINGS

The Regional Water Board finds the following:

- 2.1. Legal Authorities.** This Order serves as WDRs pursuant to California Water Code article 4, chapter 4, division 7 (commencing with § 13260). This Order is also issued pursuant to federal Clean Water Act (CWA) section 402 and implementing regulations adopted by U.S. EPA and Water Code chapter 5.5, division 7 (commencing with § 13370). It shall serve as an NPDES permit authorizing the Discharger to discharge into waters of the United States as described in Table 1 subject to the WDRs in this Order.
- 2.2. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information the Discharger submitted as part of its application, information obtained through monitoring and reporting programs, and other available information. The Fact Sheet contains background information and rationale for the requirements in this Order and is hereby incorporated into and constitutes findings for this Order. Attachments A through E, and G are also incorporated into this Order.
- 2.3. Notification of Interested Parties.** The Regional Water Board notified the Discharger and interested agencies and persons of its intent to prescribe these WDRs and has provided an opportunity to submit written comments and recommendations. Fact Sheet section 8.1 provides details regarding the notification.
- 2.4. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Fact Sheet section 8.3 provides details regarding the public hearing.

THEREFORE, IT IS HEREBY ORDERED that Order R2-2016-0023 (previous order) is rescinded upon the effective date of this Order, except for enforcement purposes, and, in order to meet the provisions contained in Water Code division 7 (commencing with § 13000) and regulations adopted thereunder and the provisions of the CWA and regulations and guidelines adopted thereunder, the Discharger shall comply with the requirements in this Order. This action in no way prevents the Regional Water Board from taking enforcement action for violations of the previous order.

### 3. DISCHARGE PROHIBITIONS

- 3.1.** Discharge of treated or partially-treated wastewater at a location or in a manner different from that described in this Order is prohibited.
- 3.2.** Bypass of untreated or partially-treated wastewater to waters of the United States is prohibited, except as provided for in Attachment D section 1.7 of this Order.

Diverting wastewater around the plant's biotower is not considered a bypass if (1) the diversion is undertaken during essential maintenance or for process control to ensure efficient operation; and (2) the discharge complies with the effluent and receiving water limits in this Order (see Attachment D, section 1.7). The Discharger shall report diversions around the biotower in its monthly monitoring reports and shall monitor ammonia as specified in the Monitoring and Reporting Program (MRP, Attachment E).

- 3.3.** Average dry weather effluent flow in excess of 3.2 million gallons per day (MGD) is prohibited. Average dry weather effluent flow shall be determined from three consecutive dry weather months each year, with compliance measured at Monitoring Location EFF-001 as described in the MRP.
- 3.4.** Any sanitary sewer overflow that results in a discharge of untreated or partially treated wastewater to waters of the United States is prohibited.

### 4. EFFLUENT LIMITATIONS

- 4.1. Effluent Limitations.** The discharge shall meet the following effluent limitations, with compliance measured at Monitoring Location EFF-001 as described in the MRP, except for ammonia, which shall be measured at Monitoring Location EFF-002:

**Table 2. Effluent Limitations**

| Parameter                              | Units          | Average Monthly        | Average Weekly | Maximum Daily          | Instantaneous Minimum | Instantaneous Maximum |
|----------------------------------------|----------------|------------------------|----------------|------------------------|-----------------------|-----------------------|
| Biochemical Oxygen Demand 5-day @ 20°C | mg/L           | 15                     | 25             | —                      | —                     | —                     |
| pH <sup>[1]</sup>                      | standard units | —                      | —              | —                      | 6.5                   | 8.5                   |
| Total Suspended Solids                 | mg/L           | 15                     | 25             | —                      | —                     | —                     |
| Ammonia, Total                         | mg/L as N      | 1.1                    | —              | 3.2                    | —                     | —                     |
| Copper, Total Recoverable              | µg/L           | 11                     | —              | 15                     | —                     | —                     |
| Cyanide, Total                         | µg/L           | 7.0                    | —              | 13                     | —                     | —                     |
| Dioxin-TEQ                             | µg/L           | 1.4 x 10 <sup>-8</sup> | —              | 2.8 x 10 <sup>-8</sup> | —                     | —                     |

Footnote:

- <sup>[1]</sup> If the Discharger monitors pH continuously, pursuant to 40 C.F.R. section 401.17 the Discharger shall be in compliance with this pH limitation provided that both of the following conditions are satisfied: (i) the total time during which the pH is outside the

required range shall not exceed 7 hours and 26 minutes in any calendar month; and (ii) no individual excursion from the required pH range shall exceed 60 minutes.

**4.2. BOD and TSS Percent Removal.** The average monthly percent removal of biochemical oxygen demand (BOD) and total suspended solids (TSS) at Discharge Point 001 shall not be less than 85 percent (i.e., in each calendar month, the arithmetic mean of BOD and TSS, by concentration, of effluent samples collected at Monitoring Location EFF-001 as described in the MRP, shall not exceed 15 percent of the arithmetic mean of BOD and TSS, by concentration, of influent samples collected at Monitoring Location INF-001 as described in the MRP, at approximately the same times during the same periods).

**4.3. *Enterococcus* Bacteria.** The discharge at Discharge Point 001 shall meet the following *Enterococcus* effluent limitations, with compliance measured at Monitoring Location EFF-001 as described in the MRP:

- 4.3.1. The six-week rolling geometric mean *Enterococcus* bacteria count shall not exceed 30 colony forming units per 100 milliliters (CFU/100 mL). Compliance with this limit shall be determined weekly by calculating the geometric mean of all *Enterococcus* sample results from the past six weeks.
- 4.3.2. No more than 10 percent of all *Enterococcus* bacteria samples collected in a calendar month shall exceed 110 CFU/100 mL. Compliance with this limit shall be determined based on measured sample results. The Discharger shall not report interpolated results. If the Discharger has nine or fewer sample results in a calendar month, compliance shall be based on the highest result. If the Discharger has 10 to 19 sample results, compliance shall be based on the second highest result, and so on.

**4.4. Acute Toxicity.** The discharge at Discharge Point 001 shall meet the following acute toxicity effluent limitations, with compliance measured at Monitoring Location EFF-001, as described in the MRP. If the Discharger can demonstrate that toxicity exceeding these acute toxicity limits is caused solely by ammonia and that the ammonia in the discharge complies with the ammonia effluent limits in Table 2 of this Order, then such toxicity shall not constitute a violation of these acute toxicity limits.

- 4.4.1. The three-sample median shall not exhibit less than 90 percent survival (i.e., a bioassay test showing survival of less than 90 percent shall represent a violation of this effluent limit if one or more of the past two bioassay tests also show less than 90 percent survival).
- 4.4.2. The single-sample maximum value shall not exhibit less than 70 percent survival (i.e., a bioassay test showing survival of less than 70 percent shall represent a violation of this effluent limitation).

## **5. RECEIVING WATER LIMITATIONS**

### **5.1. The discharge shall not cause the following conditions at any place in receiving waters:**

- 5.1.1. Floating material, including solids, liquids, foams, and scum, in concentrations that cause nuisance or adversely affect beneficial uses;
- 5.1.2. Alteration of suspended sediment in such a manner as to cause nuisance or adversely affect beneficial uses or detrimental increase in the concentrations of toxic pollutants in sediments or aquatic life;
- 5.1.3. Suspended material in concentrations that cause nuisance or adversely affect beneficial uses;
- 5.1.4. Bottom deposits or aquatic growths to the extent that such deposits or growths cause nuisance or adversely affect beneficial uses;
- 5.1.5. Alteration of temperature beyond present natural background levels unless it can be demonstrated to the satisfaction of the Regional Water Board that such alteration in temperature does not adversely affect beneficial uses;
- 5.1.6. Changes in turbidity that cause nuisance or adversely affect beneficial uses, or increases from normal background light penetration or turbidity greater than 10 percent in areas where natural turbidity is greater than 50 nephelometric turbidity units (NTU), or above 55 NTU in areas where natural turbidity is less than or equal to 50 NTU;
- 5.1.7. Coloration that causes nuisance or adversely affects beneficial uses;
- 5.1.8. Visible, floating, suspended, or deposited oil or other products of petroleum origin; or
- 5.1.9. Toxic or other deleterious substances in concentrations or quantities that cause deleterious effects on wildlife, waterfowl, or other aquatic biota, or render any of these unfit for human consumption, either at levels created in the receiving waters or as a result of biological concentration.

### **5.2. The discharge shall not cause the following limits to be exceeded at any place in receiving waters within one foot of the water surface:**

- 5.2.1. Dissolved Oxygen     5.0 mg/L, minimum

The median dissolved oxygen concentration for any three consecutive months shall not be less than 80 percent of the dissolved oxygen content at saturation. When natural factors cause concentrations less than that specified

above, the discharge shall not cause further reduction in ambient dissolved oxygen concentrations

- 5.2.2. Dissolved Sulfide      Natural background levels
- 5.2.3. pH      The pH shall not be depressed below 6.0 nor raised above 9.0. The discharge shall not cause changes greater than 0.5 pH units in normal ambient pH levels.
- 5.2.4. Nutrients      Waters shall not contain biostimulatory substances in concentrations that promote aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.

**5.3.** The discharge shall not cause a violation of any water quality standard for receiving waters adopted by the Regional Water Board or State Water Resources Control Board (State Water Board) as required by the CWA and regulations adopted thereunder beyond any mixing zone established through this Order. If more stringent water quality standards are promulgated or approved pursuant to CWA section 303, or amendments thereto, the Regional Water Board may revise or modify this Order in accordance with the more stringent standards.

## **6. PROVISIONS**

### **6.1. Standard Provisions**

- 6.1.1. The Discharger shall comply with all “Standard Provisions” in Attachment D.
- 6.1.2. The Discharger shall comply with all applicable provisions of the “Regional Standard Provisions, and Monitoring and Reporting Requirements for NPDES Wastewater Discharge Permits” in Attachment G.
- 6.1.3. If there is any conflict, duplication, or overlap between provisions in this Order, the more stringent provision shall apply.

### **6.2. Monitoring and Reporting Provisions**

The Discharger shall comply with the Monitoring and Reporting Program (MRP, Attachment E) and future revisions thereto, and applicable monitoring and reporting requirements in Attachments D and G.

### **6.3. Special Provisions**

#### **6.3.1. Reopener Provisions**

The Regional Water Board may modify or reopen this Order prior to its expiration date in any of the following circumstances as allowed by law or as otherwise authorized by law. The Discharger may request a permit modification

based on any of these circumstances. With any such request, the Discharger shall include antidegradation and anti-backsliding analyses as necessary.

- 6.3.1.1. If present or future investigations demonstrate that the discharges governed by this Order have or will have a reasonable potential to cause or contribute to adverse impacts on water quality or beneficial uses of the receiving waters;
- 6.3.1.2. If new or revised water quality objectives or total maximum daily loads (TMDLs) come into effect for San Francisco Bay or contiguous water bodies (whether statewide, regional, or site-specific). In such cases, effluent limitations in this Order may be modified as necessary to reflect the updated water quality objectives or wasteload allocations. Adoption of the effluent limitations in this Order is not intended to restrict in any way future modifications based on legally-adopted water quality objectives or TMDLs or as otherwise permitted under federal regulations governing NPDES permit modifications;
- 6.3.1.3. If translator, dilution, or other water quality studies provide a basis for determining that a permit condition should be modified;
- 6.3.1.4. If a State Water Board precedential decision, new policy, new law, or new regulation is adopted;
- 6.3.1.5. If an administrative or judicial decision on a separate NPDES permit or WDRs addresses requirements similar to this discharge; or
- 6.3.1.6. If the Discharger requests adjustments in effluent limits due to the implementation of stormwater diversion pursuant to the Municipal Regional Stormwater Permit (NPDES Permit CAS612008) for redirecting dry weather and first flush discharges from a storm drain system to the sanitary sewer system as a stormwater pollutant control strategy.

## **6.3.2. Effluent Characterization Study and Report**

- 6.3.2.1. **Study Elements.** The Discharger shall characterize and evaluate the discharge from Discharge Point 001 as required by the MRP to verify that the reasonable potential analysis conclusions of this Order remain valid and to inform the next permit reissuance. If concentrations of any of the priority pollutants listed in Attachment G, Table B, significantly increase over past performance, the Discharger shall investigate the cause of any such increase. The investigation may include, but need not be limited to, an increase in monitoring frequency, monitoring of internal process streams, and monitoring of influent sources. The Discharger shall establish remedial measures addressing any increase resulting in reasonable potential to cause or contribute to an exceedance of applicable water quality objectives. This requirement may be satisfied through identification of the constituent as a



“pollutant of concern” in the Discharger’s Pollutant Minimization Program, described in Provision 6.3.3.

**6.3.2.2. Reporting Requirements.** The Discharger shall summarize the data evaluation and any applicable source investigation in the annual self-monitoring report associated with the year in which samples were collected. The Discharger shall also report the pollutants detected at or above applicable water quality objectives (see Fact Sheet Table F-8 for the objectives) in the report’s transmittal letter. This requirement does not apply to pollutants with effluent limitations (see Table 2 of this Order).

### **6.3.3. Pollutant Minimization Program**

6.3.3.1. The Discharger shall continue to improve its existing Pollutant Minimization Program to promote minimization of pollutant loadings to the treatment plant and therefore to the receiving waters.

6.3.3.2. The Discharger shall submit an annual report no later than February 28 of each calendar year. Each annual report shall include at least the following information:

6.3.3.2.1. **Brief description of treatment plant.** The description shall include the service area and treatment plant processes.

6.3.3.2.2. **Discussion of current pollutants of concern.** Periodically, the Discharger shall analyze its circumstances to determine which pollutants are currently a problem and which pollutants may be potential future problems. This discussion shall include the reasons for choosing the pollutants.

6.3.3.2.3. **Identification of sources for pollutants of concern.** This discussion shall include how the Discharger intends to estimate and identify pollutant sources. The Discharger shall include sources or potential sources not directly within the ability or authority of the Discharger to control, such as pollutants in the potable water supply and air deposition.

6.3.3.2.4. **Identification of tasks to reduce the sources of pollutants of concern.** This discussion shall identify and prioritize tasks to address the Discharger’s pollutants of concern. The Discharger may implement the tasks by itself or participate in group, regional, or national tasks that address its pollutants of concern. The Discharger is strongly encouraged to participate in group, regional, or national tasks that address its pollutants of concern whenever it is efficient and appropriate to do so. An implementation timeline shall be included for each task.

6.3.3.2.5. **Outreach to employees.** The Discharger shall inform employees about the pollutants of concern, potential sources, and how they might be able to

help reduce the discharge of these pollutants of concern into the Facility. The Discharger may provide a forum for employees to provide input.

- 6.3.3.2.6. **Continuation of Public Outreach Program.** The Discharger shall prepare a pollution prevention public outreach program for its service area. Outreach may include participation in existing community events, such as county fairs; initiating new community events, such as displays and contests during Pollution Prevention Week; conducting school outreach programs; conducting plant tours; and providing public information in newspaper articles or advertisements, radio or television stories or spots, newsletters, utility bill inserts, or web sites. Information shall be specific to target audiences. The Discharger shall coordinate with other agencies as appropriate.
- 6.3.3.2.7. **Discussion of criteria used to measure Pollutant Minimization Program and task effectiveness.** The Discharger shall establish criteria to evaluate the effectiveness of its Pollutant Minimization Program. This discussion shall identify the specific criteria used to measure the effectiveness of each task in Provisions 6.3.3.2.3, 6.3.3.2.4, 6.3.3.2.5, and 6.3.3.2.6.
- 6.3.3.2.8. **Documentation of efforts and progress.** This discussion shall detail all of the Discharger's Pollutant Minimization Program activities during the reporting year.
- 6.3.3.2.9. **Evaluation of Pollutant Minimization Program and task effectiveness.** The Discharger shall use the criteria established in Provision 6.3.3.2.7 to evaluate the program and task effectiveness.
- 6.3.3.2.10. **Identification of specific tasks and timelines for future efforts.** Based on the evaluation, the Discharger shall explain how it intends to continue or change its tasks to more effectively reduce the amount of pollutants flowing to the treatment plant, and subsequently in its effluent.
- 6.3.3.3. The Discharger shall develop and conduct a Pollutant Minimization Program as described below when there is evidence that a priority pollutant is present in the effluent above an effluent limitation (e.g., sample results reported as detected but not quantified [DNQ] when the effluent limitation is less than the method detection limit [MDL], sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, or results of benthic or aquatic organism tissue sampling) and either:
- A sample result is reported as DNQ and the effluent limitation is less than the Reporting Level (RL); or

- A sample result is reported as not detected (ND) and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in the MRP.

6.3.3.4. If triggered for a reason set forth in Provision 6.3.3.3, above, the Discharger's Pollutant Minimization Program shall include, but not be limited to, the following actions and submittals:

- 6.3.3.4.1. An annual review and semi-annual monitoring of potential sources of the reportable priority pollutants, which may include fish tissue monitoring and other bio-uptake sampling, or alternative measures when source monitoring is unlikely to produce useful analytical data;
- 6.3.3.4.2. Quarterly monitoring for the reportable priority pollutants in the influent to the wastewater treatment system. The Executive Officer may approve alternative measures when influent monitoring is unlikely to produce useful analytical data;
- 6.3.3.4.3. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutants in the effluent at or below the effluent limitation;
- 6.3.3.4.4. Implementation of appropriate cost-effective control measures for the reportable priority pollutants, consistent with the control strategy; and
- 6.3.3.4.5. Inclusion of the following specific items within the annual report required by Provision 6.3.3.2, above:
  - 6.3.3.4.6. All Pollutant Minimization Program monitoring results for the previous year;
  - 6.3.3.4.7. List of potential sources of the reportable priority pollutants;
  - 6.3.3.4.8. Summary of all actions undertaken pursuant to the control strategy; and
  - 6.3.3.4.9. Description of actions to be taken in the following year.

#### **6.3.4. Special Provisions for Publicly-Owned Treatment Works**

##### **6.3.4.1. Sludge and Biosolids Management.**

- 6.3.4.1.1. Sludge and biosolids treatment and storage shall not create a nuisance, such as objectionable odors or flies, or result in groundwater contamination.
- 6.3.4.1.2. Sludge and biosolids treatment and storage site shall have facilities adequate to divert surface runoff from adjacent areas, to protect site boundaries from erosion, and to prevent conditions that would cause drainage from the stored materials. Adequate protection is defined as protection from at least a 100-year storm and the highest possible tidal stage that may occur.

6.3.4.1.3. This Order does not authorize permanent onsite sludge or biosolids storage or disposal. The Discharger shall file a Report of Waste Discharge and bring the site into compliance with applicable regulations prior to commencement of any such activity.

**6.3.4.2. Collection System Management.** The Discharger shall properly operate and maintain its collection system (see Attachments D and G, section 1.4), report any noncompliance with respect to its collection system (see Attachment D, section 5.5.1, and Attachment G, sections 5.5.1 and 5.5.2), and mitigate any discharges in violation of this Order associated with its collection system (see Attachments D and G, section 1.3).

State Water Board Order 2006-0003-DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems*, as amended by State Water Board Order WQ 2013-0058-EXEC (statewide WDRs), contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. The statewide WDRs clearly and specifically stipulate requirements for operation and maintenance and for reporting and mitigating sanitary sewer overflows. Implementing the requirements for operation and maintenance and mitigation of sanitary sewer overflows set forth in the statewide WDRs (and any subsequent order updating these requirements) shall satisfy the corresponding federal NPDES requirements specified in Attachments D and G of this Order for the collection systems. Following the reporting requirements set forth in the statewide WDRs (and any subsequent order updating those requirements) shall satisfy the NPDES reporting requirements for sanitary sewer overflows specified in Attachments D and G.

**6.3.4.3 Resource Recovery from Anaerobically Digestible Material.** Prior to receiving hauled-in anaerobically-digestible material for digestion in its anaerobic digesters, the Discharger shall notify the Regional Water Board of, and develop and implement Standard Operating Procedures (SOPs) for, this activity. The Discharger shall train its staff in the SOPs as part of their implementation. The Discharger's SOPs shall address:

- material transport and handling including unloading, screening, or other processing prior to anaerobic digestion;
- spill prevention and response;
- exclusion from its digesters of prohibited materials and materials that could pass through, interfere with, or upset the treatment processes;
- vector control;
- odor control;
- operation and maintenance; and
- disposal of material excluded from the digesters.

The Discharger shall maintain records of material received, and material excluded and hauled off-site, for a minimum of three years. Records of

material received shall describe the hauler, waste type, and quantity of each load. Records of material excluded and hauled off-site shall describe the disposal, location, and cumulative quantity.

### 6.3.5. Other Special Provisions

**6.3.5.1. Copper Action Plan.** The Discharger shall implement source control and pollution prevention for copper in accordance with the following tasks and time schedule:

**Table 3. Copper Action Plan**

| Task No. | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Deadline                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1        | <b>Implement Copper Control Program.</b> Continue implementing existing program to reduce identified copper sources, including, as applicable, taking the following actions: <ul style="list-style-type: none"> <li>a. Providing education and outreach to the public (e.g., focusing on proper pool and spa maintenance and plumbers' roles in reducing corrosion);</li> <li>b. If corrosion is a significant copper source, working cooperatively with local water purveyors to reduce and control water corrosivity, as appropriate, and ensuring that local plumbing contractors implement best management practices to reduce corrosion in pipes; and</li> <li>c. Educating plumbers, designers, and maintenance contractors for pools and spas to encourage best management practices that minimize copper discharges.</li> </ul> | Implementation shall be ongoing                                                                                |
| 2        | <b>Implement Additional Actions.</b> If the Regional Water Board notifies the Discharger that the three-year rolling mean dissolved copper concentration in San Pablo Bay exceeds 3.0 µg/L, then within 90 days of the notification, evaluate the effluent copper concentration trend and, if it is increasing, develop and begin implementation of additional measures to control copper discharges. Report the conclusion of the trend analysis and provide a schedule for any new actions to be taken within the next 12 months.                                                                                                                                                                                                                                                                                                     | With next annual pollution minimization program report due August 31 (at least 90 days following notification) |
| 3        | <b>Report Status.</b> Submit an annual report documenting copper control program implementation that evaluates the effectiveness of the actions taken, including any additional actions required by Task 2 above, and provides a schedule for actions to be taken within the next 12 months.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Annually with pollution minimization program report due August 31                                              |

**6.3.5.2. Cyanide Action Plan.** The Discharger shall implement monitoring and surveillance, source control, and pollution prevention for cyanide in accordance with the following tasks and time schedule:

**Table 4. Cyanide Action Plan**

| Task No. | Task                                                                                                                                                                                                                                                                                                                                                      | Deadline                                                              |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1        | <b>Review Potential Cyanide Sources.</b> Submit an up-to-date inventory of potential cyanide sources. If no cyanide source is identified, Tasks 2 and 3, below, are not required unless the Discharger receives a request to discharge detectable levels of cyanide to the sewer. In such case, notify the Executive Officer and implement Tasks 2 and 3. | With annual pollution minimization program report due August 31, 2022 |

| Task No. | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Deadline                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 2        | <p><b>Implement Cyanide Control Program.</b> Implement a control program to minimize cyanide discharges consisting, at a minimum, of the following elements:</p> <ul style="list-style-type: none"> <li>a. Inspect each potential source to assess the need to include that source in the control program.</li> <li>b. Inspect sources included in the control program annually. Inspection elements may be based on U.S. EPA guidance, such as Industrial User Inspection and Sampling Manual for POTWs (EPA 831 B 94 01).</li> <li>c. Develop and distribute educational materials regarding the need to prevent cyanide discharges to sources included in the control program.</li> <li>d. Prepare an emergency monitoring and response plan to be implemented if a significant cyanide discharge occurs.</li> </ul> <p>If the plant influent cyanide concentration exceeds 6.0 µg/L, the Discharger shall collect a follow-up sample within 5 days of becoming aware of the laboratory results. If the results of the follow-up sample also exceed 6.0 µg/L, then a “significant cyanide discharge” is occurring.</p> | Implementation shall be ongoing following Executive Officer notification under Task 1                          |
| 3        | <p><b>Implement Additional Measures.</b> If the Regional Water Board notifies the Discharger that ambient monitoring shows cyanide concentrations are 1.0 µg/L or higher in the main body of San Francisco Bay, then within 90 days of the notification, commence actions to identify and abate cyanide sources responsible for the elevated ambient concentrations, report on the progress and effectiveness of the actions taken, and provide a schedule for actions to be taken within the next 12 months..</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | With next annual pollution minimization program report due August 31 (at least 90 days following notification) |
| 4        | <p><b>Report Status of Cyanide Control Program.</b> Submit an annual report documenting cyanide control program implementation and addressing the effectiveness of actions taken, including any additional cyanide controls required by Task 3, above, and provide a schedule for actions to be taken within the next 12 months.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Annually, with pollution minimization program report due August 31 each year                                   |

**6.3.5.3. Average Annual Selenium Load.** The Discharger shall report the average annual selenium load from Discharge Point 001 with its application for permit reissuance. The average annual load shall be the arithmetic mean of the annual mass discharges for the previous permit term. Annual mass emissions shall be computed as follows:

$$\text{Annual Mass emission rate (kg/day)} = (3.785/N) \sum Q_i C_i$$

Where:

$N$  = number of samples in a year

$Q_i$  = flow rate (MGD) associated with the  $i^{\text{th}}$  sample, valid until a new sample is collected

$C_i$  = selenium concentration (mg/L) associated with the  $i^{\text{th}}$  sample, valid until a new sample is collected

When calculating selenium loads, the Discharger shall use estimated values and assume data reported below the method detection limit equal half of the detection limit.

- 6.3.5.4. Moorhen Marsh Management.** The Discharger shall implement the Moorhen Marsh Management Plan, including but not limited to all measures identified to (1) maintain and improve water quality, (2) maintain levees and other control structures, and (3) control noxious weeds and maintain and improve habitat. No later than February 1 each year, the Discharger shall review and, if necessary, update the Moorhen Marsh Management Plan. The Discharger shall summarize the changes in its annual report (see MRP § 8.2.2.2). Each update shall describe any necessary revisions, management activities completed during the previous calendar year (e.g., levee upgrades or vegetation removal), and activities planned for the next year. The updates shall incorporate plans for monitoring and managing pollutants of concern (e.g., nutrients) for Moorhen Marsh, including monitoring locations, monitoring results from the past year, and monitoring plans for the next year.

## ATTACHMENT A – DEFINITIONS AND ABBREVIATIONS

### DEFINITIONS

#### **Arithmetic Mean ( $\mu$ )**

Also called the average, sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n$$

where:  $\Sigma x$  is the sum of the measured ambient water concentrations,  
and  $n$  is the number of samples

#### **Average Monthly Effluent Limitation (AMEL)**

Highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

#### **Average Weekly Effluent Limitation (AWEL)**

Highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

#### **Bioaccumulative**

Taken up by an organism from its surrounding medium through gill membranes, through epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

#### **Carcinogenic**

Known to cause cancer in living organisms.

#### **Coefficient of Variation (CV)**

Measure of data variability calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

#### **Daily Discharge**

Either: (1) the total mass of a constituent discharged over a calendar day (12:00 a.m. through 11:59 p.m.) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit) for a constituent with limitations expressed in units of mass; or (2) the unweighted arithmetic mean measurement of a constituent over a day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.



For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period is considered the result for the calendar day in which the 24-hour period ends.

**Detected, but Not Quantified (DNQ)**

Sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

**Dilution Credit**

Amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

**Effluent Concentration Allowance (ECA)**

Value derived from the water quality criterion or objective, dilution credit, and ambient background concentration that is used, in conjunction with the CV for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as wasteload allocation (WLA) as used in U.S. EPA guidance (*Technical Support Document for Water Quality-based Toxics Control*, March 1991, second printing, EPA/505/2-90-001).

**Enclosed Bays**

Indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

**Estimated Chemical Concentration**

Concentration that results from the confirmed detection of a substance below the ML by the analytical method.

**Estuaries**

Waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters are considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220; Suisun Bay; Carquinez Strait downstream to the Carquinez Bridge; and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

### **Inland Surface Waters**

All surface waters of the state that are not the ocean, enclosed bays, or estuaries.

### **Instantaneous Maximum Effluent Limitation**

Highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

### **Instantaneous Minimum Effluent Limitation**

Lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

### **Maximum Daily Effluent Limitation (MDEL)**

Highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

### **Median**

Middle measurement in a data set. The median of a data set is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements ( $n$ ) is odd, then the median =  $X_{(n+1)/2}$ . If  $n$  is even, then the median =  $(X_{n/2} + X_{(n/2+1)})/2$  (i.e., the midpoint between  $n/2$  and  $n/2+1$ ).

### **Method Detection Limit (MDL)**

Minimum concentration of a substance that can be reported with 99 percent confidence that the measured concentration is distinguishable from method blank results, as defined in 40 C.F.R. part 136, Appendix B.

### **Minimum Level (ML)**

Concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

### **Mixing Zone**

Limited volume of receiving water allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

### **Not Detected (ND)**

Sample results less than the laboratory's MDL.

### **Persistent Pollutants**

Substances for which degradation or decomposition in the environment is nonexistent or very slow.

### **Pollutant Minimization Program**

Program of waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of a Pollutant Minimization Program is to reduce all potential sources of a priority pollutant through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. Cost effectiveness may be considered when establishing the requirements of a Pollutant Minimization Program. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), is considered to fulfill the Pollutant Minimization Program requirements.

### **Pollution Prevention**

Any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board or Regional Water Board.

### **Reporting Level (RL)**

ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. For priority pollutants, the MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from State Implementation Plan (SIP) Appendix 4 in accordance with SIP section 2.4.2 or established in accordance with SIP section 2.4.3. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

### **Source of Drinking Water**

Any water designated as municipal or domestic supply (MUN) beneficial use.

### **Standard Deviation ( $\sigma$ )**

Measure of variability calculated as follows:

$$\text{Standard deviation} = \sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:  $x$  is the observed value  
 $\mu$  is the arithmetic mean of the observed values  
 $n$  is the number of samples

### **Toxicity Reduction Evaluation (TRE)**

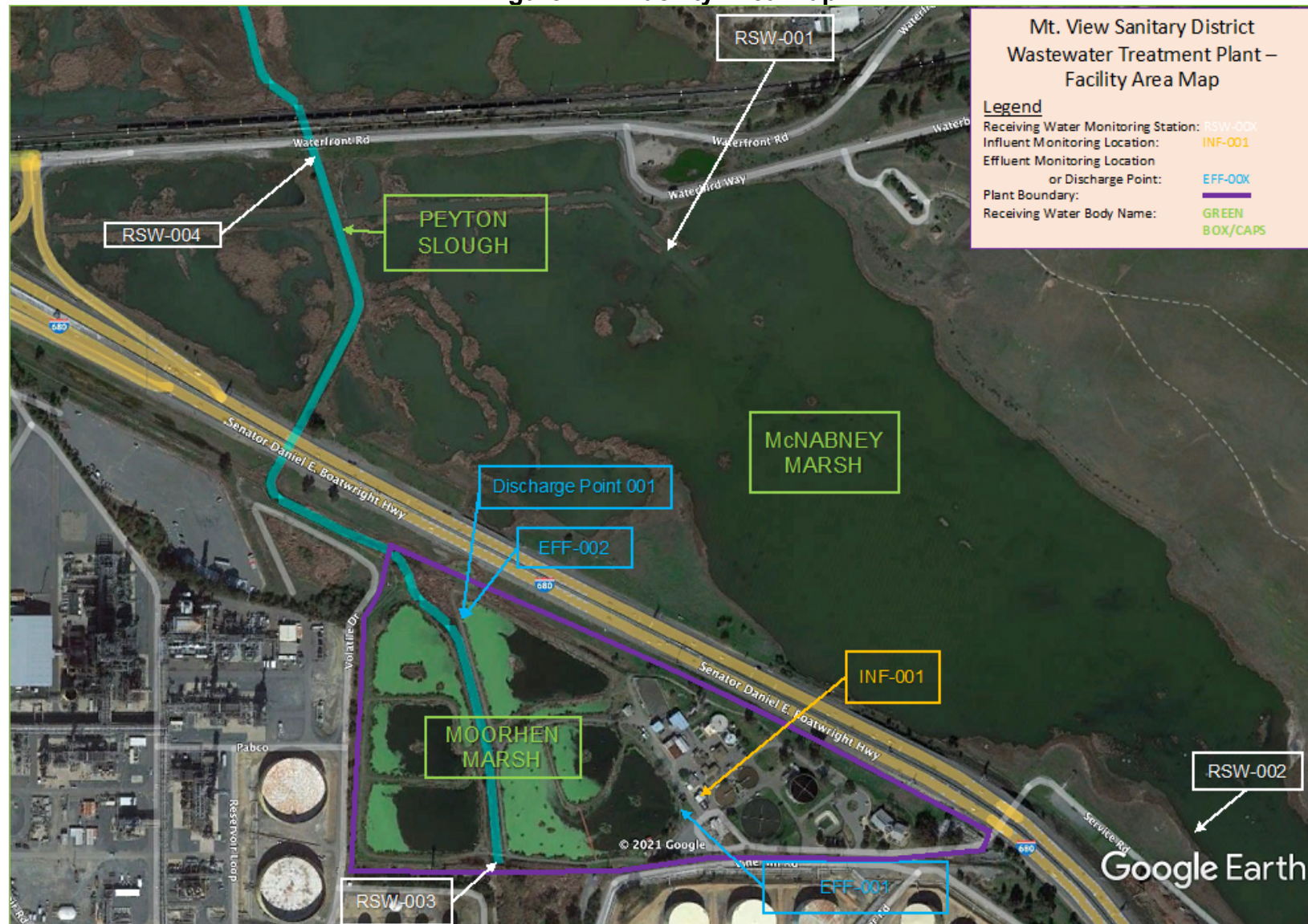
Study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. A TIE is a set of procedures to identify the specific chemicals responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.

## ABBREVIATIONS

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| <b>%</b>                | Percent                                                 |
| <b>°C</b>               | degrees Celsius                                         |
| <b>°F</b>               | degrees Fahrenheit                                      |
| <b>µg/L</b>             | Micrograms per liter                                    |
| <b>1/Blending Event</b> | Once per blending event                                 |
| <b>1/Day</b>            | Once per day                                            |
| <b>1/Month</b>          | Once per month                                          |
| <b>1/Quarter</b>        | Once per quarter                                        |
| <b>1/Week</b>           | Once per week                                           |
| <b>1/Year</b>           | Once per year                                           |
| <b>2/Month</b>          | Two times per month                                     |
| <b>2/Week</b>           | Twice per week                                          |
| <b>2/Year</b>           | Twice per year                                          |
| <b>3/Week</b>           | Three times per week                                    |
| <b>4/Week</b>           | Four times per week                                     |
| <b>5/Week</b>           | Five times per week                                     |
| <b>AMEL</b>             | Average monthly effluent limitation                     |
| <b>AWEL</b>             | Average weekly effluent limitation                      |
| <b>B</b>                | Background concentration                                |
| <b>C</b>                | Water quality criterion or objective                    |
| <b>C-24</b>             | 24-hour composite                                       |
| <b>CFU/100 mL</b>       | Colony forming units per 100 milliliters                |
| <b>Continuous</b>       | Measured continuously                                   |
| <b>Continuous/D</b>     | Measured continuously, and recorded and reported daily  |
| <b>Continuous/H</b>     | Measured continuously, and recorded and reported hourly |
| <b>CV</b>               | Coefficient of Variation                                |
| <b>DNQ</b>              | Detected, but not quantified                            |
| <b>DL</b>               | Detection level                                         |
| <b>ECA</b>              | Effluent Concentration Allowance                        |
| <b>Grab</b>             | Grab sample                                             |
| <b>MDEL</b>             | Maximum Daily Effluent Limitation                       |

|                   |                                          |
|-------------------|------------------------------------------|
| <b>MDL</b>        | Method detection limit                   |
| <b>MEC</b>        | Maximum effluent concentration           |
| <b>MG</b>         | Million gallons                          |
| <b>mg/L</b>       | Milligrams per liter                     |
| <b>mg/L as N</b>  | Milligrams per liter as nitrogen         |
| <b>MGD</b>        | Million gallons per day                  |
| <b>ML</b>         | Minimum level                            |
| <b>MPN/100 mL</b> | Most probable number per 100 milliliters |
| <b>ND</b>         | Not detected                             |
| <b>NTU</b>        | Nephelometric turbidity units            |
| <b>ppt</b>        | Parts per thousand                       |
| <b>RL</b>         | Reporting level                          |
| <b>RPA</b>        | Reasonable potential analysis            |
| <b>s.u.</b>       | Standard pH units                        |
| <b>TIE</b>        | Toxicity identification evaluation       |
| <b>TRE</b>        | Toxicity reduction evaluation            |
| <b>TUa</b>        | Acute toxicity units                     |
| <b>TUc</b>        | Chronic toxicity units                   |

**ATTACHMENT B – MAPS**  
**Figure B-1. Facility Area Map**



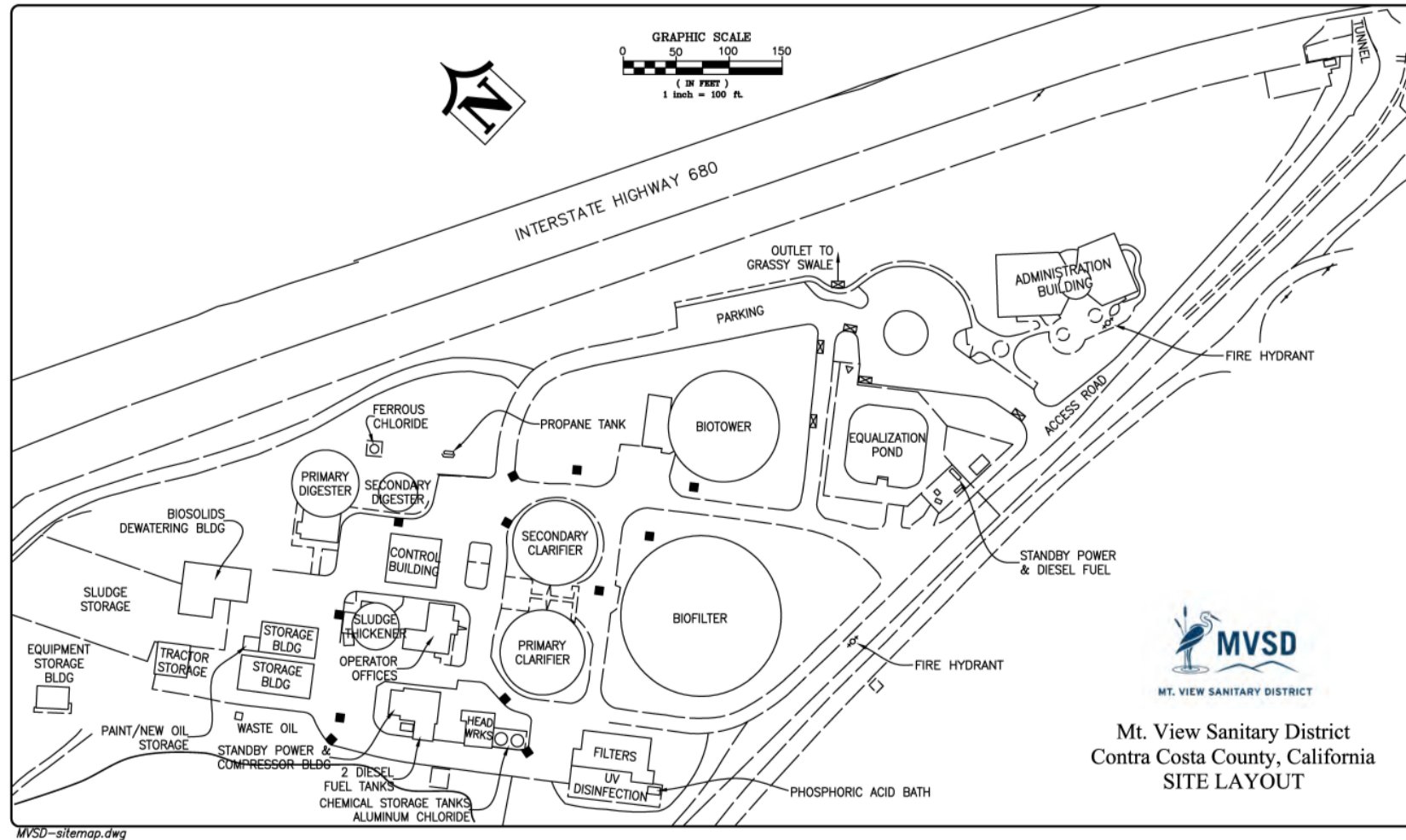


**Figure B-2. Topographic Map of Facility**

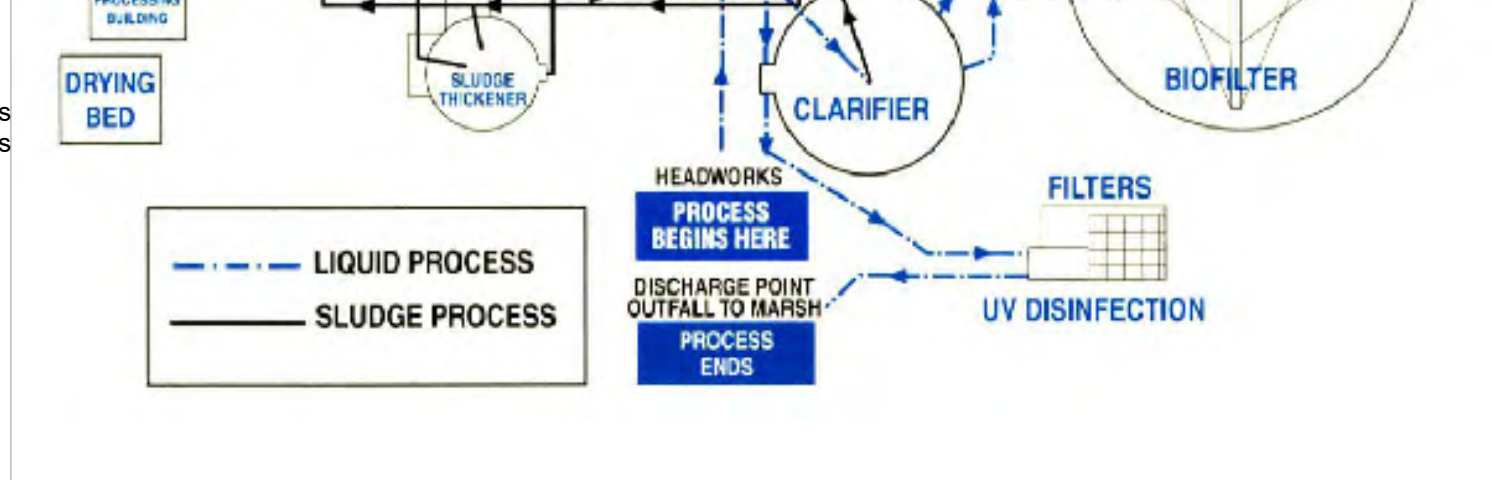




Figure B-3. Facility Layout



Mt. View Sanitary Dis  
Mt. View Sanitary Dis



## **ATTACHMENT D – STANDARD PROVISIONS**

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## **ATTACHMENT D – STANDARD PROVISIONS**

### **1. STANDARD PROVISIONS – PERMIT COMPLIANCE**

#### **1.1. Duty to Comply**

- 1.1.1. The Discharger must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action; permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 C.F.R. § 122.41(a); Wat. Code, §§ 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
- 1.1.2. The Discharger shall comply with effluent standards or prohibitions established under CWA section 307(a) for toxic pollutants within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

- 1.2. **Need to Halt or Reduce Activity Not a Defense.** It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

- 1.3. **Duty to Mitigate.** The Discharger shall take all reasonable steps to minimize or prevent any discharge in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

- 1.4. **Proper Operation and Maintenance.** The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) that are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

#### **1.5. Property Rights**

- 1.5.1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)
- 1.5.2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

**1.6. Inspection and Entry.** The Discharger shall allow the Regional Water Board, State Water Board, U.S. EPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. § 1318(a)(4)(B); 40 C.F.R. § 122.41(i); Wat. Code, §§ 13267, 13383):

- 1.6.1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. § 1318(a)(4)(B)(i); 40 C.F.R. § 122.41(i)(1); Wat. Code, §§ 13267, 13383);
- 1.6.2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(2); Wat. Code, §§ 13267, 13383);
- 1.6.3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(3); Wat. Code, §§ 13267, 13383); and
- 1.6.4. Sample or monitor, at reasonable times, for the purposes of ensuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C. § 1318(a)(4)(B); 40 C.F.R. § 122.41(i)(4); Wat. Code, §§ 13267, 13383.)

## **1.7. Bypass**

### **1.7.1. Definitions**

- 1.7.1.1. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
- 1.7.1.2. "Severe property damage" means substantial physical damage to property; damage to the treatment facilities, which causes them to become inoperable; or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
- 1.7.2. **Bypass not exceeding limitations.** The Discharger may allow any bypass to occur that does not cause exceedances of effluent limitations, but only if it is for essential maintenance to ensure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance sections 1.7.3, 1.7.4, and 1.7.5 below. (40 C.F.R. § 122.41(m)(2).)
- 1.7.3. **Prohibition of bypass.** Bypass is prohibited, and the Regional Water Board may take enforcement action against a Discharger for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):

- 1.7.3.1. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));
- 1.7.3.2. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and
- 1.7.3.3. The Discharger submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance section 1.7.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)
- 1.7.4. **Approval.** The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance section 1.7.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)
- 1.7.5. **Notice**
  - 1.7.5.1. **Anticipated bypass.** If the Discharger knows in advance of the need for a bypass, it shall submit prior notice, if possible, at least 10 days before the date of the bypass. The notice shall be sent to the Regional Water Board. As of December 21, 2025, a notice shall also be submitted electronically to the initial recipient defined in Standard Provisions – Reporting section 5.10 below. Notices shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(m)(3)(i).)
  - 1.7.5.2. **Unanticipated bypass.** The Discharger shall submit a notice of an unanticipated bypass as required in Standard Provisions – Reporting section 5.5 below (24-hour notice). The notice shall be sent to the Regional Water Board. As of December 21, 2025, a notice shall also be submitted electronically to the initial recipient defined in Standard Provisions – Reporting section 5.10 below. Notices shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(m)(3)(ii).)
- 1.8. **Upset.** Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology-based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

- 1.8.1. **Effect of an upset.** An upset constitutes an affirmative defense to an action brought for noncompliance with such technology-based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance section 1.8.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).)
- 1.8.2. **Conditions necessary for a demonstration of upset.** A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
- 1.8.2.1. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
- 1.8.2.2. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
- 1.8.2.3. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting section 5.5.2.2 below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
- 1.8.2.4. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance section 1.3 above. (40 C.F.R. § 122.41(n)(3)(iv).)
- 1.8.3. **Burden of proof.** In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

## 2. STANDARD PROVISIONS – PERMIT ACTION

- 2.1. **General.** This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)
- 2.2. **Duty to Reapply.** If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)
- 2.3. **Transfers.** This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and Water Code. (40 C.F.R. §§ 122.41(l)(3), 122.61.)

### **3. STANDARD PROVISIONS – MONITORING**

- 3.1.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- 3.2.** Monitoring must be conducted according to test procedures approved under 40 C.F.R. part 136 for the analyses of pollutants unless another method is required under 40 C.F.R. chapter 1, subchapter N. Monitoring must be conducted according to sufficiently sensitive test methods approved under 40 C.F.R. part 136 for the analysis of pollutants or pollutant parameters or as required under 40 C.F.R. chapter 1, subchapter N. For the purposes of this paragraph, a method is sufficiently sensitive when:
- 3.2.1.** The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
- 3.2.2.** The method has the lowest ML of the analytical methods approved under 40 C.F.R. part 136 or required under 40 C.F.R. chapter 1, subchapter N, for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 C.F.R. part 136 or otherwise required under 40 C.F.R. chapter 1, subchapter N, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 C.F.R. §§ 122.21(e)(3), 122.41(j)(4), 122.44(i)(1)(iv).)

### **4. STANDARD PROVISIONS – RECORDS**

- 4.1.** The Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report, or application. This period may be extended by request of the Regional Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)
- 4.2.** Records of monitoring information shall include:
- 4.2.1.** The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));



- 4.2.2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
  - 4.2.3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
  - 4.2.4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
  - 4.2.5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
  - 4.2.6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)
- 4.3. Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):**
- 4.3.1. The name and address of any permit applicant or Discharger (40 C.F.R. § 122.7(b)(1)); and
  - 4.3.2. Permit applications and attachments, permits, and effluent data. (40 C.F.R. § 122.7(b)(2).)

## **5. STANDARD PROVISIONS – REPORTING**

**5.1. Duty to Provide Information.** The Discharger shall furnish to the Regional Water Board, State Water Board, or U.S. EPA within a reasonable time, any information that the Regional Water Board, State Water Board, or U.S. EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Regional Water Board, State Water Board, or U.S. EPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, §§ 13267, 13383.)

### **5.2. Signatory and Certification Requirements**

- 5.2.1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or U.S. EPA shall be signed and certified in accordance with Standard Provisions – Reporting sections 5.2.2, 5.2.3, 5.2.4, 5.2.5, and 5.2.6 below. (40 C.F.R. § 122.41(k).)
- 5.2.2. For a corporation, all permit applications shall be signed by a responsible corporate officer. For the purpose of this section, a responsible corporate officer means: (1) a president, secretary, treasurer, or vice-president of the corporation in charge of a principal business function, or any other person who performs similar policy- or decision-making functions for the corporation, or (2) the manager of one or more manufacturing, production, or operating facilities, provided the manager is authorized to make management decisions that govern the operation of the regulated facility, including having the explicit or implicit duty of making major capital investment recommendations, and initiating and directing other comprehensive measures to ensure long term environmental

compliance with environmental laws and regulations; the manager can ensure that the necessary systems are established or actions taken to gather complete and accurate information for permit application requirements; and where authority to sign documents has been assigned or delegated to the manager in accordance with corporate procedures. (40 C.F.R. § 122.22(a)(1).)

For a partnership or sole proprietorship, all permit applications shall be signed by a general partner or the proprietor, respectively. (40 C.F.R. § 122.22(a)(2).)

For a municipal, state, federal, or other public agency, all permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes (1) the chief executive officer of the agency, or (2) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of U.S. EPA). (40 C.F.R. § 122.22(a)(3).)

- 5.2.3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or U.S. EPA shall be signed by a person described in Standard Provisions – Reporting section 5.2.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - 5.2.3.1. The authorization is made in writing by a person described in Standard Provisions – Reporting section 5.2.2 above (40 C.F.R. § 122.22(b)(1));
  - 5.2.3.2. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity, such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and
  - 5.2.3.3. The written authorization is submitted to the Regional Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
- 5.2.4. If an authorization under Standard Provisions – Reporting section 5.2.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting section 5.2.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
- 5.2.5. Any person signing a document under Standard Provisions – Reporting section 5.2.2 or 5.2.3 above shall make the following certification:

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations." (40 C.F.R. § 122.22(d).)

- 5.2.6. Any person providing the electronic signature for documents described in Standard Provisions – Reporting sections 5.2.1, 5.2.2, or 5.2.3 that are submitted electronically shall meet all relevant requirements of Standard Provisions – Reporting section 5.2, and shall ensure that all relevant requirements of 40 C.F.R. part 3 (Cross-Media Electronic Reporting) and 40 C.F.R. part 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 C.F.R. § 122.22(e).)

### **5.3. Monitoring Reports**

- 5.3.1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.41(l)(4).)
- 5.3.2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board. All reports and forms must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting section 5.10 and comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. (40 C.F.R. § 122.41(l)(4)(i).)
- 5.3.3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 C.F.R. part 136, or another method required for an industry-specific waste stream under 40 C.F.R. chapter 1, subchapter N, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or reporting form specified by the Regional Water Board or State Water Board. (40 C.F.R. § 122.41(l)(4)(ii).)
- 5.3.4. Calculations for all limitations that require averaging of measurements shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

- 5.4. Compliance Schedules.** Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

## 5.5. Twenty-Four Hour Reporting

- 5.5.1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written report shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and, if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (i.e., combined sewer overflow, sanitary sewer overflow, or bypass event), type of overflow structure (e.g., manhole, combined sewer overflow outfall), discharge volume untreated by the treatment works treating domestic sewage, types of human health and environmental impacts of the event, and whether the noncompliance was related to wet weather.

As of December 21, 2025, all reports related to combined sewer overflows, sanitary sewer overflows, or bypass events must be submitted to the Regional Water Board and must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting section 5.10. The reports shall comply with 40 C.F.R. part 3, 40 C.F.R. section 122.22, and 40 C.F.R. part 127. The Regional Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 C.F.R. § 122.41(l)(6)(i).)

- 5.5.2. The following shall be included as information that must be reported within 24 hours:
- 5.5.2.1. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)
- 5.5.2.2. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)
- 5.5.3. The Regional Water Board may waive the above required written report on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(ii)(B).)

- 5.6. Planned Changes.** The Discharger shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

- 5.6.1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40 C.F.R. section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
- 5.6.2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order unless the discharge is an existing manufacturing, commercial, mining, or silvicultural discharge as referenced in 40 C.F.R. section 122.42(a). (40 C.F.R. § 122.41(l)(1)(ii).) If the discharge is an existing manufacturing, commercial, mining, or silvicultural discharge as referenced in 40 C.F.R. section 122.42(a), this notification applies to pollutants that are subject neither to effluent limitations in this Order nor to notification requirements under 40 C.F.R. section 122.42(a)(1) (see Additional Provisions – Notification Levels section 7.1.1). (40 C.F.R. § 122.41(l)(1)(ii).)
- 5.7. Anticipated Noncompliance.** The Discharger shall give advance notice to the Regional Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 C.F.R. § 122.41(l)(2).)
- 5.8. Other Noncompliance.** The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting sections 5.3, 5.4, and 5.5 above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting section 5.5 above. For noncompliance related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports shall contain the information described in Standard Provision – Reporting section 5.5 and the applicable required data in appendix A to 40 C.F.R. part 127. The Regional Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 C.F.R. § 122.41(l)(7).)
- 5.9. Other Information.** When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or U.S. EPA, the Discharger shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)
- 5.10. Initial Recipient for Electronic Reporting Data.** The owner, operator, or duly authorized representative is required to electronically submit NPDES information specified in appendix A to 40 C.F.R. part 127 to the initial recipient defined in 40 C.F.R. section 127.2(b). U.S. EPA will identify and publish the list of initial recipients on its website and in the Federal Register, by state and by NPDES data group [see 40 C.F.R. section 127.2(c)]. U.S. EPA will update and maintain this list. (40 C.F.R. § 122.41(l)(9).)

## 6. STANDARD PROVISIONS – ENFORCEMENT

- 6.1. The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, Water Code sections 13268, 13385, 13386, and 13387.

## 7. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

- 7.1. **Non-Municipal Facilities.** Existing manufacturing, commercial, mining, and silvicultural Dischargers shall notify the Regional Water Board as soon as they know or have reason to believe (40 C.F.R. § 122.42(a)):

- 7.1.1. That any activity has occurred or will occur that would result in the discharge, on a routine or frequent basis, of any toxic pollutant that is not limited in this Order, if that discharge will exceed the highest of the following “notification levels” (40 C.F.R. § 122.42(a)(1)):
- 7.1.1.1. 100 micrograms per liter (µg/L) (40 C.F.R. § 122.42(a)(1)(i));
  - 7.1.1.2. 200 µg/L for acrolein and acrylonitrile; 500 µg/L for 2,4 dinitrophenol and 2-methyl 4,6 dinitrophenol; and 1 milligram per liter (mg/L) for antimony (40 C.F.R. § 122.42(a)(1)(ii));
  - 7.1.1.3. Five (5) times the maximum concentration reported for that pollutant in the Report of Waste Discharge (40 C.F.R. § 122.42(a)(1)(iii)); or
  - 7.1.1.4. The level established by the Regional Water Board in accordance with 40 C.F.R. section 122.44(f). (40 C.F.R. § 122.42(a)(1)(iv).)
- 7.1.2. That any activity has occurred or will occur that would result in the discharge, on a non-routine or infrequent basis, of any toxic pollutant that is not limited in this Order, if that discharge will exceed the highest of the following “notification levels” (40 C.F.R. § 122.42(a)(2)):
- 7.1.2.1. 500 micrograms per liter (µg/L) (40 C.F.R. § 122.42(a)(2)(i));
  - 7.1.2.2. 1 milligram per liter (mg/L) for antimony (40 C.F.R. § 122.42(a)(2)(ii));
  - 7.1.2.3. Ten (10) times the maximum concentration reported for that pollutant in the Report of Waste Discharge (40 C.F.R. § 122.42(a)(2)(iii)); or
  - 7.1.2.4. The level established by the Regional Water Board in accordance with 40 C.F.R. section 122.44(f). (40 C.F.R. § 122.42(a)(2)(iv).)

## 7.2 Publicly Owned Treatment Works (POTWs)

- 7.2.1. All POTWs shall provide adequate notice to the Regional Water Board of any new introduction of pollutants into the POTW from an indirect discharger that

would be subject to CWA sections 301 or 306 if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)).

- 7.2.2. All POTWs shall provide adequate notice to the Regional Water Board of any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of this Order. (40 C.F.R. § 122.42(b)(2).)
- 7.2.3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3).)

## **ATTACHMENT E – MONITORING AND REPORTING PROGRAM**

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## **ATTACHMENT E – MONITORING AND REPORTING PROGRAM**

Clean Water Act (CWA) section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code section 13383 also authorizes the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This Monitoring and Reporting Program (MRP) establishes monitoring, reporting, and recordkeeping requirements that implement the federal and State laws and regulations.

### **1. GENERAL MONITORING PROVISIONS**

- 1.1. The Discharger shall comply with this MRP. The Executive Officer may amend this MRP pursuant to 40 C.F.R. section 122.63. If any discrepancies exist between this MRP and the “Regional Standard Provisions, and Monitoring and Reporting Requirements (Supplement to Attachment D) for NPDES Wastewater Discharge Permits” (Attachment G), this MRP shall prevail.
- 1.2. The Discharger shall conduct all monitoring in accordance with Attachment D section 3, as supplemented by Attachment G. Equivalent test methods must be more sensitive than those specified in 40 C.F.R. section 136 and must be specified in this permit.
- 1.3. For the analysis of monitoring samples, the Discharger shall use laboratories certified by the State Water Resources Control Board (State Water Board) in accordance with Water Code section 13176 and shall obtain quality assurance/quality control data with laboratory reports. For any onsite field tests (e.g., turbidity, pH, temperature, dissolved oxygen, conductivity, disinfectant residual) analyzed by a noncertified laboratory, the Discharger shall implement a Quality Assurance-Quality Control Program. The Discharger shall keep a manual onsite containing the steps followed in this program and shall demonstrate sufficient capability to adequately perform these field tests (e.g., qualified and trained employees, properly calibrated and maintained field instruments). The program shall conform to U.S. EPA guidelines or other approved procedures.
- 1.4. The Discharger shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board electronically to the DMR-QA Officer via email.

### **2. MONITORING LOCATIONS**

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements of this Order:

**Table E-1. Monitoring Locations**

| Discharge Point | Monitoring Location | Monitoring Location Description                                                                                                                       |
|-----------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Influent        | INF-001             | A point in the plant headworks at which all waste tributary to the treatment system is present.                                                       |
| Effluent        | EFF-001             | A point prior to the point of discharge to Moorhen Marsh where all phases of treatment are complete and all waste tributary to this point is present. |
| Effluent        | EFF-002             | A point where treated water is discharged from Moorhen Marsh to Peyton Slough.                                                                        |
| Receiving Water | RSW-001             | A point in McNabney Marsh, south of the corner of Waterbird Way and Waterfront Road.                                                                  |
| Receiving Water | RSW-002             | A point in the southeastern part of McNabney Marsh.                                                                                                   |
| Receiving Water | RSW-003             | A point in Upper Peyton Slough, located upstream of the Pond A discharge weir.                                                                        |
| Receiving Water | RSW-004             | A point in Upper Peyton Slough, located 30 feet upstream of the culvert under Waterfront Road.                                                        |

### 3. INFLUENT MONITORING

3.1. The Discharger shall monitor plant influent at Monitoring Location INF-001 as follows:

**Table E-2. Influent Monitoring**

| Parameter                                                     | Unit | Sample Type | Minimum Sampling Frequency |
|---------------------------------------------------------------|------|-------------|----------------------------|
| Biochemical Oxygen Demand (5-day @ 20°C) (BOD) <sup>[1]</sup> | mg/L | C-24        | 1/Week                     |
| Total Suspended Solids (TSS) <sup>[1]</sup>                   | mg/L | C-24        | 1/Week                     |
| Cyanide <sup>[2]</sup>                                        | µg/L | Grab        | 1/Year                     |

Footnotes:

<sup>[1]</sup> The Discharger shall collect influent samples on the same days as effluent samples.

<sup>[2]</sup> The Discharger may, at its option, analyze for cyanide as weak acid dissociable cyanide using protocols specified in 40 C.F.R. Part 136, or an equivalent method in the latest Standard Method edition.

### 4. EFFLUENT MONITORING

4.1. The Discharger shall monitor plant effluent at Monitoring Location EFF-001, or at Monitoring Location EFF-002 where indicated, as follows:

**Table E-3. Effluent Monitoring**

| Parameter                                   | Unit       | Sample Type | Minimum Sampling Frequency |
|---------------------------------------------|------------|-------------|----------------------------|
| Flow <sup>[1]</sup>                         | MG/MGD     | Continuous  | Continuous/D               |
| Biochemical Oxygen Demand (5-day @ 20°C)    | mg/L       | C-24        | 1/Week                     |
| Total Suspended Solids                      | mg/L       | C-24        | 1/Week                     |
| <i>Enterococcus</i> Bacteria <sup>[2]</sup> | CFU/100 mL | Grab        | 3/Week                     |
| pH <sup>[3, 4]</sup>                        | s.u.       | Grab        | 5/Week <sup>[5]</sup>      |

| Parameter                          | Unit            | Sample Type          | Minimum Sampling Frequency |
|------------------------------------|-----------------|----------------------|----------------------------|
| Ammonia, Total <sup>[3]</sup>      | mg/L as N       | C-24 or Grab         | 1/Month <sup>[5]</sup>     |
| Copper, Total Recoverable          | µg/L            | C-24                 | 1/Month                    |
| Cyanide, Total <sup>[6]</sup>      | µg/L            | Grab                 | 1/Month                    |
| Dioxin-TEQ                         | µg/L            | Grab                 | 1/Year                     |
| Selenium                           | µg/L            | C-24                 | 1/Year                     |
| Temperature <sup>[3]</sup>         | °C              | Grab                 | 1/Month <sup>[5]</sup>     |
| Acute Toxicity <sup>[7]</sup>      | % survival      | Flow-through         | 2/Year                     |
| Chronic Toxicity <sup>[8]</sup>    | TU <sub>c</sub> | C-24                 | 1/Quarter                  |
| Priority Pollutants <sup>[9]</sup> | µg/L            | Grab <sup>[10]</sup> | Once                       |

Footnotes:

- <sup>[1]</sup> The following flow information shall be reported in monthly self-monitoring reports:
- Daily average flow rate (MGD)
  - Total monthly flow volume (MG)
- <sup>[2]</sup> U.S. EPA Method 1600 or an equivalent method is suggested to measure culturable enterococci. Results may be reported as either Most Probable Number (MPN)/100 mL if the laboratory method used provides results in MPN/100 mL or Colony Forming Units (CFU)/100 mL if the laboratory method used provides results in CFU/100 mL.
- <sup>[3]</sup> Ammonia, temperature, and pH monitoring shall occur concurrently and at the same frequency at Monitoring Location EFF-002 to allow for calculation of the un-ionized ammonia fraction. The Discharger may monitor for other nutrient parameters (e.g., nitrate-nitrite and total phosphorus) required by NPDES Permit CA0038873 at Monitoring Location EFF-001 or EFF-002.
- <sup>[4]</sup> pH monitoring for compliance with the effluent limits specified in Table 2 of this Order shall occur at Monitoring Location EFF-001.
- <sup>[5]</sup> The monitoring frequency for ammonia, pH, and temperature shall be once per day when the biotower is not operating.
- <sup>[6]</sup> The Discharger may, at its option, analyze for cyanide as weak acid dissociable cyanide using protocols specified in 40 C.F.R. Part 136 or an equivalent method in the latest Standard Method edition.
- <sup>[7]</sup> Acute toxicity tests shall be performed in accordance with MRP section 5.1.
- <sup>[8]</sup> Chronic toxicity tests shall be performed in accordance with MRP section 5.2.
- <sup>[9]</sup> The Discharger shall monitor for the pollutants listed in Attachment G, Table B.
- <sup>[10]</sup> The Discharger shall collect C-24 samples for metals.

## 5. TOXICITY MONITORING

### 5.1. Acute Toxicity

- 5.1.1. Compliance with the acute toxicity effluent limitations shall be evaluated at Monitoring Location EFF-001 by measuring survival of test organisms exposed to 96-hour continuous flow-through bioassays.
- 5.1.2. Test organisms shall be fathead minnow (*Pimephales promelas*). Alternatively, the Executive Officer may specify a more sensitive organism or, if testing a particular organism proves unworkable, the most sensitive organism available.
- 5.1.3. All bioassays shall be performed according to the most up-to-date protocols in 40 C.F.R. part 136, currently *Methods for Measuring the Acute Toxicity of Effluents and Receiving Water to Freshwater and Marine Organisms*, 5<sup>th</sup> Edition (EPA-821-R-02-012). If these protocols prove unworkable, the Executive Officer and the Environmental Laboratory Accreditation Program may grant exceptions in writing upon the Discharger's request with justification.

- 5.1.4. If the Discharger demonstrates that specific identifiable substances in the discharge are rapidly rendered harmless upon discharge to the receiving water, compliance with the acute toxicity limit may be determined after test samples are adjusted to remove the influence of those substances. Written acknowledgement that the Executive Officer concurs with the Discharger's demonstration and that the adjustment will not remove the influence of other substances must be obtained prior to any such adjustment. The Discharger is authorized to adjust the effluent pH in order to suppress the level of un-ionized (free) ammonia. This adjustment shall be achieved by continuously monitoring test tank pH and automatic addition of analytical grade acid as needed, using a combination of continuous pH-sensor/analyzer and pump.
- 5.1.5. Bioassay monitoring shall include, on a daily basis, pH, dissolved oxygen, ammonia (if toxicity is observed), temperature, hardness, and alkalinity. These results shall be reported. If final or intermediate results of an acute bioassay test indicate a violation or threatened violation (e.g., the percentage of surviving test organisms is less than 70 percent), the Discharger shall initiate a new test as soon as practical and shall investigate the cause of the mortalities and report its findings in the next self-monitoring report. The Discharger shall repeat the test until a test fish survival rate of 90 percent or greater is observed. If the control fish survival rate is less than 90 percent, the bioassay test shall be restarted with new fish and shall continue as soon as practical until an acceptable test is completed (i.e., control fish survival rate is 90 percent or greater).

## 5.2. Chronic Toxicity

### 5.2.1. Monitoring Requirements

- 5.2.1.1. **Sampling.** The Discharger shall collect 24-hour composite effluent samples at Monitoring Location EFF-001 for critical life stage toxicity testing as indicated below. For toxicity tests requiring renewals, the Discharger shall collect 24-hour composite samples on consecutive or alternating days.
- 5.2.1.2. **Test Species.** The test species shall be fathead minnow (*Pimephales promelas*). The Discharger shall conduct a screening chronic toxicity test as described in Appendix E-1, or as described in applicable State Water Board plan provisions that become effective after adoption of this Order, following any significant change in the nature of the effluent. If there is no significant change in the nature of the effluent, the Discharger shall conduct a screening test and submit the results with its application for permit reissuance. Upon completion of the chronic toxicity screening, the Discharger shall use the most sensitive species to conduct subsequent monitoring.
- 5.2.1.3. **Frequency.** Chronic toxicity monitoring shall be as specified below:
- 5.2.1.3.1. The Discharger shall monitor routinely once per quarter.

- 5.2.1.3.2. The Discharger shall accelerate monitoring to monthly after exceeding a three-sample median of 1.0 TUC or a single sample maximum of 2.0 TUC. Based on the TUC results, the Executive Officer may specify a different frequency for accelerated monitoring to ensure that accelerated monitoring provides useful information.
- 5.2.1.3.3. The Discharger shall return to routine monitoring if accelerated monitoring does not exceed the trigger in section 5.2.1.3.2, above.
- 5.2.1.3.4. If accelerated monitoring confirms consistent toxicity in excess of the trigger in section 5.2.1.3.2, above, the Discharger shall continue accelerated monitoring and initiate toxicity reduction evaluation (TRE) procedures in accordance with section 5.2.3, below.
- 5.2.1.3.5. The Discharger shall return to routine monitoring after implementing appropriate elements of the TRE, and either the toxicity drops below the trigger in 5.2.1.3.2, above, or, based on the TRE results, the Executive Officer determines that accelerated monitoring would no longer provide useful information.
- 5.2.1.3.6. Monitoring conducted pursuant to a TRE shall satisfy the requirements for routine and accelerated monitoring while the TRE is underway.
- 5.2.1.4. **Methodology.** Sample collection, handling, and preservation shall be in accordance with U.S. EPA protocols. Bioassays shall be conducted in compliance with the most recently promulgated test methods, as shown in Appendix E-2. These are *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to West Coast Marine and Estuarine Organisms*, currently 1st edition (EPA/600/R-95-136). If these protocols prove unworkable, the Executive Officer and the Environmental Laboratory Accreditation Program may grant exceptions in writing upon the Discharger's request with justification.

If the Discharger demonstrates that specific identifiable substances in the discharge are rapidly rendered harmless upon discharge to the receiving water, compliance with the chronic toxicity limit may be determined after test samples are adjusted to remove the influence of those substances. The adjustment shall not remove the influence of other substances. Written acknowledgement that the Executive Officer concurs with the Discharger's demonstration must be obtained prior to any such adjustment.

- 5.2.1.5. **Dilution Series.** The Discharger shall conduct tests with a control and at 100%, 50%, 25%, 10%, and 5% effluent. The "%" represents percent effluent as discharged. Test sample pH may be controlled to the level of the effluent sample as received prior to being salted up.

5.2.2. **Reporting Requirements.** The Discharger shall provide toxicity test results with self-monitoring reports and shall include the following, at a minimum, for each test:

- 5.2.2.1. Sample date;
- 5.2.2.2. Test initiation date;
- 5.2.2.3. Test species;
- 5.2.2.4. End point values for each dilution (e.g., number of young, growth rate, percent survival);
- 5.2.2.5. No Observable Effect Level (NOEL) values in percent effluent. The NOEL shall equal the IC25 or EC25 (see MRP Appendix E-1). If the IC25 or EC25 cannot be statistically determined, the NOEL shall equal the No Observable Effect Concentration (NOEC) derived using hypothesis testing. The NOEC is the maximum percent effluent concentration that causes no observable effect on test organisms based on a critical life stage toxicity test;
- 5.2.2.6. IC15, IC25, IC40, and IC50 values (or EC15, EC25, EC40, and EC50) as percent effluent;
- 5.2.2.7. TU values (100/NOEL and upper and lower confidence intervals, where NOEL = IC25, EC25, or NOEC);
- 5.2.2.8. Mean percent mortality ( $\pm$ standard deviation) after 96 hours in 100% effluent (if applicable);
- 5.2.2.9. IC50 or EC50 values for reference toxicant tests;
- 5.2.2.10. Available water quality measurements for each test (e.g., pH, dissolved oxygen, temperature, conductivity, hardness, salinity, ammonia); and
- 5.2.2.11. The results of the three most recent chronic toxicity tests and the three-sample median for these results.

5.2.3. **Toxicity Reduction Evaluation (TRE)**

- 5.2.3.1. The Discharger shall prepare a generic TRE work plan within 90 days of the effective date of this Order to be ready to respond to toxicity events. The Discharger shall review and update the work plan as necessary so that it remains current and applicable to the discharge and discharge facilities.
- 5.2.3.2. Within 30 days of exceeding the chronic toxicity trigger in section 5.2.1.3.2, above, the Discharger shall submit a TRE work plan, which shall be the generic work plan revised as appropriate for this toxicity event after consideration of available discharge data.

- 5.2.3.3. Within 30 days of completing an accelerated monitoring test observed to exceed the trigger in section 5.2.1.3.2, above, the Discharger shall initiate a TRE in accordance with a TRE work plan that incorporates any and all Executive Officer comments.
- 5.2.3.4. The TRE shall be specific to the discharge and be in accordance with current technical guidance and reference materials, including U.S. EPA guidance materials. The Discharger shall conduct the TRE as a tiered evaluation as summarized below:
  - 5.2.3.4.1. Tier 1 shall consist of basic data collection (routine and accelerated monitoring).
  - 5.2.3.4.2. Tier 2 shall consist of evaluation of treatment process, including operational practices and in-plant process chemicals.
  - 5.2.3.4.3. Tier 3 shall consist of a toxicity identification evaluation (TIE).
  - 5.2.3.4.4. Tier 4 shall consist of a toxicity source evaluation.
  - 5.2.3.4.5. Tier 5 shall consist of a toxicity control evaluation, including options for modifications of in-plant treatment processes.
  - 5.2.3.4.6. Tier 6 shall consist of implementation of selected toxicity control measures, and follow-up monitoring and confirmation of implementation success.
- 5.2.3.5. The Discharger may end the TRE at any stage if monitoring finds there is no longer consistent toxicity (i.e., compliance with the trigger in section 5.2.1.3.2, above).
- 5.2.3.6. The objective of the TIE shall be to identify the substance or combination of substances causing the observed toxicity. The Discharger shall employ all reasonable efforts using currently available TIE methodologies.
- 5.2.3.7. As toxic substances are identified or characterized, the Discharger shall continue the TRE by determining the sources and evaluating alternative strategies for reducing or eliminating the toxic substances from the discharge. The Discharger shall take all reasonable steps to reduce toxicity to levels below the chronic toxicity trigger.
- 5.2.3.8. Many recommended TRE elements parallel required or recommended efforts related to source control, pollution prevention, and stormwater control programs. TRE efforts should be coordinated with such efforts. To prevent duplication of efforts, evidence of complying with requirements or recommended efforts of such programs may be acceptable to demonstrate compliance with TRE requirements.

## 6. RECEIVING WATER MONITORING

- 6.1** The Discharger shall continue to participate in the Regional Monitoring Program (RMP), which involves collection of data on pollutants and toxicity in water, sediment and biota of the San Francisco Bay. The Discharger's participation and support of the RMP is used in consideration of the level of receiving water monitoring required by this Order.
- 6.2** The Discharger shall monitor receiving waters at Monitoring Locations RSW-001 through RSW-004 as follows:

**Table E-4. Receiving Water Monitoring**

| Parameter                            | Unit                      | Sample Type | Minimum Sampling Frequency |
|--------------------------------------|---------------------------|-------------|----------------------------|
| pH <sup>[1]</sup>                    | standard units            | Grab        | 1/Quarter                  |
| Ammonia, Total <sup>[1]</sup>        | mg/L as N                 | Grab        | 1/Quarter                  |
| Dissolved Oxygen                     | mg/L                      | Grab        | 1/Quarter                  |
| Dissolved Sulfides                   | mg/L                      | Grab        | 1/Quarter <sup>[2]</sup>   |
| Hardness                             | mg/L as CaCO <sub>3</sub> | Grab        | 1/Quarter                  |
| Nitrate-nitrite                      | mg/L as N                 | Grab        | 1/Quarter                  |
| Phosphorus, Total                    | mg/L                      | Grab        | 1/Quarter                  |
| Salinity <sup>[1]</sup>              | ppt                       | Grab        | 1/Quarter                  |
| Temperature <sup>[1]</sup>           | °C                        | Grab        | 1/Quarter                  |
| Turbidity                            | NTU                       | Grab        | 1/Quarter                  |
| Standard Observations <sup>[3]</sup> | —                         | —           | 1/Quarter                  |

Footnotes:

<sup>[1]</sup> Ammonia, pH, salinity, and temperature shall be monitored on the same day.

<sup>[2]</sup> Sulfide samples shall be run when dissolved oxygen is less than 2.0 mg/L.

<sup>[3]</sup> Standard Observations are specified in Attachment G § 3.2.1.

## 7. RECYCLED WATER MONITORING

The Discharger shall comply with the following recycled water monitoring requirements. The Executive Officer may modify these requirements to reflect any changes made to the requirements of State Water Board Order WQ 2019-0037 EXEC (Amending Monitoring and Reporting Programs for Waste Discharge Requirements, National Pollutant Discharge Elimination System Permits, Water Reclamation Requirements, Master Recycling Permits, and General Waste Discharge Requirements).

- 7.1. Influent Monitoring.** The Discharger shall monitor the monthly volume of influent to the treatment plant.
- 7.2. Production Monitoring.** The Discharger shall monitor the monthly volume for each level of treated effluent (e.g., secondary or tertiary) from the treatment plant at Monitoring Location REC-001.



**7.3. Discharge Monitoring.** The Discharger shall monitor the monthly volumes discharged to each of the following, for each level of treated effluent (e.g., secondary or tertiary) from the treatment plant:

- 7.3.1. Inland surface waters, specifying volumes required to maintain minimum instream flow;
- 7.3.2. Enclosed bays, estuaries and coastal lagoons, and ocean waters;
- 7.3.3. Natural systems, such as wetlands, wildlife habitats, and duck clubs, where augmentation or restoration has occurred, and that are not part of a wastewater treatment or water recycling treatment plant;
- 7.3.4. Underground injection wells, such as those classified by U.S. EPA's Underground Injection Control Program, excluding groundwater recharge via subsurface application intended to reduce seawater intrusion into a coastal aquifer with a seawater interface; and
- 7.3.5. Land, where beneficial use is not taking place, including evaporation or percolation ponds, overland flow, or spray irrigation disposal, excluding pastures or fields with harvested crops.

**7.4. Reuse Monitoring.** The Discharger shall monitor the following in compliance with California Code of Regulations, title 22:

- 7.4.1. Monthly volume of recycled water distributed; and
- 7.4.2. Annual volumes of treated wastewater distributed for use in each of the use categories listed below:
  - 7.4.2.1. Agricultural irrigation: pasture or crop irrigation;
  - 7.4.2.2. Landscape irrigation: irrigation of parks, greenbelts, and playgrounds; school yards; athletic fields; cemeteries; residential landscaping, common areas; commercial landscaping; industrial landscaping; and freeway, highway, and street landscaping;
  - 7.4.2.3. Golf course irrigation: irrigation of golf courses, including water used to maintain aesthetic impoundments within golf courses;
  - 7.4.2.4. Commercial application: commercial facilities, business use (such as laundries and office buildings), car washes, retail nurseries, and appurtenant landscaping that is not separately metered;
  - 7.4.2.5. Industrial application: manufacturing facilities, cooling towers, process water, and appurtenant landscaping that is not separately metered;
  - 7.4.2.6. Geothermal energy production: augmentation of geothermal fields;

- 7.4.2.7. Other non-potable uses: including but not limited to dust control, flushing sewers, fire protection, fill stations, snow making, and recreational impoundments;
- 7.4.2.8. Groundwater recharge: the planned use of recycled water for replenishment of a groundwater basin or an aquifer that has been designated as a source of water supply for a public water system. Includes surface or subsurface application, except for seawater intrusion barrier use;
- 7.4.2.9. Seawater intrusion barrier: groundwater recharge via subsurface application intended to reduce seawater intrusion into a coastal aquifer with a seawater interface;
- 7.4.2.10. Reservoir water augmentation: the planned placement of recycled water into a raw surface water reservoir used as a source of domestic drinking water supply for a public water system, as defined in Health and Safety Code section 116275, or into a constructed system conveying water to such a reservoir (Water Code § 13561);
- 7.4.2.11. Raw water augmentation: the planned placement of recycled water into a system of pipelines or aqueducts that delivers raw water to a drinking water treatment plant that provides water to a public water system as defined in Health and Safety Code section 116275 (Water Code § 13561); and
- 7.4.2.12. Other potable uses: both indirect and direct potable reuse other than for groundwater recharge, seawater intrusion barrier, reservoir water augmentation, or raw water augmentation.

## 8. REPORTING REQUIREMENTS

**8.1. General Monitoring and Reporting Requirements.** The Discharger shall comply with all Standard Provisions (Attachments D and G) related to monitoring, reporting, and recordkeeping.

### 8.2. Self-Monitoring Reports (SMRs)

8.2.1. **SMR Format.** The Discharger shall electronically submit SMRs using the State Water Board's [California Integrated Water Quality System \(CIWQS\) Program website](#). The CIWQS website will provide additional information for SMR submittal in the event of a planned service interruption for electronic submittal.

8.2.2. **SMR Due Dates and Contents.** The Discharger shall submit SMRs by the due dates, and with the contents, specified below:

8.2.2.1. **Monthly SMRs** — Monthly SMRs shall be due 30 days after the end of each calendar month, covering that calendar month. Each SMR shall contain the applicable items described in Provision 6.3.2 (Effluent Characterization Study and Report) of the Order, Attachment D section 5.2, and Attachment G

section 5.3. Each SMR shall include all new monitoring results obtained since the last SMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the Discharger shall include the results of such monitoring in the calculations and reporting for the SMR.

8.2.2.2. **Annual SMR** — Annual SMRs shall be due February 1 each year, covering the previous calendar year. The annual SMR shall contain the applicable items described in Provisions 6.3.2 (Effluent Characterization Study and Report) and 6.3.4.3 (Anaerobically-Digestible Material) of the Order, and Attachment G section 5.3.1.6.

8.2.3. **Specifications for Submitting SMRs to CIWQS.** The Discharger shall submit analytical results and other information using one of the following methods:

**Table E-5. CIWQS Reporting**

| Parameter                                                                                                                                                                                                                                           | Method of Reporting:<br>EDF/CDF data upload                            | Attached File                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------|
| All parameters identified in influent, effluent, and receiving water monitoring tables (except Dissolved Oxygen and Temperature)                                                                                                                    | Required for all results                                               | —                                                              |
| Dissolved Oxygen, Temperature                                                                                                                                                                                                                       | Required for monthly maximum and minimum results only <sup>[1]</sup>   | Discharger may use this method for all results or keep records |
| Antimony, Arsenic, Beryllium, Cadmium, Chromium, Copper, Cyanide, Lead, Mercury, Nickel, Selenium, Silver, Thallium, Zinc, Dioxins & Furans (by U.S. EPA Method 1613), Other Pollutants (by U.S. EPA Methods 601, 602, 608, 610, 614, 624, and 625) | Required for all results <sup>[2]</sup>                                | —                                                              |
| Volume and Duration of Blended Discharge <sup>[3]</sup>                                                                                                                                                                                             | Required for all blended effluent discharges                           | —                                                              |
| Analytical Method                                                                                                                                                                                                                                   | Not required (Discharger may select “data unavailable”) <sup>[1]</sup> | —                                                              |
| Collection Time, Analysis Time                                                                                                                                                                                                                      | Not required                                                           | —                                                              |

Footnotes:

- <sup>[1]</sup> The Discharger shall continue to monitor at the minimum frequency specified in this MRP, keep records of the measurements, and make the records available upon request.
- <sup>[2]</sup> These parameters require EDF/CDF data upload or manual entry regardless of whether monitoring is required by this MRP or other provisions of this Order (except for biosolids, sludge, or ash provisions).
- <sup>[3]</sup> The requirement for volume and duration of blended discharge applies only if this Order authorizes the Discharger to discharge blended effluent.

The Discharger shall arrange all reported data in a tabular format and summarize data to clearly illustrate whether the Facility is operating in compliance with effluent limitations. The Discharger is not required to duplicate the submittal of data entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format, the Discharger shall electronically submit the data in a tabular format as an attachment.

8.2.4. **Monitoring Periods.** Monitoring periods for all required monitoring shall be as set forth below unless otherwise specified:

**Table E-6. Monitoring Periods**

| Sampling Frequency                   | Monitoring Period Begins On...                                                                       | Monitoring Period                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous/D                         | Order effective date                                                                                 | All times                                                                                                                                          |
| 1/Day                                | Order effective date                                                                                 | Any 24-hour period that reasonably represents a calendar day for sampling purposes (e.g., beginning at midnight and continuing through 11:59 p.m.) |
| 1/Week<br>2/Week<br>3/Week<br>5/Week | First Sunday following or on Order effective date                                                    | Sunday through Saturday                                                                                                                            |
| 1/Month                              | First day of calendar month following or on Order effective date                                     | First day of calendar month through last day of calendar month                                                                                     |
| 1/Quarter                            | Closest January 1, April 1, July 1, or October 1 before or after Order effective date <sup>[1]</sup> | January 1 through March 31<br>April 1 through June 30<br>July 1 through September 30<br>October 1 through December 31                              |
| 1/Year                               | Closest January 1 before or after Order effective date <sup>[1]</sup>                                | January 1 through December 31                                                                                                                      |
| Once                                 | Order effective date                                                                                 | Once during the term of the Order within 12 months prior to applying for permit reissuance                                                         |

Footnote:

<sup>[1]</sup> Monitoring performed during the previous order term may be used to satisfy monitoring required by this Order.

8.2.5. **RL and MDL Reporting.** The Discharger shall report with each sample result the Reporting Level (RL) and Method Detection Limit (MDL) as determined by the procedure in 40 C.F.R. part 136. The Discharger shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- 8.2.5.1. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- 8.2.5.2. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For purposes of data collection, the Discharger shall require the laboratory to write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ( $\pm$  a percentage of the reported value), numerical ranges (low to high), or any other means the laboratory considers appropriate.

- 8.2.5.3. Sample results less than the laboratory's MDL shall be reported as "Not Detected", or ND.
- 8.2.5.4. The Discharger shall instruct laboratories to establish calibration standards so that the minimum level (ML) value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
- 8.2.6. **Compliance Determination.** Compliance with effluent limitations shall be determined using sample reporting protocols defined above, in the Fact Sheet, and in Attachments A, D, and G. For purposes of reporting and administrative enforcement by the Regional Water Board and State Water Board, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the pollutant in the monitoring sample is greater than the effluent limitation and, if applicable, greater than or equal to the RL.
- 8.3. **Discharge Monitoring Reports (DMRs).** DMRs are U.S. EPA reporting requirements. The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or the latest upgraded version. Electronic DMR submittal shall be in addition to electronic SMR submittal. Information about electronic DMR submittal is available at the [DMR website](https://waterboards.ca.gov/water_issues/programs/discharge_monitoring) (waterboards.ca.gov/water\_issues/programs/discharge\_monitoring).
- 8.4. **Biosolids Monitoring Reports.** For biosolids applied to land, placed on a surface disposal site, or fired in a biosolids incinerator as defined in 40 CFR § 503, the Discharger shall:
1. submit an annual report to U.S. EPA's CDX electronic reporting system containing monitoring results and pathogen and vector attraction reduction requirements as specified by 40 CFR § 503, by February 19 of each year, for the period covering the previous calendar year.
  2. submit a copy of its annual biosolids report to U.S. EPA to the Regional Water Board.
  3. notify U.S. EPA and the Regional Water Board within 30 days of changing its biosolids use or disposal practices.
- 8.5. **Recycled Water Reports.** The Discharger shall electronically submit annual reports to the State Water Board by April 30 each year covering the previous calendar year using the [State Water Board's GeoTracker website](https://geotracker.waterboards.ca.gov) (geotracker.waterboards.ca.gov) under a site-specific global identification number. For the 2021 calendar year, the Discharger shall submit a report by April 30, 2022, covering January through December 2021. The annual report shall include the elements specified in section 7, above.

Information for setting up and using the GeoTracker system can be found in the *ESI Guide for Responsible Parties* document on the [State Water Board's website](https://waterboards.ca.gov/ust/electronic_submittal/index.html) (waterboards.ca.gov/ust/electronic\_submittal/index.html).

## APPENDIX E-1

### CHRONIC TOXICITY DEFINITION OF TERMS AND SCREENING PHASE REQUIREMENTS

#### 1. Definition of Terms

- 1.1. **No observed effect level (NOEL).** For compliance determination, the NOEL is equal to IC25 or EC25. If the IC25 or EC25 cannot be statistically determined, the NOEL shall be equal to the NOEC derived using hypothesis testing.
- 1.2. **Effective concentration (EC).** The EC is a point estimate of the toxicant concentration that would cause an adverse effect on a quantal, "all or nothing," response (such as death, immobilization, or serious incapacitation) in a given percent of the test organisms. If the effect is death or immobility, the term lethal concentration (LC) may be used. EC values may be calculated using point estimation techniques such as probit, logit, and Spearman-Kärber. EC25 is the concentration of toxicant (in percent effluent) that causes a response in 25 percent of the test organisms.
- 1.3. **Inhibition concentration (IC).** The IC is a point estimate of the toxicant concentration that would cause a given percent reduction in a nonlethal, nonquantal biological measurement, such as growth. For example, an IC25 is the estimated concentration of toxicant that would cause a 25 percent reduction in average young per female or growth. IC values may be calculated using a linear interpolation method such as U.S. EPA's Bootstrap Procedure.
- 1.4. **No observed effect concentration (NOEC).** The NOEC is the highest tested concentration of an effluent or a toxicant at which no adverse effects are observed on the aquatic test organisms at a specific time of observation. It is determined using hypothesis testing.

#### 2. Chronic Toxicity Screening Phase Requirements

- 2.1. The Discharger shall perform screening phase monitoring:
  - 2.1.1. Subsequent to any significant change in the nature of the effluent discharged through changes in sources or treatment, except those changes resulting from reductions in pollutant concentrations attributable to source control efforts, or
  - 2.1.2. Prior to permit reissuance. Screening phase monitoring data shall be included in the NPDES permit application for reissuance. The information shall be as recent as possible, but may be based on screening phase monitoring conducted within 5 years before the permit expiration date.

**2.2** Design of the screening phase shall, at a minimum, consist of the following elements:

2.2.1. Use of test species specified in Appendix E-2, attached, and use of the protocols referenced in those tables.

2.2.2. Two stages:

2.2.2.1. Stage 1 shall consist of a minimum of one battery of tests conducted concurrently. Selection of the type of test species and minimum number of tests shall be based on Appendix E-2 (attached).

2.2.2.2. Stage 2 shall consist of a minimum of two test batteries conducted at a monthly frequency using the three most sensitive species based on the Stage 1 test results.

2.2.3. Appropriate controls.

2.2.4. Concurrent reference toxicant tests.

2.2.5. Dilution series of 100%, 50%, 25%, 12.5%, 6.25%, and 0.0%, where “%” is percent effluent as discharged, or as otherwise approved by the Executive Officer if different dilution ratios are needed to reflect discharge conditions.

**2.3.** The Discharger shall submit a screening phase proposal. The proposal shall address each of the elements listed above. If within 30 days, the Executive Officer does not comment, the Discharger shall commence with screening phase monitoring.



## APPENDIX E-2 SUMMARY OF TOXICITY TEST SPECIES REQUIREMENTS

**Table AE-1. Critical Life Stage Toxicity Tests for Estuarine Waters**

| Species                               | Scientific Name                                                                                         | Effect                                             | Test Duration         | Reference |
|---------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|-----------|
| Alga                                  | <i>Skeletonema costatum</i> ,<br><i>Thalassiosira pseudonana</i>                                        | Growth rate                                        | 4 days                | 1         |
| Red alga                              | <i>Champia parvula</i>                                                                                  | Number of<br>cystocarps                            | 7–9 days              | 3         |
| Giant kelp                            | <i>Macrocystis pyrifera</i>                                                                             | Percent<br>germination;<br>germ tube length        | 48 hours              | 2         |
| Abalone                               | <i>Haliotis rufescens</i>                                                                               | Abnormal shell<br>development                      | 48 hours              | 2         |
| Oyster<br>Mussel                      | <i>Crassostrea gigas</i> ,<br><i>Mytilus edulis</i>                                                     | Abnormal shell<br>development;<br>percent survival | 48 hours              | 2         |
| Echinoderms<br>Urchins<br>Sand dollar | <i>Strongylocentrotus<br/>purpuratus</i> ,<br><i>S. franciscanus</i> ,<br><i>Dendraster excentricus</i> | Percent fertilization<br>or larval<br>development  | 1 hour or<br>72 hours | 2         |
| Shrimp                                | <i>Americamysis bahia</i>                                                                               | Percent survival;<br>growth                        | 7 days                | 3         |
| Shrimp                                | <i>Holmesimysis costata</i>                                                                             | Percent survival;<br>growth                        | 7 days                | 2         |
| Topsmelt                              | <i>Atherinops affinis</i>                                                                               | Percent survival;<br>growth                        | 7 days                | 2         |
| Silverside                            | <i>Menidia beryllina</i>                                                                                | Larval growth rate;<br>percent survival            | 7 days                | 3         |

Toxicity Test References:

1. American Society for Testing Materials (ASTM). 1990. Standard Guide for Conducting Static 96-Hour Toxicity Tests with Microalgae. Procedure E 1218-90. ASTM, Philadelphia, PA.
2. Short-term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to West Coast Marine and Estuarine Organisms. EPA/600/R-95/136. August 1995.
3. Short-term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to Marine and Estuarine Organisms. EPA/821/R-02/014. October 2002.

**Table AE-2. Critical Life Stage Toxicity Tests for Fresh Waters**

| Species           | Scientific Name                      | Effect                       | Test Duration | Reference |
|-------------------|--------------------------------------|------------------------------|---------------|-----------|
| Fathead<br>minnow | <i>Pimephales promelas</i>           | Survival;<br>growth rate     | 7 days        | 4         |
| Water flea        | <i>Ceriodaphnia dubia</i>            | Survival;<br>number of young | 7 days        | 4         |
| Alga              | <i>Selenastrum<br/>capricornutum</i> | Final cell density           | 4 days        | 4         |

Toxicity Test Reference:

1. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, fourth Edition Chronic manual (EPA-821-R-02-013, October 2002).

**Table AE-3. Toxicity Test Requirements for Stage One Screening Phase**

| <b>Requirements</b>                                                 | <b>Receiving Water Characteristics:<br/>Discharges to Ocean</b> | <b>Receiving Water Characteristics:<br/>Discharges to Marine/<br/>Estuarine Waters <sup>[1]</sup></b> | <b>Receiving Water Characteristics:<br/>Discharges to<br/>Freshwater</b> |
|---------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Taxonomic diversity                                                 | 1 plant<br>1 invertebrate<br>1 fish                             | 1 plant<br>1 invertebrate<br>1 fish                                                                   | 1 plant<br>1 invertebrate<br>1 fish                                      |
| Number of tests of each salinity type:<br>Freshwater <sup>[2]</sup> | 0                                                               | 1 or 2                                                                                                | 3                                                                        |
| Marine/Estuarine                                                    | 4                                                               | 3 or 4                                                                                                | 0                                                                        |
| Total number of tests                                               | 4                                                               | 5                                                                                                     | 3                                                                        |

Footnotes:

- <sup>[1]</sup> (a) Marine refers to receiving water salinities greater than 10 parts per thousand (ppt) at least 95 percent of the time during a normal water year.  
(b) Freshwater refers to receiving water with salinities less than 1 ppt at least 95 percent of the time during a normal water year.  
(c) Estuarine refers to receiving water salinities that fall between those of marine and freshwater, as described above.
- <sup>[2]</sup> The freshwater species may be substituted with marine species if:  
(a) The salinity of the effluent is above 1 ppt greater than 95 percent of the time, or  
(b) The ionic strength (TDS or conductivity) of the effluent at the test concentration used to determine compliance is documented to be toxic to the test species.

## ATTACHMENT F – FACT SHEET

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## ATTACHMENT F – FACT SHEET

This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order. As described in section 2.2 of the Order, the Regional Water Board incorporates this Fact Sheet as findings supporting the issuance of the Order.

### 1. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

**Table F-1. Facility Information**

|                                              |                                                                                            |
|----------------------------------------------|--------------------------------------------------------------------------------------------|
| WDID                                         | 2 0710290001                                                                               |
| CIWQS Place ID                               | 241815                                                                                     |
| Discharger                                   | Mt. View Sanitary District                                                                 |
| Facility Name                                | Mt. View Sanitary District Wastewater Treatment Plant and its wastewater collection system |
| Facility Address                             | 3800 Arthur Road<br>Martinez, California 94553<br>Contra Costa County                      |
| Facility Contact, Title, and Phone           | Lilia Corona, District Manager,<br>(925) 228-5635 x18                                      |
| Authorized Person to Sign and Submit Reports | Same as Facility Contact                                                                   |
| Mailing Address                              | P.O. Box 2757<br>Martinez, CA 94553                                                        |
| Billing Address                              | Same as Mailing Address                                                                    |
| Facility Type                                | Publicly-Owned Treatment Works (POTW)                                                      |
| Major or Minor Facility                      | Major                                                                                      |
| Water Quality Threat                         | 2                                                                                          |
| Complexity                                   | A                                                                                          |
| Pretreatment Program                         | No                                                                                         |
| Recycling Requirements                       | No                                                                                         |
| Mercury and PCBs Requirements                | NPDES Permit CA0038849                                                                     |
| Nutrients Requirements                       | NPDES Permit CA0038873                                                                     |
| Facility Permitted Flow                      | 3.2 MGD – average dry weather design flow                                                  |
| Facility Design Flow                         | 10.9 MGD – peak wet weather design capacity                                                |
| Watershed                                    | Suisun                                                                                     |
| Receiving Water                              | Peyton Slough, a tributary to McNabney Marsh and Carquinez Strait                          |
| Receiving Water Type                         | Estuarine                                                                                  |

- 1.1.** The Mt. View Sanitary District (Discharger) owns the Mt. View Sanitary District Wastewater Treatment Plant and its collection system (collectively, the Facility). The plant provides advanced secondary treatment of wastewater collected from its service area and discharges it to Peyton Slough, a tributary of McNabney Marsh and Carquinez Strait. All are waters of the United States within the Suisun watershed.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and State laws, regulations, plans, and policies are held to be equivalent to references to the Discharger herein.

- 1.2. The Discharger is regulated pursuant to National Pollutant Discharge Elimination System (NPDES) Permit CA0037770. The Discharger was previously subject to Order R2-2016-0023 (previous order). The Discharger filed a Report of Waste Discharge and submitted an application for reissuance of its Waste Discharge Requirements (WDRs) and NPDES permit on November 30, 2020.
- 1.3. The Discharger is authorized to discharge subject to the WDRs in this Order at the discharge location described in Table 1 of this Order. Regulations in 40 C.F.R. section 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. Accordingly, this Order limits the effective period for the discharge authorization. Pursuant to 40 C.F.R. section 122.6(d) and California Code of Regulations, title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Discharger complies with all requirements for continuation of expired permits.
- 1.4. Order R2-2016-0008 amended the previous order to provide for an alternate monitoring program and remains in effect with this Order. The discharge is also regulated under NPDES Permits CA0038849 and CA0038873, which establish requirements for mercury, polychlorinated biphenyls (PCBs), and nutrients from wastewater discharges to San Francisco Bay. This Order does not affect those permits.
- 1.5. When applicable, State law requires dischargers to file a petition with the State Water Resources Control Board (State Water Board), Division of Water Rights, and receive approval for any change in the point of discharge, place of use, or purpose of use of treated wastewater that decreases the flow in any portion of a watercourse. The State Water Board retains separate jurisdictional authority to enforce such requirements under Water Code section 1211. This is not an NPDES permit requirement.

## 2. FACILITY DESCRIPTION

### 2.1. Wastewater and Biosolids Treatment

- 2.1.1. **Service Area and Collection System.** The plant provides advanced secondary treatment for domestic and non-industrial commercial wastewater from unincorporated areas near the City of Martinez and portions of the City of Martinez, serving a population of about 21,000 people. The collection system consists of approximately 73 miles of gravity sewer main, two miles of pressure sewer main, and four pump stations. Attachment B contains maps of the Facility area.
- 2.1.2. **Wastewater Treatment.** The plant has an advanced secondary treatment capacity of 3.2 million gallons per day (MGD) as a dry-weather daily average.

The treatment system consists of screening, primary clarification, trickling filtration, ammonia removal nitrification (accomplished in a bio-tower), secondary sedimentation, sand filtration, and ultraviolet light (UV) disinfection. Attachment C provides a plant flow schematic.

Following disinfection, the wastewater flows through a 21-acre constructed wetland (Moorhen Marsh) for “polishing” treatment to remove ammonia and other pollutants before being discharged to Peyton Slough at Discharge Point 001. The average discharge flow for July 2016 through April 2021 was 1.3 MGD.

- 2.1.3. **Sludge and Biosolids Management.** Sludge is thickened, anaerobically digested, dewatered in a centrifuge, and placed as alternative daily cover at a landfill. Seasonally, sludge volume is further reduced in drying beds prior to placement in the landfill. Runoff from the drying beds is collected and pumped back to the headworks.
- 2.1.4. **Stormwater Management.** The Discharger is not required to be covered under the statewide NPDES permit for stormwater discharges associated with industrial activities (NPDES General Permit CAS000001) because all stormwater flows that contact equipment or sewage at the plant and the pump stations serving the plant are collected and directed to the plant headworks for treatment.
- 2.1.5. **Changes During Previous Order Term.** The Discharger made the following plant upgrades during the previous order term:
- **Biotower Feed Pump Improvements:** The Discharger replaced one pump, rehabilitated another, replaced all three pumps’ guide rails, and replaced all three discharge elbows and associated piping in the Biotower wet well. Construction began in August 2020 and was largely complete by November 2020.
  - **SCADA System Upgrades:** The Discharger replaced many of the hardware and software components that comprise its SCADA system, significantly upgrading existing graphics screens and developing new ones to improve plant operations and control, and remote access capabilities. These upgrades began in August 2020 and were completed in January 2021.
- 2.2. **Discharge Points and Receiving Waters.** The plant discharges treated wastewater into Peyton Slough through a shallow water outfall (Discharge Point 001). Peyton Slough is part of a marsh complex composed of Peyton Slough, McNabney Marsh, and other marsh segments along Peyton Slough. This marsh complex is a tidally-influenced estuarine wetland connected to Carquinez Strait through a restricted channel controlled by a tide gate downstream of

McNabney Marsh and about one mile from Carquinez Strait. Peyton Slough flows through McNabney Marsh and through the tide gate, ultimately to Carquinez Strait.

McNabney Marsh comprises 138 acres upstream of the Union Pacific Railroad railway and Waterfront Road (see Attachment B, Figure B-1). McNabney Marsh is co-owned by the Discharger and the East Bay Regional Parks District, which are the two agencies primarily responsible for managing it. However, water quality in the Peyton Slough marsh complex, including in McNabney Marsh, is affected by the actions of other stakeholders in the marsh complex. In particular, the marshes and tide gate downstream of Waterfront Road are on the property of and are primarily managed by Eco Services Operations, LLC, and TransMontaigne Operating Company LP. In addition, several other stakeholders, with concerns related to water quality, habitat, odors, mosquito control, flooding, and sea level rise, have interests in the Peyton Slough marsh complex. The actions of these stakeholders affect McNabney Marsh and vice versa; however, no single stakeholder has overall control or authority over the marsh complex.

The Discharger has agreed to work with the other landowning stakeholders (Eco Services, EBRPD, and TransMontaigne) regarding tide gate operation in 2022, with the goal of improving flushing and circulation in McNabney Marsh while maintaining pickleweed habitat in Eco-Services' 17-acre mitigation marsh (Rhodia Marsh), located downstream and just northwest of McNabney Marsh. Eco-Services currently operates the tide gates to facilitate tidal inundation of Rhodia Marsh.

**2.3. Previous Requirements and Monitoring Data.** The table below presents the previous order's effluent limitations and representative monitoring data from the previous order term (July 2016 – April 2021):

**Table F-2. Previous Effluent Limitations and Monitoring Data**

| Parameter                                      | Unit       | Average Monthly Limit | Average Weekly Limit | Maximum Daily Limit | Other Limit                 | Long-Term Average | Highest Daily Value      |
|------------------------------------------------|------------|-----------------------|----------------------|---------------------|-----------------------------|-------------------|--------------------------|
| Biochemical Oxygen Demand (5 day @ 20°C) (BOD) | mg/L       | 15                    | 25                   | —                   | —                           | 6.0               | 15                       |
| BOD percent removal                            | %          | 85 (minimum)          | —                    | —                   | —                           | 98                | 95 <sup>[1]</sup>        |
| Total Suspended Solids (TSS)                   | mg/L       | 15                    | 25                   | —                   | —                           | 9.1               | 54                       |
| TSS percent removal                            | %          | 85 (minimum)          | —                    | —                   | —                           | 97                | 93 <sup>[1]</sup>        |
| Oil and Grease                                 | mg/L       | 10                    | —                    | 20                  | —                           | <1.4              | <1.4                     |
| pH                                             | s.u.       | —                     | —                    | —                   | 6.5 – 8.5 <sup>[2]</sup>    | —                 | 6.3 – 8.2 <sup>[3]</sup> |
| Enterococcus Bacteria                          | MPN/100 mL | —                     | —                    | —                   | 35 (monthly geometric mean) | —                 | 38 <sup>[4]</sup>        |



| Parameter                 | Unit       | Average Monthly Limit                           | Average Weekly Limit   | Maximum Daily Limit | Other Limit                               | Long-Term Average | Highest Daily Value |
|---------------------------|------------|-------------------------------------------------|------------------------|---------------------|-------------------------------------------|-------------------|---------------------|
| Ammonia, Total            | mg/L as N  | 1.6                                             | —                      | 4.7                 | —                                         | 0.75              | 7.7                 |
| Copper, Total Recoverable | µg/L       | 11                                              | —                      | 15                  | —                                         | 6.0               | 10                  |
| Cyanide                   | µg/L       | 7.0                                             | --                     | 13                  | --                                        |                   | 2.6                 |
| Benzo(a)Anthracene        | µg/L       | 0.049                                           | —                      | 0.098               | —                                         | <0.020            | 0.11                |
| Dioxin-TEQ                | µg/L       | 1.4 x 10 <sup>-8</sup>                          | 2.4 x 10 <sup>-8</sup> | 13                  | —                                         | ND                | ND                  |
| Acute Toxicity            | % survival | —                                               | —                      | —                   | Not less than 70% (Single-Sample Maximum) | 99                | 95 <sup>[5]</sup>   |
| Chronic Toxicity          | TUc        | No detectable chronic toxicity in the discharge |                        |                     |                                           | <1.0              | <1.0                |

Footnotes:

<sup>[1]</sup> Lowest percent removal value on a monthly basis.

<sup>[2]</sup> Instantaneous minimum and instantaneous maximum.

<sup>[3]</sup> Range of lowest to highest pH values.

<sup>[4]</sup> This was not a violation because the monthly geometric mean was below 35 MPN/100mL.

<sup>[5]</sup> Lowest percent survival.

## 2.4. Compliance Summary

2.4.1. **Treatment Plant.** The Discharger violated its numeric effluent limitations ten times during the previous order term as listed in the table below:

**Table F-3. Violations of Order R2-2016-0023's Effluent Limits**

| Date       | Parameter            | Units | Limit Type            | Limit | Result |
|------------|----------------------|-------|-----------------------|-------|--------|
| 12/22/2017 | pH                   | s.u.  | Instantaneous Minimum | 6.5   | 6.3    |
| 12/23/2017 | pH                   | s.u.  | Instantaneous Minimum | 6.5   | 6.4    |
| 6/5/2018   | Total Ammonia (as N) | mg/L  | Daily Maximum         | 4.7   | 5.5    |
| 6/6/2018   | Total Ammonia (as N) | mg/L  | Daily Maximum         | 4.7   | 7.7    |
| 1/27/2021  | Total Ammonia (as N) | mg/L  | Daily Maximum         | 4.7   | 4.8    |
| 1/31/2021  | Total Ammonia (as N) | mg/L  | Monthly Average       | 1.6   | 2.8    |
| 2/13/2021  | TSS                  | mg/L  | Weekly Average        | 25    | 32     |
| 2/18/2021  | Total Ammonia (as N) | mg/L  | Daily Maximum         | 4.7   | 4.8    |
| 2/28/2021  | TSS                  | mg/L  | Monthly Average       | 15    | 18     |
| 3/31/2021  | TSS                  | mg/L  | Monthly Average       | 15    | 16     |

The Discharger violated its instantaneous minimum pH limit on December 22 and 23, 2017. The Discharger had increased the rate at which it fed biosolids into the digester, which increased the use of the biosolids dewatering centrifuge, which in turn increased the amount of ammonia returned to the plant headworks in the centrate side stream. The increased ammonia load to the biotower removed more alkalinity during nitrification, which in turn reduced the effluent pH. The Discharger has since modified its standard operating

procedure to now add alkalinity (as sodium bicarbonate) when the biosolids centrifuge is run more than four hours per day. Neither violation was “serious” or “chronic” as defined under California Water Code section 13385, and the Regional Water Board took no further enforcement.

The Discharger violated its daily maximum total ammonia limit on June 5 and 6, 2018, during essential biotower maintenance that took place from June 4 through June 8, 2018. Construction work associated with the Moorhen Marsh Western Pond Turtle Habitat Enhancement Project took one pond (Pond A) out of service. The Discharger intended to use the four remaining ponds (Ponds B, C, D, and E) during the biotower maintenance. However, on June 6, the Discharger discovered a distribution pipe to Ponds C, D, and E had ruptured, causing all flow to go to Pond B. Pond B was insufficient to treat the flow of wastewater not treated by the biotower, causing the ammonia violations. The Discharger repaired the ruptured pipe by June 7, 2018, thereby ensuring sufficient treatment capacity in Moorhen Marsh for all future biotower maintenance. The Regional Water Board issued Order R2-2019-1004 on May 16, 2019, assessing the Discharger a \$3,000 mandatory minimum penalty for these violations.

In January 2021, the Discharger violated the daily maximum and monthly average ammonia limits due to operator error while flushing the biotower. Specifically, on two separate occasions, an operator mistakenly turned on a second pump during a biotower flush. The Discharger was in the process of upgrading its SCADA system at the time, which may have contributed to a longer than necessary flushing cycle. These errors flushed out enough of the biotower’s microbes to reduce the biotower’s treatment efficiency, leading to the ammonia violations. The Discharger began daily monitoring after receiving the laboratory report confirming ammonia violations. The average result at Monitoring Location EFF-002, just before the discharge enters Peyton Slough, was 0.30 mg/L total ammonia (as N), less than the total ammonia effluent limits of 1.6 mg/L (average monthly) and 4.7 mg/L (maximum daily), so there was likely no impact to the receiving waters. (The average at EFF-002 is also less than this Order’s total ammonia effluent limits of 1.1 mg/L [average monthly] and 3.2 mg/L [maximum daily].) Nevertheless, the Discharger suspended biotower flushing to allow microbial regrowth and re-trained its operators in biotower flushing procedures. The Regional Water Board has not yet taken enforcement for these violations.

The Discharger has yet to determine a cause for the remaining TSS and ammonia violations that occurred in February and March 2021. The Discharger decreased its frequency of biotower flushing and flow recirculation, and ensured that all valves and connections between process units are stable and in good condition. The Discharger also hired an engineering firm with wastewater process expertise to further investigate what happened and propose additional corrective actions. The Regional Water Board has not yet taken enforcement for these violations.

2.4.2. **Collection System.** The table below summarizes the Discharger's Category 1 sanitary sewer overflow (SSO) rates for the last five years, along with the primary causes of these discharges. Category 1 SSOs are those that reach waters of the United States and thus may violate Prohibition 3.6 of this Order.

**Table F-4. Collection System and Category 1 SSO Rates (SSO per 100 miles)**

(Values based on CIWQS data analysis completed in July 2021) <sup>[1]</sup>

|                          | <b>Length<br/>(miles)</b> | <b>Average Pipe<br/>Age (years)</b> | <b>2016</b> | <b>2017</b> | <b>2018</b> | <b>2019</b> | <b>2020</b> |
|--------------------------|---------------------------|-------------------------------------|-------------|-------------|-------------|-------------|-------------|
| Discharger               | 75 <sup>[2]</sup>         | 41                                  | 1.3         | 6.4         | 3.8         | 5.3         | 6.7         |
| San Francisco Bay Region | 17,700                    | 46 <sup>[3]</sup>                   | 1.2         | 1.7         | 0.71        | 1.4         | 0.67        |
| State of California      | 89,100                    | 44 <sup>[3]</sup>                   | 0.48        | 0.69        | 0.40        | 0.58        | 0.33        |

Footnotes:

<sup>[1]</sup> The State Water Board's *Enrollees's Guide to the SSO Database* defines "Total number of SSOs per 100 miles of Sewer" as "...the number of SSOs, for which the reporting enrollee is responsible, for every 100 miles of pipe or sewer lines in an enrollee's sanitary sewer system. Due to the large variation in facility specific characteristics, this metric should only be viewed as a rough comparison of the operation and maintenance performance of enrollees and their sanitary sewer systems."

<sup>[2]</sup> Lengths shown are based on 2020 data.

<sup>[3]</sup> The average pipe age for the State of California is estimated based on the percentages of piping constructed during each decade as reported by enrollees under State Water Board Order 2006-0003-DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, as amended by State Water Board Order WQ 2013-0058-EXEC.

The Discharger's SSO rates were higher than those for the San Francisco Bay Region and State, but, at less than 100 miles, its collection system is relatively short. On November 28, 2017, the Regional Water Board inspected the Discharger's collection system and reviewed its SSO recordkeeping and reporting. The inspection found that the Discharger had reported 31 Category 1 SSOs during the decade from May 2, 2007, through December 1, 2017, with a combined volume of more than 700,000 gallons, most of which reached surface waters. The inspection also found that several SSO reports were late, not timely certified (within 15 days), missing, or had discrepancies with the field records they were based on. The Regional Water Board issued a Notice of Violation on December 9, 2017, that required the Discharger to take corrective action and submit a completion report by February 28, 2018. The Discharger submitted the completion report, which described its corrective actions as follows:

- Increasing its cleaning and maintenance frequency to 100 percent of the gravity system every three years;
- Updating its Trouble Spot Program to inspect, reassess, and prioritize known trouble spots for rehabilitation and repair;
- Renewing its Root Foaming Program and follow-up line cleaning;
- Revising its Fats, Oil, and Grease (FOG) Prevention Plan to include food service establishment inspections and a new public outreach campaign;

- Purchasing additional SSO control equipment (gutter guards, rubber mats, sandbags, plastic sheeting, in-line plugs) and training response staff in their use;
- Revising its SSO Response Plan to emphasize preventing SSOs from reaching storm drains and surface waters;
- Providing CIWQS training to its personnel and establishing a 6-hour standard for its CIWQS data submitter to provide initial SSO reports;
- Increasing the number of its personnel able to certify reports from three to five; and
- Revising its reporting and certification to avoid reporting discrepancies by requiring two persons to review the reports for accuracy.

During the previous order term, the Discharger also performed the following activities as part of its Sewer System Master Plan:

- Manhole repair and replacement from December 2020 through summer 2021, consisting of abandoning, replacing, exposing and raising, improving access to, and installing or rehabilitating manholes at 39 locations, as well as making small pipe repairs;
- Pipeline cleaning and televising of every pipeline in the collection system in two phases, the first from December 2020 through summer 2021, and the second beginning in late 2021; and
- Pipeline condition assessment, immediately following each phase of the pipeline cleaning and televising project.

The Discharger also plans miscellaneous pump station rehabilitation, including pump and piping replacements, coating rehabilitations, and electrical upgrades at all four pump stations, for 2022.

**2.5. Sea Level Rise.** Sea level rise is projected to affect Discharge Point 001, which discharges by gravity, and portions of the collection system. The Discharger plans to conduct a study to estimate potential climate change, sea level rise, and groundwater rise impacts on the Facility for its 2022 Capital Improvement Plan. The Discharger will use the results of the study to inform its Plant Master Plan update, which it expects to complete within the next several years.

**2.6. Planned Changes.** The Discharger plans or is in the process of making the following operational changes at the treatment plant:

- **UV Disinfection Replacement:** The Discharger plans to replace the existing UV disinfection system equipment and controls with new equipment and

controls. Design work is currently in progress and is scheduled to conclude in fall 2021, with the bid period and construction to immediately follow. Project completion is expected by spring 2023.

- **Centrifuge Replacement:** The Discharger plans to replace the existing centrifuge equipment and controls with new sludge dewatering equipment and controls. Project design is planned to begin in fall 2021, with the bid period and construction to follow in spring 2023.
- **Headworks Automatic Screening Rebuild:** The Discharger plans to rebuild the headworks automatic screening mechanism. Project design is planned to begin in fall 2021, with the bid period and construction to follow in spring 2023.
- **Nutrient Removal in Moorhen Marsh:** The Discharger plans to install additional Floating Treatment Islands™ in Moorhen Marsh to enhance nutrient removal. Study design began in summer 2021, and a nutrient removal strategy is expected to be completed by summer 2022. Construction is expected by 2023.

### 3. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

**3.1. Legal Authorities.** This Order serves as WDRs pursuant to California Water Code article 4, chapter 4, division 7 (commencing with § 13260). This Order is also issued pursuant to federal Clean Water Act (CWA) section 402 and implementing regulations adopted by the U.S. EPA, and Water Code chapter 5.5, division 7 (commencing with § 13370). It serves as an NPDES permit authorizing the Discharger to discharge into waters of the United States at the discharge locations described in Table 1 subject to the WDRs in this Order.

**3.2. California Environmental Quality Act (CEQA).** Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of the California Environmental Quality Act (CEQA), Public Resources Code division 13, chapter 3 (commencing with § 21100).

#### 3.3. State and Federal Laws, Regulations, Policies, and Plans

**3.3.1. Water Quality Control Plan.** The Regional Water Board adopted the *Water Quality Control Plan for the San Francisco Bay Basin* (Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan. Beneficial uses applicable to Carquinez Strait and Peyton Slough are listed below. Beneficial uses of Peyton Slough apply to McNabney Marsh by the Tributary Rule:

**Table F-5. Beneficial Uses**

| Discharge Points | Receiving Water  | Beneficial Uses                                                                                                                                                                                                                                                                                                                           |
|------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 001              | Carquinez Strait | Industrial Service Supply (IND)<br>Ocean, Commercial, and Sport Fishing (COMM)<br>Fish Spawning (SPWN)<br>Estuarine Habitat (EST)<br>Wildlife Habitat (WILD)<br>Water Contact Recreation (REC1)<br>Non-Contact Water Recreation (REC2)<br>Fish Migration (MIGR)<br>Preservation of Rare and Endangered Species (RARE)<br>Navigation (NAV) |
|                  | Peyton Slough    | Industrial Service Supply (IND)<br>Ocean, Commercial, and Sport Fishing (COMM)<br>Estuarine Habitat (EST)<br>Wildlife Habitat (WILD)<br>Water Contact Recreation (REC1)<br>Non-Contact Water Recreation (REC2)<br>Fish Migration (MIGR)<br>Preservation of Rare and Endangered Species (RARE)                                             |

State Water Board Resolution No. 88-63 establishes as State policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply (MUN). Because of the marine influence in Suisun Bay, total dissolved solids in this section of Peyton Slough sometimes exceed 3,000 mg/L and thereby meet an exception to State Water Board Resolution No. 88-63. The MUN designation therefore does not apply to this section of Peyton Slough.

- 3.3.2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** The NTR and CTR contain federal water quality criteria for priority pollutants. U.S. EPA adopted the NTR on December 22, 1992, and amended it on May 4, 1995, and November 9, 1999. About 40 NTR criteria apply in California. U.S. EPA adopted the CTR on May 18, 2000. The CTR promulgated new toxics criteria for California and incorporated the NTR criteria that applied in the State. U.S. EPA amended the CTR on February 13, 2001.
- 3.3.3. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the *Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California* (State Implementation Policy or SIP). The SIP establishes implementation provisions for priority pollutant criteria and objectives, and provisions for chronic toxicity control. The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria U.S. EPA promulgated for California through the NTR and the priority pollutant objectives the Regional Water Board established through the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria U.S. EPA promulgated through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. Requirements of this Order implement the SIP.

- 3.3.4. **Bacteria Objectives.** The State Water Board adopted the *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Part 3, Bacteria Provisions and a Water Quality Standards Variance Policy* on August 7, 2018, and it became effective on March 22, 2019. This plan establishes *Enterococcus* bacteria water quality objectives and related implementation provisions for discharges to marine and estuarine waters that support the water contact recreation beneficial use.
- 3.3.5. **Sediment Quality.** The State Water Board adopted the *Water Quality Control Plan for Enclosed Bays and Estuaries – Part 1, Sediment Quality* on September 16, 2008, and it became effective on August 25, 2009. The State Water Board adopted amendments to the plan on June 5, 2018, that became effective on March 11, 2019. This plan establishes sediment quality objectives and related implementation provisions for specifically defined sediments in most bays and estuaries.
- 3.3.6. **Antidegradation Policy.** Federal regulations at 40 C.F.R. section 131.12 require that state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy through State Water Board Resolution 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California*, which incorporates the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Basin Plan implements, and incorporates by reference, both the State and federal antidegradation policies. Permitted discharges must be consistent with the antidegradation provisions of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16.
- 3.3.7. **Anti-Backsliding Requirements.** CWA sections 402(o) and 303(d)(4) and 40 C.F.R. section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.
- 3.3.8. **Endangered Species Act Requirements.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code §§ 2050 to 2097) or Federal Endangered Species Act (16 U.S.C.A. §§ 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the State, including protecting rare, threatened, or endangered species. The Discharger is responsible for meeting all applicable Endangered Species Act requirements.
- 3.3.9. **Sewage Sludge and Biosolids.** U.S. EPA administers 40 C.F.R. part 503, Standards for the Use or Disposal of Sewage Sludge, which regulates the final

use or disposal of sewage sludge generated during the treatment of domestic sewage in a municipal wastewater treatment facility. The Discharger is responsible for complying with the requirements of 40 C.F.R. part 503. This Order does not authorize any act that violates those requirements.

**3.4. Impaired Water Bodies on CWA section 303(d) List.** On April 6, 2018, U.S. EPA approved a revised list of impaired waters pursuant to CWA section 303(d), which requires identification of specific water bodies where it is expected that water quality standards will not be met after implementation of technology-based effluent limitations on point sources. Where it has not done so already, the Regional Water Board plans to adopt total maximum daily loads (TMDLs) for pollutants on the 303(d) list. TMDLs establish wasteload allocations for point sources and load allocations for nonpoint sources and are established to achieve water quality standards.

Peyton Slough is not listed as impaired by any pollutants. Carquinez Strait is listed as impaired by chlordane, DDT, dieldrin, dioxin compounds, furan compounds, mercury, PCBs, dioxin-like PCBs, selenium, and invasive species. U.S. EPA approved separate TMDLs for mercury and PCBs in San Francisco Bay on February 12, 2008, and March 29, 2010, respectively. The TMDLs for mercury and PCBs apply to this discharge but are implemented through NPDES Permit CA0038849, not this Order. U.S. EPA also approved a TMDL for selenium in North San Francisco Bay, which includes Carquinez Strait, on August 23, 2016. The selenium TMDL does not require effluent limits for municipal wastewater dischargers because these discharges have an insignificant impact on North San Francisco Bay water quality. As shown in Fact Sheet section 4.3.3, the discharge is not a significant source of chlordane, DDT, or dieldrin because these pollutants have not been detected in the discharge. The discharge is not a source of invasive species because it is disinfected. The discharge is not a significant source of dioxins and furans because discharge concentrations of these pollutants are consistently below the water quality objectives or are estimated concentrations that cannot be reliably quantified.

#### **4. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS**

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants discharged into waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations: 40 C.F.R. section 122.44(a) requires that permits include applicable technology-based limitations and standards, and 40 C.F.R. section 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of receiving waters.



## 4.1. Discharge Prohibitions

### 4.1.1. Prohibitions in this Order

- 4.1.1.1. **Discharge Prohibition 3.1 (No discharge other than as described):** This prohibition is based on 40 C.F.R. section 122.21(a) and Water Code section 13260, which require filing an application and Report of Waste Discharge before a discharge can occur. Discharges not described in the application and Report of Waste Discharge, and subsequently in this Order, are prohibited.
- 4.1.1.2. **Discharge Prohibition 3.2 (No bypass to waters of the United States):** This prohibition is based on 40 C.F.R. section 122.41(m) (see Attachment D section I.G). It does not apply to wastewater diverted around the biotower during essential maintenance or for process control to ensure efficient operation as long as the discharge complies with the effluent and receiving water limits in this Order. When the biotower is not operating, the treatment marshes still provide biological treatment.
- 4.1.1.3. **Discharge Prohibition 3.3 (No average dry weather flow above design capacity):** This Order prohibits average dry weather effluent flows greater than the plant's average dry weather flow design treatment capacity of 3.2 MGD. Exceeding this flow could result in lower treatment reliability and greater potential to violate water quality requirements.
- 4.1.1.4. **Discharge Prohibition 3.4 (No sanitary sewer overflows to waters of the United States):** Basin Plan Table 4-1 (Discharge Prohibition 15) and the CWA prohibit the discharge of wastewater to surface waters, except as authorized under an NPDES permit. Publicly-owned treatment works must achieve secondary treatment at a minimum and any more stringent limitations necessary to meet water quality standards. A sanitary sewer overflow that results in the discharge to waters of the United States of raw sewage or wastewater not meeting this Order's effluent limitations is therefore prohibited under the Basin Plan and CWA.
- 4.1.2 Basin Plan Discharge Prohibition 1.** This Order allows discharge of treated wastewater at Discharge Point 001 without a minimum initial dilution of at least 10:1. Basin Plan Discharge Prohibition 1 (Basin Plan Table 4-1) prohibits the discharge of any wastewater that has particular characteristics of concern to beneficial uses at any point where the discharge does not receive an initial dilution of at least 10:1. However, Basin Plan section 4.2 allows exceptions to Discharge Prohibition 1 when:
- An inordinate burden would be placed on the Discharger relative to the beneficial uses protected, and an equivalent level of environmental protection can be achieved by alternate means;

- Net environmental benefits will be derived as a result of the discharge; or
- A discharge is approved as part of a reclamation project;

The Basin Plan further states:

In reviewing requests for exceptions, the Water Board will consider the reliability of the discharger's system in preventing inadequately treated wastewater from being discharged to the receiving water and the environmental consequence of such discharges.

This Order grants an exception to Basin Plan Discharge Prohibition 1 for the discharges from Discharge Point 001 because construction of a deep water outfall to provide dilution would be inordinately burdensome and the Discharger provides an equivalent level of protection.

Construction of a deep water outfall would be inordinately burdensome, because the treatment plant is located 1.3 miles from the nearest deep water channel, Carquinez Strait. The Discharger provides an equivalent level of protection by providing advanced-secondary treatment that includes ammonia removal in the biotower and Moorhen Marsh, and sand filtration of secondary-treated wastewater. The Facility thus treats the discharge to the advanced secondary treatment level reflected by the BOD and TSS effluent limits in Table 4 of this Order. Moorhen Marsh also provides enhanced treatment reliability by providing storage capacity that can be used to contain peak wet weather flows or for emergency storage in the event of a plant upset.

In addition, the Discharger collaborates with other stakeholders in McNabney Marsh and the Peyton Slough marsh complex to maintain and improve water quality. The Discharger currently participates in regular meetings with other marsh complex stakeholders that are being facilitated by contractors provided by the State Water Board's Office of Public Participation to discuss marsh management and tide gate operation. The Discharger intends to continue participating in these and subsequent meetings with other marsh complex stakeholders. Funding allowing, the Discharger may pursue other collaborative projects aimed at improving understanding of the marsh complex.

The Discharger also continues to provide net environmental benefits by providing high quality freshwater to 40 acres of constructed wetland habitat (Moorhen Marsh) that would otherwise not exist. The marsh provides a diverse freshwater habitat and refuge for an array of migratory and year-round wildlife and provides opportunities for bird watching, photography, and other wildlife-related activities.

## 4.2. Technology-Based Effluent Limitations

### 4.2.1. Scope and Authority

CWA section 301(b) and 40 C.F.R. section 122.44 require that permits include conditions meeting technology-based requirements, at a minimum, and any more stringent effluent limitations necessary to meet water quality standards. The discharges authorized by this Order must meet minimum federal technology-based requirements based on the Secondary Treatment Standards at 40 C.F.R. section 133 as summarized below. Basin Plan Table 4-2 contains additional requirements for certain pollutants.

**Table F-6. Secondary Treatment Standards**

| Parameter                                                      | Monthly Average                          | Weekly Average |
|----------------------------------------------------------------|------------------------------------------|----------------|
| Biochemical Oxygen Demand (BOD) <sup>[1,2]</sup>               | 30 mg/L                                  | 45 mg/L        |
| Carbonaceous Biochemical Oxygen Demand (CBOD) <sup>[1,2]</sup> | 25 mg/L                                  | 40 mg/L        |
| Total Suspended Solids TSS <sup>[2]</sup>                      | 30 mg/L                                  | 45 mg/L        |
| pH                                                             | 6.0 – 9.0 standard units (instantaneous) |                |

Footnotes:

<sup>[1]</sup> CBOD effluent limitations may be substituted for BOD limitations.

<sup>[2]</sup> The monthly average percent removal, by concentration, is not to be less than 85 percent.

### 4.2.2. Technology-Based Effluent Limitations

4.2.2.1. **BOD and TSS.** The average weekly and average monthly BOD and TSS effluent limitations are more stringent than the Secondary Treatment Standards and the limits set forth in Basin Plan Table 4-2. The more stringent limits are comparable to the limitations of other advanced-secondary treatment plants in the San Francisco Bay region and provide part of the basis for the exception to Basin Plan Prohibition 1. Effluent monitoring indicates that the Discharger can meet these limits. The BOD and TSS 85 percent removal requirement is based on the Secondary Treatment Standards and Basin Plan Table 4-2.

4.2.2.2. **pH.** The pH effluent limitations are based on the Secondary Treatment Standards and Basin Plan Table 4-2.

## 4.3. Water Quality-Based Effluent Limitations

### 4.3.1. Scope and Authority

CWA section 301(b) and 40 C.F.R. section 122.44(d) require permits to include limitations more stringent than federal technology-based requirements where necessary to achieve water quality standards. According to 40 C.F.R. section 122.44(d)(1)(i), permits must include effluent limitations for all pollutants that are or may be discharged at levels that have a reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and

narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective, water quality-based effluent limitations (WQBELs) must be established using (1) U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting a narrative criterion, supplemented with relevant information. The process for determining reasonable potential and calculating WQBELs when necessary is intended to achieve applicable water quality objectives and criteria, and thereby protect designated beneficial uses of receiving waters.

#### 4.3.2. **Beneficial Uses and Water Quality Criteria and Objectives**

Discharge Point 001 discharges to Peyton Slough. Fact Sheet section 3.3.1 identifies the beneficial uses of this receiving water. Water quality criteria and objectives to protect these beneficial uses are described below.

- 4.3.2.1. **Basin Plan Objectives.** The Basin Plan specifies numerous water quality objectives, including numeric objectives for 10 priority pollutants, un-ionized ammonia, and temperature. The Basin Plan also contains various narrative objectives, including those for bioaccumulation and toxicity.
- 4.3.2.1.1. **Un-ionized Ammonia.** Basin Plan section 3.3.20 establishes acute and chronic water quality objectives for un-ionized ammonia of 0.16 mg/L (as nitrogen [N]) as a maximum for Central San Francisco Bay and upstream waters, and 0.025 mg/L (as N) as an annual median. The un-ionized ammonia fraction in the effluent and receiving water can be derived from pH, salinity, and temperature data from Monitoring Locations EFF-002 and RSW-004 as discussed in Fact Sheet section 4.3.3.3.2.
- 4.3.2.1.2. **Temperature.** Peyton Slough supports warm water and cold water habitat beneficial uses; therefore, the temperature water quality objectives in Basin Plan section 3.3.17 apply:
- The natural receiving water temperature of inland surface waters shall not be altered unless it can be demonstrated to the satisfaction of the Regional Board that such alteration in temperature does not adversely affect beneficial uses.
  - The temperature of any cold or warm freshwater habitat shall not be increased by more than 5°F (2.8°C) above natural receiving water temperature.
- 4.3.2.1.3. **Dioxin-TEQ.** The narrative bioaccumulation objective (Basin Plan § 3.3.2) states, “Many pollutants can accumulate on particulates, in sediments, or bioaccumulate in fish and other aquatic organisms. Controllable water quality factors shall not cause a detrimental increase in concentrations of

toxic substances found in bottom sediments or aquatic life. Effects on aquatic organisms, wildlife, and human health will be considered.” Because it is the consensus of the scientific community that dioxins and furans associate with particulates, accumulate in sediments, and bioaccumulate in the fatty tissue of fish and other organisms, the Basin Plan’s narrative bioaccumulation water quality objective applies to these pollutants. Elevated levels of dioxins and furans in San Francisco Bay fish tissue demonstrate that the narrative bioaccumulation water quality objective is not being met. U.S. EPA has therefore placed San Pablo Bay on its 303(d) list of receiving waters where water quality objectives are not being met after imposition of applicable technology-based requirements.

When the CTR was promulgated, U.S. EPA stated its support for the regulation of dioxin and dioxin-like compounds through the use of toxicity equivalencies (TEQs). U.S. EPA stated, “For California waters, if the discharge of dioxin or dioxin-like compounds has reasonable potential to cause or contribute to a violation of a narrative criterion, numeric water quality-based effluent limits for dioxin or dioxin-like compounds should be included in NPDES permits and should be expressed using a TEQ scheme” (Fed. Reg. Vol. 65, No. 97, pages 31695-31696, May 18, 2000). This Order uses a TEQ scheme based on a set of toxicity equivalency factors (TEFs) the World Health Organization developed in 2005, and a set of bioaccumulation equivalency factors (BEFs) U.S. EPA developed for the Great Lakes region (40 C.F.R. § 132, Appendix F) to convert the concentration of any congener of dioxin or furan into an equivalent concentration of 2,3,7,8-tetrachlorinated dibenzo-p-dioxin (2,3,7,8-TCDD). Although the 2005 World Health Organization scheme includes TEFs for dioxin-like PCBs, they are not included in this Order’s TEQ scheme. The CTR has established a specific water quality criterion for PCBs, and dioxin-like PCBs are included in the analysis of total PCBs.

The CTR establishes a numeric water quality objective for 2,3,7,8-TCDD of  $1.3 \times 10^{-8}$  µg/L for the protection of human health when water and aquatic organisms are consumed. This CTR criterion is used as a criterion for dioxin-TEQ because dioxin-TEQ represents a toxicity-weighted concentration equivalent to 2,3,7,8 TCDD, thus translating the narrative bioaccumulation objective into a numeric criterion.

- 4.3.2.1.4. **Toxicity.** The narrative toxicity objective (Basin Plan § 3.3.18) states, “All waters shall be maintained free of toxic substances in concentrations that are lethal to or that produce other detrimental responses in aquatic organisms... Attainment of this objective will be determined by analyses of indicator organisms, species diversity, population density, growth anomalies, or toxicity tests... or other methods selected by the Water Board.”

For this Order, the narrative toxicity objective is translated into a numeric criterion of 1.0 chronic toxicity unit (TUc). Toxic units are an indicator of the pollutant of concern (i.e., toxicity). At 1.0 TUc, there is no observable detrimental effect when an indicator organism is exposed to 100 percent effluent; therefore, 1.0 TUc is a direct translation of the narrative objective into a number. Moreover, in *Technical Support Document for Water Quality-based Toxics Control* (Technical Support Document) (EPA/505/2-90-001, section 3.3.3, *Step 3: Decision Criteria for Permit Limit Development*, page 60), U.S. EPA recommends that 1.0 TUc be used as a criterion continuous concentration, which generally implies a four-day average (Technical Support Document, Appendix D, page D-2).

- 4.3.2.2. **CTR Criteria.** The CTR specifies numeric aquatic life and human health criteria for numerous priority pollutants. These criteria apply to inland surface waters and enclosed bays and estuaries. Some human health criteria are for consumption of “water and organisms” and others are for consumption of “organisms only.” The criteria applicable to “organisms only” apply to Peyton Slough because it is not a source of drinking water.
- 4.3.2.3. **NTR Criteria.** The NTR establishes numeric aquatic life and human health criteria for a number of toxic pollutants for San Francisco Bay waters upstream to and including Suisun Bay and the Sacramento-San Joaquin Delta. The NTR criteria apply to Peyton Slough.
- 4.3.2.4. **Bacteria Objectives.** The *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Part 3, Bacteria Provisions and a Water Quality Standards Variance Policy* establishes *Enterococcus* bacteria water quality objectives to limit cases of gastrointestinal illness from water contact recreation. The *Enterococcus* bacteria objectives apply to marine and estuarine waters.
- 4.3.2.5. **Sediment Quality Objectives.** The *Water Quality Control Plan for Enclosed Bays and Estuaries – Part 1, Sediment Quality* contains the following narrative water quality objectives:
- 4.3.2.5.1. “Pollutants in sediments shall not be present in quantities that, alone or in combination, are toxic to benthic communities in bays and estuaries of California.” This objective is to be implemented by integrating three lines of evidence: sediment toxicity, benthic community condition, and sediment chemistry. The policy requires that if the Regional Water Board determines that a discharge has reasonable potential to cause or contribute to an exceedance of this objective, it is to impose the objective as a receiving water limit.
- 4.3.2.5.2. “Pollutants shall not be present in sediments at levels that will bioaccumulate in aquatic life to levels that are harmful to human health in bays and estuaries of California.” This objective is to be implemented by a

three-tiered procedure based on pollutant concentrations in sediment and fish tissue.

- 4.3.2.5.3. “Pollutants shall not be present in sediment at levels that alone or in combination are toxic to wildlife and resident finfish by direct exposure or bioaccumulate in aquatic life at levels that are harmful to wildlife or resident finfish by indirect exposure in bays and estuaries of California.” This objective is to be implemented on a case-by-case basis, based upon an ecological risk assessment.
- 4.3.2.6. **Receiving Water Salinity.** Basin Plan section 4.6.2 (like the CTR and NTR) states that the salinity characteristics (i.e., freshwater vs. saltwater) of the receiving water are to be considered in determining the applicable water quality objectives. Freshwater criteria apply to discharges to waters with salinities equal to or less than one part per thousand (ppt) at least 95 percent of the time. Saltwater criteria apply to discharges to waters with salinities equal to or greater than 10 ppt at least 95 percent of the time in a normal water year. For discharges to water with salinities in between these two categories, or tidally-influenced freshwaters that support estuarine beneficial uses, the water quality objectives are the lower of the salt or freshwater criteria (the latter calculated based on ambient hardness) for each substance.
- Peyton Slough is a tidally-influenced slough. Therefore, the more stringent of the freshwater or saltwater water quality objectives from the Basin Plan, NTR, and CTR apply to this discharge.
- 4.3.2.7. **Receiving Water Hardness.** Ambient hardness data were used to derive freshwater quality objectives that are hardness-dependent. Receiving water monitoring for hardness was conducted in Peyton Slough at Monitoring Locations RSW-003 and -004 from July 2016 through February 2021. Hardness ranged from 110 to 4,000 mg/L as calcium carbonate ( $\text{CaCO}_3$ ). Hardness results above 400 mg/L were censored, and the adjusted geometric mean of the remaining data (218 mg/L as  $\text{CaCO}_3$ ) was used to calculate the water quality objectives for hardness-dependent pollutants.
- 4.3.2.8. **Metals Translators.** Effluent limitations for metals must be expressed as total recoverable metal (40 C.F.R. § 122.45[c]). Since the water quality objectives for metals are typically expressed as dissolved metal, translators must be used to convert metals concentrations from dissolved to total recoverable and vice versa. The CTR contains default translators; however, site-specific conditions, such as water temperature, pH, total suspended solids, and organic carbon may affect the form of metal (dissolved, non-filterable, or otherwise) present and therefore available to cause toxicity. In general, dissolved metals are more available and more toxic to aquatic life than other forms. Site-specific translators can account for site-specific

conditions, thereby preventing overly stringent or under-protective water quality objectives.

This Order's reasonable potential analysis uses CTR default translators for all metals except copper and nickel. The site-specific translators used for copper and nickel are shown in Table F-7.

The site-specific translators for copper are based on the Discharger's *Mt. View Sanitary District Copper Translator Study* (August 2012). The Discharger assessed geographic and seasonal variability by comparing receiving water samples collected at three receiving water locations over a 10-month period (July 2011 – April 2012). Final translators were calculated from the fractional dissolved data for 38 samples. The chronic translator is the median value, and the acute translator is the 90th percentile. These site-specific translators were used to calculate the site-specific water quality objectives that this Order uses to evaluate reasonable potential and calculate effluent limits for copper.

The site-specific translators for nickel are from the Clean Estuary Partnership's *North of Dumbarton Bridge Copper and Nickel Development and Selection of Final Translators* report (March 2005).

**Table F-7. Site-Specific Translators**

| Parameter | Site Specific Translators –<br>Acute | Site Specific Translators -<br>Chronic |
|-----------|--------------------------------------|----------------------------------------|
| Copper    | 0.64                                 | 0.51                                   |
| Nickel    | 0.57                                 | 0.27                                   |

#### 4.3.3. Reasonable Potential Analysis

- 4.3.3.1. **Available Information.** This Order's reasonable potential analysis is based on effluent data the Discharger collected from July 2016 through April 2021 and ambient background data collected through the RMP, supplemented by ambient background data from Bay Area Clean Water Agencies (BACWA). RMP data are summarized in the RMP's *San Francisco Bay California Toxics Rule Priority Pollutant Ambient Water Monitoring Report* (2017), which includes data collected at the Yerba Buena station (BC10) from 1993 through 2017. BACWA data are from BACWA reports *San Francisco Bay Ambient Water Monitoring Interim Report* (2003) and *Ambient Water Monitoring: Final CTR Sampling Update* (2004).

SIP section 1.4.3 requires that background water quality data be representative of the ambient receiving water that will mix with the discharge. RMP monitoring station BC10, relative to other RMP stations, fits SIP guidance for establishing priority pollutant background conditions at Discharge Point 001. For ammonia, receiving water monitoring data collected



from July 2016 through February 2021 at Monitoring Locations EFF-002 and RSW-004 were used.

This Order does not contain WQBELs for constituents that do not demonstrate reasonable potential; however, the MRP still requires monitoring for those pollutants. If effluent concentrations are found to have increased significantly, Provision 6.3.2 of the Order requires the Discharger to investigate the sources of the increases and implement remedial measures if the increases pose a threat to receiving water quality.

4.3.3.2. **Priority Pollutants and Dioxin-TEQ.** SIP section 1.3 sets forth the methodology used to assess whether priority pollutants have reasonable potential to exceed CTR and NTR water quality objectives. Here, SIP section 1.3 is also used as guidance for dioxin-TEQ. The analysis begins with identifying the maximum effluent concentration (MEC) observed for each pollutant based on available effluent concentration data and the ambient background concentration (B). SIP section 1.4.3 states that ambient background concentrations are either the maximum ambient concentration observed or, for water quality objectives intended to protect human health, the arithmetic mean of observed concentrations. There are three triggers in determining reasonable potential:

- **Trigger 1** is activated if the maximum effluent concentration is greater than or equal to the lowest applicable water quality objective ( $MEC \geq$  water quality objective).
- **Trigger 2** is activated if the ambient background concentration observed in the receiving water is greater than the lowest applicable water quality objective ( $B >$  water quality objective) *and* the pollutant is detected in any effluent sample.
- **Trigger 3** is activated if a review of other information indicates that a WQBEL is needed to protect beneficial uses.

The maximum effluent concentrations, most stringent applicable water quality criteria and objectives, and ambient background concentrations used in the analysis are presented in the following table, along with the reasonable potential analysis results (yes, no, or unknown) for each pollutant. Based on this analysis, dioxin-TEQ has reasonable potential by Trigger 1; copper and cyanide have reasonable potential by Trigger 3 because Basin Plan sections 7.2.1.2 and 4.7.2.2 require copper and cyanide WQBELs.

**Table F-8. Reasonable Potential Analysis for Priority Pollutants and Dioxin-TEQ**

| CTR No.   | Pollutant                     | C or Governing Criterion or Objective (µg/L) | MEC or Minimum DL (µg/L) <sup>[1][2]</sup> | B or Minimum DL (µg/L) <sup>[1][2]</sup> | RPA Result <sup>[3]</sup> |
|-----------|-------------------------------|----------------------------------------------|--------------------------------------------|------------------------------------------|---------------------------|
| 1         | Antimony                      | 4,300                                        | 4.0                                        | 18                                       | No                        |
| 2         | Arsenic                       | 36                                           | 2.4                                        | 2.5                                      | No                        |
| 3         | Beryllium                     | No Criterion                                 | <0.90                                      | 0.22                                     | U                         |
| 4         | Cadmium                       | 2.1                                          | <0.050                                     | 0.13                                     | No                        |
| 5a        | Chromium (III) <sup>[4]</sup> | 392                                          | 0.29                                       | 4.4                                      | No                        |
| 5b        | Chromium (VI) <sup>[4]</sup>  | 11                                           | <1.5                                       | 4.4                                      | No                        |
| <b>6</b>  | <b>Copper</b>                 | <b>14</b>                                    | <b>10</b>                                  | <b>2.5</b>                               | <b>Yes</b>                |
| 7         | Lead                          | 8.5                                          | 0.12                                       | 0.80                                     | No                        |
| 8         | Mercury <sup>[5]</sup>        | —                                            | —                                          | —                                        | —                         |
| 9         | Nickel                        | 30                                           | 2.9                                        | 3.7                                      | No                        |
| 10        | Selenium <sup>[5]</sup>       | —                                            | —                                          | —                                        | —                         |
| 11        | Silver                        | 2.2                                          | <0.020                                     | 0.52                                     | No                        |
| 12        | Thallium                      | 6.3                                          | 0.060                                      | 0.023                                    | No                        |
| 13        | Zinc                          | 86                                           | 26                                         | 5.1                                      | No                        |
| <b>14</b> | <b>Cyanide</b>                | <b>2.9</b>                                   | <b>2.6</b>                                 | <b>0.52</b>                              | <b>Yes</b>                |
| 15        | Asbestos (fibers/L)           | No Criterion                                 | —                                          | —                                        | U                         |
| 16        | 2,3,7,8-TCDD                  | 1.4E-08                                      | <2.1E-04                                   | <2.7E-8                                  | U                         |
|           | <b>Dioxin-TEQ</b>             | <b>1.4E-08</b>                               | <b>8.0E-08</b>                             | <b>4.1E-08</b>                           | <b>Yes</b>                |
| 17        | Acrolein                      | 780                                          | <1.7                                       | <0.050                                   | No                        |
| 18        | Acrylonitrile                 | 0.66                                         | <1.8                                       | 0.030                                    | U                         |
| 19        | Benzene                       | 71                                           | <0.18                                      | <0.050                                   | No                        |
| 20        | Bromoform                     | 360                                          | <0.15                                      | <0.15                                    | No                        |
| 21        | Carbon Tetrachloride          | 4.4                                          | <0.16                                      | 0.060                                    | No                        |
| 22        | Chlorobenzene                 | 21,000                                       | <0.18                                      | <0.18                                    | No                        |
| 23        | Chlorodibromomethane          | 34                                           | <0.17                                      | <0.050                                   | No                        |
| 24        | Chloroethane                  | No Criterion                                 | <0.38                                      | <0.38                                    | U                         |
| 25        | 2-Chloroethylvinyl ether      | No Criterion                                 | <0.28                                      | <0.28                                    | U                         |
| 26        | Chloroform                    | No Criterion                                 | <0.19                                      | <0.19                                    | U                         |
| 27        | Dichlorobromomethane          | 46                                           | <0.16                                      | <0.16                                    | No                        |
| 28        | 1,1-Dichloroethane            | No Criterion                                 | <0.19                                      | <0.19                                    | U                         |
| 29        | 1,2-Dichloroethane            | 99                                           | <0.18                                      | 0.040                                    | No                        |
| 30        | 1,1-Dichloroethylene          | 3.2                                          | <0.21                                      | <0.21                                    | No                        |
| 31        | 1,2-Dichloropropane           | 39                                           | <0.18                                      | <0.050                                   | No                        |
| 32        | 1,3-Dichloropropylene         | 1,700                                        | <0.16                                      | <0.16                                    | No                        |
| 33        | Ethylbenzene                  | 29,000                                       | <0.26                                      | <0.26                                    | No                        |
| 34        | Methyl Bromide                | 4,000                                        | <0.30                                      | <0.30                                    | No                        |
| 35        | Methyl Chloride               | No Criterion                                 | <0.30                                      | <0.30                                    | U                         |
| 36        | Methylene Chloride            | 1,600                                        | <0.40                                      | 22                                       | No                        |
| 37        | 1,1,2,2-Tetrachloroethane     | 11                                           | <0.15                                      | <0.050                                   | No                        |
| 38        | Tetrachloroethylene           | 8.9                                          | <0.19                                      | <0.050                                   | No                        |

| CTR No. | Pollutant                   | C or Governing Criterion or Objective (µg/L) | MEC or Minimum DL (µg/L) <sup>[1][2]</sup> | B or Minimum DL (µg/L) <sup>[1][2]</sup> | RPA Result <sup>[3]</sup> |
|---------|-----------------------------|----------------------------------------------|--------------------------------------------|------------------------------------------|---------------------------|
| 39      | Toluene                     | 200,000                                      | <0.19                                      | <0.19                                    | No                        |
| 40      | 1,2-Trans-Dichloroethylene  | 140,000                                      | <0.22                                      | <0.22                                    | No                        |
| 41      | 1,1,1-Trichloroethane       | No Criterion                                 | <0.19                                      | <0.19                                    | U                         |
| 42      | 1,1,2-Trichloroethane       | No Criterion                                 | <0.40                                      | <0.16                                    | U                         |
| 43      | Trichloroethylene           | 42                                           | <0.20                                      | <0.20                                    | No                        |
| 44      | Vinyl Chloride              | 81                                           | <0.25                                      | <0.25                                    | No                        |
| 45      | 2-Chlorophenol              | 525                                          | <0.40                                      | <0.70                                    | No                        |
| 46      | 2,4-Dichlorophenol          | 400                                          | <0.40                                      | <0.90                                    | No                        |
| 47      | 2,4-Dimethylphenol          | 790                                          | <0.40                                      | <0.80                                    | No                        |
| 48      | 2-Methyl- 4,6-Dinitrophenol | 2,300                                        | <0.30                                      | <0.60                                    | No                        |
| 49      | 2,4-Dinitrophenol           | 765                                          | <0.20                                      | <0.70                                    | No                        |
| 50      | 2-Nitrophenol               | 14,000                                       | <0.40                                      | <0.80                                    | No                        |
| 51      | 4-Nitrophenol               | No Criterion                                 | <0.50                                      | <0.50                                    | U                         |
| 52      | 3-Methyl 4-Chlorophenol     | No Criterion                                 | <0.50                                      | <0.80                                    | U                         |
| 53      | Pentachlorophenol           | 7.9                                          | <0.40                                      | <0.60                                    | U                         |
| 54      | Phenol                      | 4,600,000                                    | <0.30                                      | <0.50                                    | No                        |
| 55      | 2,4,6-Trichlorophenol       | 6.5                                          | <0.50                                      | <0.97                                    | No                        |
| 56      | Acenaphthene                | 2,700                                        | <0.020                                     | 0.0020                                   | No                        |
| 57      | Acenaphthylene              | No Criterion                                 | <0.020                                     | 0.0010                                   | U                         |
| 58      | Anthracene                  | 110,000                                      | <0.020                                     | 0.0010                                   | No                        |
| 59      | Benzidine                   | 0.00054                                      | <4.0                                       | <0.00030                                 | U                         |
| 60      | Benzo(a)Anthracene          | 0.049                                        | <0.020                                     | 0.0050                                   | No                        |
| 61      | Benzo(a)Pyrene              | 0.049                                        | <0.020                                     | 0.0020                                   | No                        |
| 62      | Benzo(b)Fluoranthene        | 0.049                                        | <0.020                                     | 0.0050                                   | No                        |
| 63      | Benzo(ghi)Perylene          | No Criterion                                 | <0.020                                     | 0.0030                                   | U                         |
| 64      | Benzo(k)Fluoranthene        | 0.049                                        | <0.020                                     | 0.0020                                   | No                        |
| 65      | Bis(2-Chloroethoxy)Methane  | No Criterion                                 | <0.50                                      | <0.30                                    | U                         |
| 66      | Bis(2-Chloroethyl)Ether     | 1.4                                          | <0.40                                      | <0.30                                    | No                        |
| 67      | Bis(2-Chloroisopropyl)Ether | 170,000                                      | <0.40                                      | <0.60                                    | No                        |
| 68      | Bis(2-Ethylhexyl)Phthalate  | 5.9                                          | 1.0                                        | <0.50                                    | No                        |
| 69      | 4-Bromophenyl Phenyl Ether  | No Criterion                                 | 0.50                                       | <0.23                                    | U                         |
| 70      | Butylbenzyl Phthalate       | 5,200                                        | <0.50                                      | <0.50                                    | No                        |
| 71      | 2-Chloronaphthalene         | 4,300                                        | <0.40                                      | <0.30                                    | No                        |
| 72      | 4-Chlorophenyl Phenyl Ether | No Criterion                                 | <0.50                                      | <0.30                                    | U                         |
| 73      | Chrysene                    | 0.049                                        | <0.020                                     | 0.0020                                   | No                        |
| 74      | Dibenzo(a,h)Anthracene      | 0.049                                        | <0.020                                     | 0.0010                                   | No                        |
| 75      | 1,2-Dichlorobenzene         | 17,000                                       | <0.27                                      | <0.27                                    | No                        |
| 76      | 1,3-Dichlorobenzene         | 2,600                                        | <0.18                                      | <0.18                                    | No                        |
| 77      | 1,4-Dichlorobenzene         | 2,600                                        | <0.18                                      | <0.18                                    | No                        |
| 78      | 3,3 Dichlorobenzidine       | 0.077                                        | <5.0                                       | <0.00020                                 | U                         |
| 79      | Diethyl Phthalate           | 120,000                                      | <0.50                                      | <0.20                                    | No                        |

| CTR No. | Pollutant                 | C or Governing Criterion or Objective (µg/L) | MEC or Minimum DL (µg/L) <sup>[1][2]</sup> | B or Minimum DL (µg/L) <sup>[1][2]</sup> | RPA Result <sup>[3]</sup> |
|---------|---------------------------|----------------------------------------------|--------------------------------------------|------------------------------------------|---------------------------|
| 80      | Dimethyl Phthalate        | 2,900,000                                    | <0.50                                      | <0.20                                    | No                        |
| 81      | Di-n-Butyl Phthalate      | 12,000                                       | <0.40                                      | <0.50                                    | No                        |
| 82      | 2,4-Dinitrotoluene        | 9.1                                          | <0.40                                      | <0.27                                    | No                        |
| 83      | 2,6-Dinitrotoluene        | No Criterion                                 | <0.40                                      | <0.29                                    | U                         |
| 84      | Di-n-Octyl Phthalate      | No Criterion                                 | <0.40                                      | <0.38                                    | U                         |
| 85      | 1,2-Diphenylhydrazine     | 0.54                                         | Unavailable                                | 0.0040                                   | U                         |
| 86      | Fluoranthene              | 370                                          | <0.020                                     | 0.11                                     | No                        |
| 87      | Fluorene                  | 14,000                                       | <0.010                                     | .0020                                    | No                        |
| 88      | Hexachlorobenzene         | 0.00077                                      | <0.40                                      | 0.000020                                 | U                         |
| 89      | Hexachlorobutadiene       | 50                                           | <0.40                                      | <0.30                                    | No                        |
| 90      | Hexachlorocyclopentadiene | 17,000                                       | <0.30                                      | <0.30                                    | No                        |
| 91      | Hexachloroethane          | 8.9                                          | <0.40                                      | <0.20                                    | No                        |
| 92      | Indeno(1,2,3-cd)Pyrene    | 0.049                                        | <0.020                                     | 0.040                                    | No                        |
| 93      | Isophorone                | 600                                          | <0.50                                      | <0.30                                    | No                        |
| 94      | Naphthalene               | No Criterion                                 | <0.020                                     | .0090                                    | U                         |
| 95      | Nitrobenzene              | 1,900                                        | <0.50                                      | <0.25                                    | No                        |
| 96      | N-Nitrosodimethylamine    | 8.1                                          | <0.30                                      | <0.30                                    | No                        |
| 97      | N-Nitrosodi-n-Propylamine | 1.4                                          | <0.50                                      | <0.00020                                 | No                        |
| 98      | N-Nitrosodiphenylamine    | 16                                           | <0.30                                      | <0.0010                                  | No                        |
| 99      | Phenanthrene              | No Criterion                                 | <0.20                                      | 0.00060                                  | U                         |
| 100     | Pyrene                    | 11,000                                       | <0.20                                      | 0.019                                    | No                        |
| 101     | 1,2,4-Trichlorobenzene    | No Criterion                                 | <0.40                                      | <0.30                                    | U                         |
| 102     | Aldrin                    | 0.00014                                      | <0.0020                                    | <0.0000085                               | U                         |
| 103     | Alpha-BHC                 | 0.013                                        | <0.0030                                    | 0.00050                                  | No                        |
| 104     | Beta-BHC                  | 0.046                                        | <0.0030                                    | 0.00040                                  | No                        |
| 105     | Gamma-BHC                 | 0.063                                        | <0.0030                                    | 0.0010                                   | No                        |
| 106     | Delta-BHC                 | No Criterion                                 | <0.0030                                    | 0.00010                                  | U                         |
| 107     | Chlordane                 | 0.00059                                      | <0.020                                     | 0.00010                                  | U                         |
| 108     | 4,4'-DDT                  | 0.00059                                      | <0.0030                                    | 0.000040                                 | U                         |
| 109     | 4,4'-DDE                  | 0.00059                                      | <0.0030                                    | 0.00020                                  | U                         |
| 110     | 4,4'-DDD                  | 0.00084                                      | <0.0030                                    | 0.0010                                   | U                         |
| 111     | Dieldrin                  | 0.00014                                      | <0.0040                                    | 0.00030                                  | U                         |
| 112     | Alpha-Endosulfan          | 0.0087                                       | <0.0030                                    | 0.00030                                  | No                        |
| 113     | beta-Endosulfan           | 0.0087                                       | <0.0030                                    | 0.00010                                  | No                        |
| 114     | Endosulfan Sulfate        | 240                                          | <0.0030                                    | 0.00010                                  | No                        |
| 115     | Endrin                    | 0.0023                                       | <0.0030                                    | 0.00010                                  | U                         |
| 116     | Endrin Aldehyde           | 0.81                                         | <0.0040                                    | 0.000040                                 | No                        |
| 117     | Heptachlor                | 0.00021                                      | <0.0030                                    | <0.0050                                  | U                         |
| 118     | Heptachlor Epoxide        | 0.00011                                      | <0.0030                                    | 0.000020                                 | U                         |
| 119-125 | PCBs sum <sup>[5]</sup>   | —                                            | —                                          | —                                        | —                         |
| 126     | Toxaphene                 | 0.00020                                      | <0.20                                      | <0.30                                    | No                        |

Footnotes:

- [1] The MEC and ambient background concentration are the actual detected concentrations unless preceded by a "<" sign, in which case the value shown is the minimum detection level (MDL).
- [2] The MEC or ambient background concentration is "Unavailable" when there are no monitoring data for the constituent.
- [3] RPA Results = Yes, if MEC  $\geq$  WQC, B > WQC and MEC is detected, or Trigger 3  
= No, if MEC and B are < WQC or all effluent data are undetected  
= Unknown (U) if no criteria have been promulgated or data are insufficient.
- [4] The maximum effluent chromium (III) and ambient background chromium (III) and (VI) concentrations are the total chromium concentration. The chromium (III) and (VI) concentrations are unknown but less than these values.
- [5] SIP section 1.3 excludes from its reasonable potential analysis procedure priority pollutants for which a TMDL has been developed. TMDLs have been developed for mercury and PCBs in San Francisco Bay. Mercury and PCBs from wastewater discharges are regulated by NPDES Permit No. CA0038849, which implements the San Francisco Bay Mercury and PCBs TMDLs. A TMDL has also been developed for selenium in North San Francisco Bay. Basin Plan section 7.2.4.5 finds that municipal wastewater dischargers have no reasonable potential to cause or contribute to the selenium impairment in San Francisco Bay segments and, therefore, are not required to have numeric effluent limitations.

### 4.3.3.3. Ammonia

4.3.3.3.1. **Methodology.** Ammonia is a toxic pollutant but not a priority pollutant as defined by the CTR; therefore, the procedure outlined in the Technical Support Document was used to determine if ammonia in the discharge has reasonable potential to cause a water quality objective to be exceeded in the receiving water. According to the Technical Support Document, the reasonable potential analyses can be performed based on the receiving water concentrations projected using effluent data or measured receiving water concentrations. Both values may be compared directly with the Basin Plan un-ionized ammonia objectives. The following summarizes the steps using effluent data:

- **Step 1.** Determine the total number of samples (n) and the maximum effluent concentration (MEC) in the dataset.
- **Step 2.** Determine the coefficient of variation (CV). For a data set where  $n < 10$ , the CV is estimated to equal 0.6. For a dataset where  $n \geq 10$ , the CV is calculated as the standard deviation divided by the mean.
- **Step 3.** Determine a ratio (R) for projecting the upper bound concentration based on a selected confidence interval (e.g., 95<sup>th</sup> or 99<sup>th</sup> percentile) and assuming a lognormal distribution as follows:

Calculate the percentile ( $P_n$ ) represented by the MEC in the data set of n samples based on the selected confidence level.

$$P_n = (1 - \text{confidence interval})^{1/n}$$

Calculate the concentration-multiplying factors ( $C_P$ ) for the MEC percentile ( $P_n$ ) and the chosen upper bound percentile ( $P_{\text{upper bound}}$ ), typically the 99<sup>th</sup> percentile, using the following equation:

$$C_P = \exp(Z_P\sigma - 0.5\sigma^2)$$

Where:

$$\sigma^2 = \ln(CV^2 + 1)$$

P is the percentile (either  $P_n$  or the selected  $P_{\text{upper bound}}$ )

$Z_p$  is the standard normal distribution value for the percentile P  
(available from statistical references)

Finally, calculate R as:

$$R = C_{\text{upper bound}} / C_{Pn}$$

- **Step 4.** Calculate the projected maximum receiving water concentration (RWC) as follows:

$$RWC = (MEC \times R) / \text{dilution ratio}$$

The dilution ratio can be defined as:

$$\text{Dilution ratio} = (D + [1 \text{ part effluent}]) / (1 \text{ part effluent})$$

Where D is the parts receiving water available to dilute 1 part effluent

Therefore:

$$RWC = (MEC \times R) / (D + 1)$$

- **Step 5.** Compare the RWC to the most stringent water quality objective for the pollutant. There is reasonable potential if the RWC is greater than or equal to the lowest applicable water quality objective.

4.3.3.3.2. **Data Translation.** Effluent and receiving water data are available for total ammonia, but not un-ionized ammonia, because (1) sampling and laboratory methods are unavailable to analyze for un-ionized ammonia, and (2) the fraction of total ammonia that exists in the toxic un-ionized form depends on pH, salinity, and temperature of the receiving water. Total ammonia concentrations (as N) were translated into un-ionized ammonia concentrations for comparison with the Basin Plan un-ionized ammonia objectives based on the following equations (U.S. EPA, 1989, *Ambient Water Quality Criteria for Ammonia (Saltwater)*–1989, EPA Publication 440/5-88-004, 1989):

$$\text{Fraction of un-ionized ammonia} = (1 + 10^{[pK - pH]})^{-1}$$

Where, for salinity less than 1 ppt:

$$pK = 0.09018 + 2,729.92/T$$

T = temperature in Kelvin

Where, for salinity greater than 10 ppt:

$$pK = 9.245 + 0.116(I) + 0.0324(298-T) + 0.0415(P)/T$$

I = molal ionic strength of saltwater =  $19.9273(S)/(1,000-1.005109S)$   
S = salinity (parts per thousand)  
T = temperature in Kelvin  
P = pressure (one atmosphere)

- 4.3.3.3.3. **Analysis Based on Effluent Data.** Concurrent effluent total ammonia, pH, and temperature data, collected at Monitoring Location EFF-002 from June 2018 through February 2021, were used to evaluate reasonable potential based on the acute and chronic ammonia objectives. Steps 1 through 5 as listed in Fact Sheet section 4.3.3.3.1 were used to project a RWC to compare to the acute objective. The annual median did not need to be projected for comparison to the chronic objective.

Based on the 28 effluent data points (n=28) with a calculated un-ionized ammonia MEC of 0.75 mg/L (as N),  $P_n$  at 95 percent confidence was 0.90, indicating the MEC represents the 90<sup>th</sup> percentile of effluent concentrations. With the upper bound set at the 99<sup>th</sup> percentile, R is 1.9 ( $C_{Pn}$  and  $C_{upper\ bound}$  are 1.8 and 3.6). D is assumed to be zero (no dilution); thus the projected RWC is 1.5 mg/L, which is greater than the Basin Plan's acute un-ionized ammonia water quality objective of 0.16 mg/L.

The highest annual median effluent concentration (0.15 mg/L) was compared with the chronic objective (0.025 mg/L as N). Because the projected RWC is greater than the acute water quality objective and the maximum annual median concentration is greater than the chronic objective, there is reasonable potential for ammonia.

- 4.3.3.3.4. **Analysis Based on Receiving Water Data.** Total ammonia, pH, salinity, and temperature data collected concurrently at Monitoring Location RSW-004 from October 2016 through February 2021 were used to evaluate reasonable potential for ammonia. The maximum calculated un-ionized ammonia concentration of 0.027 mg/L was less than the acute water quality objective of 0.16 mg/L, and the highest annual median of 0.010 mg/L was less than the chronic water quality objective of 0.025 mg/L. Based on receiving water data, there is not reasonable potential for ammonia.

- 4.3.3.3.5. **Conclusion.** There is reasonable potential for ammonia to exceed the Basin Plan's acute and chronic water quality objectives based on effluent data. Although receiving water data do not indicate reasonable potential, the discharge occurs in a sensitive shallow-water habitat and receives little dilution. Without regulatory assurance that nitrification will continue, receiving water un-ionized ammonia could increase and cause or contribute to toxicity outside of the mixing zone. Therefore, this Order

contains ammonia WQBELs to ensure that nitrification will continue, and that the discharge will not cause or contribute to such an exceedance.

- 4.3.3.4. **Temperature.** Basin Plan section 3.3.17 prohibits the alteration of natural receiving water temperatures such that beneficial uses are adversely affected and temperature increases greater than 2.8°C above natural receiving water temperatures. Receiving water monitoring from July 2016 through February 2021 showed the effluent did not significantly affect receiving water temperatures. Therefore, the discharge does not have reasonable potential to exceed the Basin Plan water quality objective for temperature.
- 4.3.3.5. **Acute Toxicity.** Basin Plan section 4.5.5.3.1 requires acute toxicity monitoring and limitations, implying there is reasonable potential for the discharge to cause or contribute to exceedances of the acute toxicity water quality objective.
- 4.3.3.6. **Chronic Toxicity.** The Discharger monitored effluent chronic toxicity annually with no toxicity observed; all results were reported as <1.0 TUc. Therefore, there is no reasonable potential for the discharge to cause or contribute to exceedances of the chronic toxicity objective.
- 4.3.3.7. ***Enterococcus Bacteria.*** The *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Part 3, Bacteria Provisions and a Water Quality Standards Variance Policy* requires *Enterococcus* bacteria effluent limitations for discharges to marine and estuarine receiving waters that support the water contact recreation (REC1) beneficial use.
- 4.3.3.8. **Sediment Quality.** Pollutants in some receiving water sediments may be present in quantities that alone or in combination are toxic to benthic communities. However, to date there is no evidence directly linking compromised sediment conditions to the discharges subject to this Order; therefore, the Regional Water Board cannot draw a conclusion about reasonable potential for these discharges to cause or contribute to exceedances of the sediment quality objectives. Nevertheless, the Discharger continues to participate in the Regional Monitoring Program, which routinely monitors San Francisco Bay sediment and seeks to identify stressors responsible for degraded sediment quality.

#### 4.3.4. **Water Quality-Based Effluent Limitations**

WQBELs were developed for the pollutants determined to have reasonable potential to cause or contribute to exceedances of water quality objectives. With the exception of acute toxicity and *Enterococcus* bacteria (discussed below), the WQBEL calculations are based on the procedures in SIP section 1.4.



- 4.3.4.1. **WQBEL Expression.** NPDES regulations at 40 C.F.R. section 122.45(d) require that permit limits for publicly-owned treatment works be expressed as average weekly and average monthly limits, unless impracticable. This Order contains daily limits instead of weekly limits because daily limits better protect against acute water quality effects and are necessary to prevent fish kills or mortality to aquatic organisms. Weekly limits could allow acute and chronic toxicity to occur over shorter periods (acute and chronic aquatic life criteria are typically expressed as one-hour and four-day averages).
- 4.3.4.2. **Mixing Zones and Dilution Credits.** The Order provides a dilution credit for cyanide based on Basin Plan Table 4-6. The cyanide dilution credit is 3.25:1 (3.25 parts combined effluent and receiving water to 1 part effluent, or D=2.25).
- 4.3.4.3. **WQBEL Calculations.** The following table shows the copper, cyanide, ammonia, and dioxin-TEQ WQBEL calculations in accordance with SIP section 1.4.

**Table F-9. WQBEL Calculations**

| Pollutant                                      | Copper                             | Cyanide                            | Total Ammonia (Acute)   | Total Ammonia (Chronic) | Dioxin-TEQ           |
|------------------------------------------------|------------------------------------|------------------------------------|-------------------------|-------------------------|----------------------|
| Units                                          | µg/L                               | µg/L                               | mg/L                    | mg/L                    | µg/L                 |
| Basis and Criteria type                        | Basin Plan Site-Specific Objective | Basin Plan Site-Specific Objective | Basin Plan Aquatic Life | Basin Plan Aquatic Life | Basin Plan Narrative |
| Criteria - Acute                               | —                                  | —                                  | 3.2 <sup>[1]</sup>      | —                       | —                    |
| Criteria - Chronic                             | —                                  | —                                  | —                       | 3.0 <sup>[1]</sup>      | —                    |
| Site-Specific Objective Criteria - Acute       | 15                                 | 9.4                                | —                       | —                       | —                    |
| Site-Specific Objective Criteria - Chronic     | 12                                 | 2.9                                | —                       | —                       | —                    |
| Water Effects Ratio (WER)                      | 1                                  | 1                                  | 1                       | 1                       | 1                    |
| Lowest WQO                                     | 12                                 | 2.9                                | 3.2                     | 3.0                     | 1.4E-08              |
| Site Specific Translator – MDEL                | <sup>[2]</sup>                     | —                                  | —                       | —                       | —                    |
| Site Specific Translator – AMEL                | <sup>[2]</sup>                     | —                                  | —                       | —                       | —                    |
| Dilution Factor (D)                            | —                                  | 2.25                               | —                       | —                       | —                    |
| No. of samples per month                       | 4                                  | 4                                  | 4                       | 30 <sup>[3]</sup>       | 4                    |
| Aquatic life criteria analysis required? (Y/N) | Y                                  | Y                                  | Y                       | Y                       | N                    |
| HH criteria analysis required? (Y/N)           | N                                  | Y                                  | N                       | N                       | Y                    |
|                                                |                                    |                                    |                         |                         |                      |
| Applicable Acute WQO                           | 15                                 | 9.4                                | 3.2                     | —                       | —                    |
| Applicable Chronic WQO                         | 12                                 | 2.9                                | —                       | 3.0                     | —                    |
| HH Criteria                                    | —                                  | 220,000                            | —                       | —                       | 1.4E-08              |

| Pollutant                                                                 | Copper | Cyanide   | Total Ammonia (Acute) | Total Ammonia (Chronic) | Dioxin-TEQ |
|---------------------------------------------------------------------------|--------|-----------|-----------------------|-------------------------|------------|
| Background (Maximum Concentration for Aquatic Life Calc.)                 | 2.5    | 0.52      | 0.68                  | 0.24 <sup>[4]</sup>     | —          |
| Background (Average Concentration for Human Health Calc.)                 | —      | 0.90      | —                     | —                       | 4.1E-08    |
| Is the pollutant on the 303d list and/or bioaccumulative (Y/N)?           | N      | N         | N                     | N                       | Y          |
| ECA Acute                                                                 | 15     | 29        | 3.2                   | —                       | —          |
| ECA Chronic                                                               | 12     | 8.3       | —                     | 3.0                     | —          |
| ECA HH                                                                    | —      | 710,000   | —                     | —                       | 1.4E-08    |
| No. of data points <10 or at least 80% of data reported non-detect? (Y/N) | N      | N         | N                     | N                       | Y          |
| Avg of effluent data points                                               | 6.0    | 1.1       | 0.75                  | 0.75                    | —          |
| Std Dev of effluent data points                                           | 1.4    | 0.55      | 1.4                   | 1.4                     | —          |
| CV Calculated                                                             | 0.23   | 0.48      | 1.9                   | 1.9                     | —          |
| CV (Selected) – Final                                                     | 0.23   | 0.48      | 1.9                   | 1.9                     | 0.60       |
| ECA Acute Mult99                                                          | 0.60   | 0.38      | 0.12                  | 0.12                    | —          |
| ECA Chronic Mult99                                                        | 0.77   | 0.59      | 0.22                  | 0.49                    | —          |
| LTA Acute                                                                 | 8.8    | 11        | 0.39                  | —                       | —          |
| LTA Chronic                                                               | 9.0    | 4.9       | —                     | 1.5                     | —          |
| Minimum of LTAs                                                           | 8.8    | 4.9       | 0.39                  | 1.5                     | —          |
| AMEL Mult95                                                               | 1.2    | 1.4       | 2.7                   | 1.6                     | 1.6        |
| MDEL Mult99                                                               | 1.7    | 2.6       | 8.1                   | 8.1                     | 3.1        |
| AMEL (Aquatic Life)                                                       | 11     | 7.0       | 1.1                   | 2.4                     | —          |
| MDEL (Aquatic Life)                                                       | 15     | 13        | 3.2                   | 12                      | —          |
| MDEL/AMEL Multiplier                                                      | 1.4    | 1.8       | 3.0                   | 5.0                     | 2.0        |
| AMEL (Human Health)                                                       | —      | 710,000   | —                     | —                       | 1.4E-08    |
| MDEL (Human Health)                                                       | —      | 1,300,000 | —                     | —                       | 2.8E-08    |
| Minimum of AMEL for Aq. Life vs HH                                        | 11     | 7.0       | 1.1                   | 2.4                     | 1.4E-08    |
| Minimum of MDEL for Aq. Life vs HH                                        | 15     | 13        | 3.2                   | 12                      | 2.8E-08    |
| Previous Order Limit – AMEL                                               | 11     | 7.0       | 1.6                   | 1.6                     | 1.4E-08    |

| Pollutant                   | Copper    | Cyanide    | Total Ammonia (Acute) | Total Ammonia (Chronic) | Dioxin-TEQ     |
|-----------------------------|-----------|------------|-----------------------|-------------------------|----------------|
| Previous Order Limit – MDEL | 15        | 13         | 4.7                   | 4.7                     | 2.8E-08        |
|                             |           |            |                       |                         |                |
| <b>Final Limit – AMEL</b>   | <b>11</b> | <b>7.0</b> | <b>1.1</b>            | <b>1.6</b>              | <b>1.4E-08</b> |
| <b>Final Limit – MDEL</b>   | <b>15</b> | <b>13</b>  | <b>3.2</b>            | <b>4.7</b>              | <b>2.8E-08</b> |

Footnotes:

- [1] The Basin Plan water quality objectives for un-ionized ammonia were converted to total ammonia using the un-ionized ammonia fraction calculated based on pH, temperature, and salinity data collected at Monitoring Location RSW-004 during the term of the previous order. The equations for calculating the un-ionized ammonia fraction are shown in Fact Sheet § 4.3.3.3.2
- [2] The site-specific translators for copper shown in Table F-7 were used to calculate the site-specific objectives for copper shown above.
- [3] Number of samples per month estimated as 30 because the Basin Plan chronic ammonia water quality objective is based on an annual median instead of a 4-day average; thus the maximum possible number of daily samples over the averaging period but still within the month is 30 instead of 4.
- [4] The median background ammonia concentration is used for the ammonia effluent limit calculation based on the chronic criterion.

**4.3.4.4. Acute Toxicity.** This Order includes acute toxicity effluent limitations based on Basin Plan Table 4-3. Based on Basin Plan section 3.3.20, if the Discharger can demonstrate that acute toxicity that exceeds this Order's acute toxicity limitations is caused solely by ammonia, and that the ammonia in the discharge complies with this Order's effluent limitations for ammonia, then such toxicity does not constitute a violation of this Order's effluent limitations for acute toxicity.

**4.3.4.5. *Enterococcus* Bacteria.** The *Enterococcus* effluent limitations are based on the *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Part 3, Bacteria Provisions and a Water Quality Standards Variance Policy*, which requires weekly geometric mean and 90th percentile limitations for discharges to receiving waters with the water contact recreation (REC-1) beneficial use.

#### **4.4. Discharge Requirement Considerations**

**4.4.1. Anti-backsliding.** This Order complies with the anti-backsliding provisions of CWA sections 402(o) and 303(d)(4), and 40 C.F.R. section 122.44(l), which generally require effluent limitations in a reissued permit to be as stringent as those in the previous order. The requirements of this Order are at least as stringent as those in the previous order.

This Order does not retain effluent limits on benzo(a)anthracene from the previous order because data no longer indicate reasonable potential for the pollutant to exceed water quality objectives. This is consistent with State Water Board Order WQ 2001-16.

This Order removes oil and grease effluent limitations pursuant to Resolution R2-2020-0031. Although Clean Water Act section 402(o) prohibits backsliding, with certain exceptions, Clean Water Act section 303(d)(4) allows backsliding

when consistent with antidegradation policies. In this case, removing the oil and grease effluent limitations is consistent with antidegradation policies as explained below.

The Order contains new *Enterococcus* bacteria effluent limits based on the *Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California – Part 3, Bacteria Provisions and a Water Quality Standards Variance Policy*. The new effluent limitations are expressed using different averaging periods than the previous limitations and are therefore not directly comparable.

- 4.4.2. **Antidegradation.** This Order complies with the antidegradation provisions of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16. It does not authorize lowering water quality as compared to the level of discharge authorized in the previous order, which is the baseline by which to measure whether degradation will occur. This Order does not allow for an increased flow, a reduced level of treatment, or increased effluent limitations relative to the previous order. The elimination of oil and grease effluent limits is consistent with antidegradation policies because it is not expected to result in an increased volume or concentration of oil and grease in the discharge or violate oil and grease water quality objectives. The existing oil and grease limits do not drive the advanced secondary treatment performance of the plant, thus removing them will not reduce treatment. The receiving waters here attain oil and grease water quality objectives.
- 4.4.3. **Stringency of Requirements for Individual Pollutants.** This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based requirements implement minimum, applicable federal technology-based requirements. In addition, this Order contains more stringent effluent limitations as necessary to meet water quality standards. Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement CWA requirements.

This Order's WQBELs have been derived to implement water quality objectives that protect beneficial uses. The beneficial uses and water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 C.F.R. section 131.38. The procedures for calculating these WQBELs are based on the CTR, as implemented in accordance with the SIP, which U.S. EPA approved on May 18, 2000. U.S. EPA approved most Basin Plan beneficial uses and water quality objectives prior to May 30, 2000. Beneficial uses and water quality objectives submitted to U.S. EPA prior to May 30, 2000, but not approved by U.S. EPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 C.F.R. section 131.21(c)(1). U.S. EPA approved the remaining beneficial uses and water quality objectives, so they

are also applicable water quality standards pursuant to 40 C.F.R. section 131.21(c)(2).

## **5. RATIONALE FOR RECEIVING WATER LIMITATIONS**

The receiving water limitations in sections 5.1 and 5.2 of the Order are based on Basin Plan narrative and numeric water quality objectives. The receiving water limitation in section 5.3 of the Order requires compliance with federal and State water quality standards in accordance with the CWA and regulations adopted thereunder.

The receiving waters for this discharge include McNabney Marsh, i.e., the 138 acres upstream of the Union Pacific Railroad railway and Waterfront Road (see Attachment B, Figure B-1). McNabney Marsh is co-owned by the Discharger and the East Bay Regional Parks District, which are the two agencies primarily responsible for managing it. However, water quality in the Peyton Slough marsh complex, including in McNabney Marsh, is affected by the actions of other stakeholders in the marsh complex. In particular, the marshes and tide gate downstream of Waterfront Road are on the property of and are primarily managed by Eco-Services Operations, LLC, and TransMontaigne Operating Company LP. In addition, several other stakeholders, with concerns related to water quality, habitat, odors, mosquito control, flooding, and sea level rise, have interests in the Peyton Slough marsh complex. The actions of these stakeholders affect McNabney Marsh and vice versa; however, no single stakeholder has overall control or authority over the marsh complex. Thus, a collaborative approach is needed to ensure that receiving water limits are met.

The Discharger has agreed to work with the other landowning stakeholders (Eco-Services, EBRPD, and TransMontaigne) regarding tide gate operations in 2022, with the goal of improving flushing and circulation in McNabney Marsh while maintaining pickleweed habitat in Eco-Services' 17-acre mitigation marsh (Rhodia Marsh), located downstream and just northwest of McNabney Marsh.

## **6. RATIONALE FOR PROVISIONS**

### **6.1. Standard Provisions**

Attachment D contains standard provisions that apply to all NPDES permits in accordance with 40 C.F.R. section 122.41 and additional conditions applicable to specific categories of permits in accordance with 40 C.F.R. section 122.42. The Discharger must comply with these provisions. The conditions set forth in 40 C.F.R. sections 122.41(a)(1) and (b) through (n) apply to all state-issued NPDES permits and must be incorporated into permits either expressly or by reference.

In accordance with 40 C.F.R. section 123.25(a)(12), states may omit or modify conditions to impose more stringent requirements. Attachment G contains standard provisions that supplement the provisions in Attachment D. This Order omits the federal conditions that address enforcement authority specified in

40 C.F.R. sections 122.41(j)(5) and (k)(2) because the State's enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates Water Code section 13387(e) by reference.

## **6.2. Monitoring and Reporting Provisions**

CWA section 308 and 40 C.F.R. sections 122.41(h), 122.41(j)-(l), 122.44(i), and 122.48 require that NPDES permits specify monitoring and reporting requirements. Water Code section 13383 also authorizes the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The MRP establishes monitoring, reporting, and recordkeeping requirements that implement federal and State requirements. For more information, see Fact Sheet section 7. The Regional Water Board Order No. R2-2016-0008 allows the Discharger to opt into certain alternative monitoring requirements.

## **6.3. Special Provisions**

### **6.3.1. Reopener Provisions**

These provisions are based on 40 C.F.R. sections 122.62 and 122.63 and allow modification of this Order and its effluent limitations as necessary in response to updated water quality objectives, regulations, or other new and relevant information that may become available in the future, and other circumstances as allowed by law.

### **6.3.2. Effluent Characterization Study and Report**

This Order does not include WQBELs for pollutants that do not demonstrate reasonable potential, but this provision requires the Discharger to evaluate monitoring data to verify that the reasonable potential analysis conclusions of this Order remain valid. This requirement is authorized pursuant to 40 C.F.R. section 122.41(h) and Water Code section 13383, and is necessary to inform the next permit reissuance and to ensure that the Discharger takes timely steps in response to any unanticipated change in effluent quality during the term of this Order.

### **6.3.3. Pollutant Minimization Program**

This provision is based on Basin Plan section 4.13.2 and SIP section 2.4.5.

### **6.3.4. Special Provisions for Publicly-Owned Treatment Works**

- 6.3.4.1. Sludge and Biosolids Management.** This provision is based on Basin Plan section 4.17. "Sludge" refers to the solid, semisolid, and liquid residue removed during primary, secondary, and advanced wastewater treatment processes. "Biosolids" refers to sludge that has been treated and may be beneficially reused.

6.3.4.2. **Collection System Management.** The Discharger's collection system is part of the Facility regulated through this Order. This provision requires compliance with Attachments D and G and states that these requirements may be satisfied by separately complying with State Water Board Order 2006-0003-DWQ, Statewide General Waste Discharge Requirements for Sanitary Sewer Systems, as amended by State Water Board Order WQ 2013-0058-EXEC and any subsequent order updating these requirements. These statewide WDRs require public agencies that own or operate sanitary sewer systems with one or more miles of sewer lines to enroll for coverage and comply with requirements to develop sanitary sewer management plans and report sanitary sewer overflows, among other provisions and prohibitions. The statewide WDRs contain requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows that are more extensive and, therefore, more stringent than the standard provisions in Attachments D and G. Compliance with the statewide WDRs will satisfy the corresponding requirements in Attachments D and G.

6.3.4.3. **Resource Recovery from Anaerobically Digestible Material.** Standard Operating Procedures are required for publicly-owned treatment works that accept hauled waste food, fats, oil, and grease for injection into anaerobic digesters. The development and implementation of Standard Operating Procedures for management of these materials is intended to allow the California Department of Resources Recycling and Recovery to exempt this activity from separate and redundant permitting programs.

Some POTWs choose to accept organic material, such as waste food, fats, oils, and grease, into their anaerobic digesters to increase production of methane and other biogases for energy production and to prevent such materials from being discharged into the collection system and potentially causing sanitary sewer overflows. The California Department of Resources Recycling and Recovery has proposed to exempt publicly-owned treatment works from Process Facility/Transfer Station permit requirements when the same activity is regulated under WDRs or NPDES permits. The proposed exemption is restricted to anaerobically digestible materials that have been prescreened, slurried, processed, and conveyed in a closed system for co-digestion with regular sewage sludge. The exemption requires that the publicly-owned treatment works develop Standard Operating Procedures for proper handling, processing, tracking, and management of anaerobically digestible material.

### 6.3.5. **Other Special Provisions**

6.3.5.1. **Copper Action Plan.** This provision is based on Basin Plan section 7.2.1.2 and is necessary to ensure that use of copper site-specific objectives is consistent with antidegradation policies. This Order requires the Discharger to implement source control and pollution prevention for identified sources.

Additional actions may be necessary depending on the three-year rolling mean copper concentration in San Pablo Bay. Data the San Francisco Estuary Institute compiled for 2015-2019 indicate no degradation of San Francisco Bay water quality with respect to copper ([SFEI: Copper Site Specific Objective 3-year Rolling Averages](#)).

- 6.3.5.2. **Cyanide Action Plan.** This provision is based on Basin Plan section 4.7.2.2 and is necessary to ensure that use of cyanide site-specific objectives is consistent with antidegradation policies. The threshold for considering influent cyanide concentrations to indicate a possible “significant cyanide discharge” in the Discharger’s service area is set at 6.0 µg/L. This concentration is more than twice the maximum cyanide concentration (2.9 µg/L) found in the treatment plant influent during the previous order term. Because the Discharger has observed no influent cyanide concentrations greater than 2.9 µg/L during the previous permit term, if influent concentrations twice this level were observed, there could be a significant cyanide source.
- 6.3.5.3. **Average Selenium Load.** This provision is based on Basin Plan section 7.2.4.5. The information will be used to confirm that selenium loads are consistent with wasteload allocations. The requirements regarding treatment of estimated and non-detect values are consistent with the load calculations performed for the North San Francisco Bay Selenium TMDL.
- 6.3.5.4. **Moorhen Marsh Management.** This requirement is necessary to ensure that Moorhen Marsh is operated in a way that prevents nuisance conditions.

## 7. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

The following provides the rationale for the monitoring and reporting requirements in the MRP. Regional Water Board Order R2-2016-0008 allows the Discharger to opt into certain alternative monitoring requirements.

### 7.1 Monitoring Requirements Rationale

- 7.1.1. **Influent Monitoring.** Influent BOD and TSS monitoring is necessary to evaluate compliance with this Order’s 85 percent removal requirements. Basin Plan section 4.7.2.2 requires cyanide monitoring because this Order is based on site-specific cyanide water quality objectives.
- 7.1.2. **Effluent Monitoring.** Effluent monitoring at Monitoring Location EFF-001 is necessary to understand Facility operations, evaluate compliance with this Order’s requirements, and conduct future reasonable potential analyses. Effluent monitoring for ammonia, pH, and temperature at Monitoring Location EFF-002 is necessary to understand Facility operations related to nitrification and ammonia removal, to evaluate compliance with this Order’s ammonia limits, and to conduct future reasonable potential analyses for un-ionized ammonia. pH monitoring at Monitoring Location EFF-001 is needed to evaluate



compliance with this Order's effluent limits for pH, and at Monitoring Location EFF-002 to calculate the effluent un-ionized ammonia concentrations for comparison to the Basin Plan's water quality objectives for un-ionized ammonia.

- 7.1.3. **Toxicity Monitoring.** Acute and chronic toxicity tests are necessary to conduct future reasonable potential analyses. Acute toxicity tests are also necessary to evaluate compliance with this Order's acute toxicity effluent limitations.

The Discharger conducted a screening chronic toxicity test at test conditions defined in the State Policy for Water Quality Control: Toxicity Provisions from October 2019 to September 2020. The Tier 1 species fathead minnow (*Pimephales promelas*), red abalone (*Haliotis rufescens*), and green algae (*Selenastrum capricornutum*) were screened using four effluent samples (one collected in each quarter during the testing period). All species and test events provided "pass" results using the Test of Significant Toxicity at percent effects ranging from 0.0 percent (red abalone, development endpoint) to 8.1 percent (fathead minnow, growth endpoint). Fathead minnow was the most sensitive species.

- 7.1.4. **Receiving Water Monitoring.** Monitoring for most receiving water parameters is necessary to evaluate compliance with the receiving water limits. Monitoring for hardness, ammonia, temperature, pH, and salinity are necessary to complete future reasonable potential analyses. Also, this Order requires the Discharger to continue to participate in the RMP, which collects data on pollutants and toxicity in water, sediment, and estuary biota.

Further study of the receiving water is necessary to understand marsh dynamics, tide gate operations, and how effluent quality interacts with those dynamics. The Regional Water Board anticipates that additional information on the marsh will be developed collaboratively through the stakeholder planning process during this permit term.

- 7.1.5. **Recycled Water Monitoring.** The recycled water monitoring and reporting requirements incorporate the existing requirements of State Water Board Order WQ 2019-0037-EXEC (Amending Monitoring and Reporting Programs for Waste Discharge Requirements, National Pollutant Discharge Elimination System Permits, Water Reclamation Requirements, Master Recycling Permits, and General Waste Discharge Requirements), issued on July 24, 2019, pursuant to Water Code sections 13267 and 13383. The Notice of Applicability issued on April 8, 2020, for enrollment under State Water Board Order WQ 2016-0068-DDW (Water Reclamation Requirements for Recycled Water Use) contains additional recycled water monitoring and reporting requirements not imposed through this Order.

- 7.2. **Monitoring Requirements Summary.** The table below summarizes routine monitoring requirements. This table is for informational purposes only. The actual requirements are specified in the MRP and elsewhere in this Order. In addition to

undertaking the monitoring below, the Discharger must also conduct receiving water monitoring by continuing to participate in the Regional Monitoring Program.

**Table F-10. Monitoring Requirements Summary**

| Parameter <sup>[1]</sup>  | Influent<br>INF-001 | Effluent<br>EFF-001 | Effluent<br>EFF-002       | Receiving Water<br>RSW-001, RSW-002,<br>RSW-003, RSW-004 |
|---------------------------|---------------------|---------------------|---------------------------|----------------------------------------------------------|
| Flow                      | —                   | Continuous/D        | —                         | —                                                        |
| BOD <sub>5</sub>          | 1/Week              | 1/Week              | —                         | —                                                        |
| TSS                       | 1/Week              | 1/Week              | —                         | —                                                        |
| Enterococcus              | —                   | 3/Week              | —                         | Support RMP                                              |
| pH                        | —                   | 5/Week              | 1/Month <sup>[1, 2]</sup> | 1/Quarter <sup>[1]</sup>                                 |
| Ammonia, Total            | —                   | —                   | 1/Month <sup>[1, 2]</sup> | 1/Quarter <sup>[1]</sup>                                 |
| Copper, Total Recoverable | —                   | 1/Month             | —                         | Support RMP                                              |
| Cyanide, Total            | 1/Year              | 1/Month             | —                         | Support RMP                                              |
| Dioxin-TEQ                | —                   | 1/Year              | —                         | Support RMP                                              |
| Selenium                  | —                   | Once                | —                         | Support RMP                                              |
| Dissolved Oxygen          | —                   | —                   | —                         | 1/Quarter                                                |
| Dissolved Sulfides        | —                   | —                   | —                         | 1/Quarter                                                |
| Hardness                  | —                   | —                   | —                         | 1/Quarter                                                |
| Nitrate-nitrite           | —                   | —                   | —                         | 1/Quarter                                                |
| Phosphorus                | —                   | —                   | —                         | 1/Quarter                                                |
| Salinity                  | —                   | —                   | —                         | 1/Quarter <sup>[1]</sup>                                 |
| Temperature               | —                   | —                   | 1/Month <sup>[1, 2]</sup> | 1/Quarter <sup>[1]</sup>                                 |
| Turbidity                 | —                   | —                   | —                         | 1/Quarter                                                |
| Acute Toxicity            | —                   | 2/Year              | —                         | —                                                        |
| Chronic Toxicity          | —                   | 1/Quarter           | —                         | —                                                        |
| Standard Observations     | —                   | —                   | —                         | 1/Quarter                                                |
| Other Priority Pollutants | —                   | Once                | —                         | Support RMP                                              |

**Footnotes:**

<sup>[1]</sup> Ammonia, pH, salinity (for receiving water), and temperature monitoring must occur concurrently to allow for calculation of the un-ionized ammonia fraction. Effluent ammonia, pH, and temperature must be monitored at Monitoring Location EFF-002.

<sup>[2]</sup> The monitoring frequency for ammonia, pH, salinity, and temperature must be once per day when the biotower is not operating.

## 8. PUBLIC PARTICIPATION

The Regional Water Board considered the issuance of WDRs that will serve as an NPDES permit for the Facility. As a step in the WDR adoption process, Regional Water Board staff developed tentative WDRs and encouraged public participation in the WDR adoption process.

**8.1. Notification of Interested Parties.** The Regional Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge, and provided an opportunity to submit written comments and recommendations. The public had access to the agenda and any changes in dates and locations through the [Regional Water Board's website](#).

- 8.2. Written Comments.** Interested persons were invited to submit written comments concerning the tentative WDRs as explained through the notification process. Comments were to be submitted either in person or by mail to the Executive Officer at the Regional Water Board at 1515 Clay Street, Suite 1400, Oakland, California 94612, to the attention of John Madigan.

For full staff response and Regional Water Board consideration, written comments were due at the Regional Water Board office by 5:00 p.m. on November 19, 2021.

- 8.3. Public Hearing.** The Regional Water Board held a public hearing on the tentative WDRs during its meeting at the following date and time:

Date: December 15, 2021  
Time: 9:00 a.m.

Contact: John H. Madigan, (510) 622-2405,  
[John.Madigan@waterboards.ca.gov](mailto:John.Madigan@waterboards.ca.gov)

Interested persons were provided notice of the hearing and information on how to participate. At the public hearing, the Regional Water Board heard testimony pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested to be in writing.

Dates and venues can change. The [Regional Water Board's website](#) is where one can access the current agenda for changes.

- 8.4. Reconsideration of Waste Discharge Requirements.** Any person aggrieved by this Regional Water Board action may petition the State Water Board to review the action in accordance with Water Code section 13320 and California Code of Regulations, title 23, sections 2050. The State Water Board must receive the petition at the following address within 30 calendar days of the date of Regional Water Board action:

State Water Resources Control Board  
Office of Chief Counsel  
P.O. Box 100, 1001 I Street  
Sacramento, CA 95812-0100

A petition may also be filed by email at [waterqualitypetitions@waterboards.ca.gov](mailto:waterqualitypetitions@waterboards.ca.gov).

For instructions on how to file a water quality petition for review, see the [Water Board's petition instructions](#).

- 8.5. Information and Copying.** The Report of Waste Discharge, related supporting documents, and comments received are on file. To review these documents, contact Melinda Wong the Regional Water Board's custodian of records by calling (510) 622-2300 or emailing [Melinda.Wong@waterboards.ca.gov](mailto:Melinda.Wong@waterboards.ca.gov). Document copying may be arranged.

- 8.6. Register of Interested Persons.** Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference the Facility, and provide a name, address, and phone number.
- 8.7. Additional Information.** Requests for additional information or questions regarding this Order should be directed to John H. Madigan, (510) 622-2405, [John.Madigan@waterboards.ca.gov](mailto:John.Madigan@waterboards.ca.gov).

**ATTACHMENT G – REGIONAL STANDARD PROVISIONS,  
AND MONITORING AND REPORTING REQUIREMENTS  
(SUPPLEMENT TO ATTACHMENT D)**

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**ATTACHMENT G – REGIONAL STANDARD PROVISIONS,  
AND MONITORING AND REPORTING REQUIREMENTS  
(SUPPLEMENT TO ATTACHMENT D)**

**APPLICABILITY**

This document supplements the requirements of Federal Standard Provisions (Attachment D). For clarity, these provisions are arranged using the same headings as those used in Attachment D.

**1. STANDARD PROVISIONS – PERMIT COMPLIANCE**

**1.1. Duty to Comply – Not Supplemented**

**1.2. Need to Halt or Reduce Activity Not a Defense – Not Supplemented**

**1.3. Duty to Mitigate – Supplement to Attachment D, Provision 1.3.**

1.3.1. **Contingency Plan.** The Discharger shall maintain a Contingency Plan as prudent in accordance with current facility emergency planning. The Contingency Plan shall describe procedures to ensure that existing facilities remain in, or are rapidly returned to, operation in the event of a process failure or emergency incident, such as employee strike, strike by suppliers of chemicals or maintenance services, power outage, vandalism, earthquake, or fire. The Discharger may combine the Contingency Plan and Spill Prevention Plan (see Provision 1.3.2, below) into one document. In accordance with Regional Water Board Resolution No. 74-10, discharge in violation of the permit where the Discharger has failed to develop and implement a Contingency Plan as described below may be the basis for considering the discharge a willful and negligent violation of the permit pursuant to California Water Code section 13387. The Contingency Plan shall, at a minimum, provide for the following:

- 1.3.1.1. Sufficient personnel for continued facility operation and maintenance during employee strikes or strikes against contractors providing services;
- 1.3.1.2. Maintenance of adequate chemicals or other supplies, and spare parts necessary for continued facility operations;
- 1.3.1.3. Emergency standby power;
- 1.3.1.4. Protection against vandalism;
- 1.3.1.5. Expedient action to repair failures of, or damage to, equipment, including any sewer lines;

- 1.3.1.6. Reporting of spills and discharges of untreated or inadequately treated wastes, including measures taken to clean up the effects of such discharges; and
- 1.3.1.7. Maintenance, replacement, and surveillance of physical condition of equipment and facilities, including any sewer lines.
- 1.3.2. **Spill Prevention Plan.** The Discharger shall maintain a Spill Prevention Plan to prevent accidental discharges and to minimize the effects of any such discharges. The Spill Prevention Plan shall do the following:
  - 1.3.2.1. Identify the possible sources of accidental discharge, untreated or partially-treated waste bypass, and polluted drainage;
  - 1.3.2.2. State when current facilities and procedures became operational and evaluate their effectiveness; and
  - 1.3.2.3. Predict the effectiveness of any proposed facilities and procedures and provide an implementation schedule with interim and final dates when the proposed facilities and procedures will be constructed, implemented, or operational.
- 1.4. **Proper Operation and Maintenance** – Supplement to Attachment D, Provision 1.4
  - 1.4.1. **Operation and Maintenance Manual.** The Discharger shall maintain an Operation and Maintenance Manual to provide the plant and regulatory personnel with a source of information describing all equipment, recommended operational strategies, process control monitoring, and maintenance activities. To remain a useful and relevant document, the Operation and Maintenance Manual shall be kept updated to reflect significant changes in treatment facility equipment and operational practices. The Operation and Maintenance Manual shall be maintained in usable condition and be available for reference and use by all relevant personnel and Regional Water Board staff.
  - 1.4.2. **Wastewater Facilities Status Report.** The Discharger shall maintain a Wastewater Facilities Status Report and regularly review, revise, or update it, as necessary. This report shall document how the Discharger operates and maintains its wastewater collection, treatment, and disposal facilities to ensure that all facilities are adequately staffed, supervised, financed, operated, maintained, repaired, and upgraded as necessary to provide adequate and reliable transport, treatment, and disposal of all wastewater from both existing and planned future wastewater sources under the Discharger's service responsibilities.
  - 1.4.3. **Proper Supervision and Operation of Publicly-Owned Treatment Works (POTWs).** POTWs shall be supervised and operated by persons possessing

certificates of appropriate grade pursuant to Title 23, section 3680, of the California Code of Regulations.

**1.5. Property Rights – Not Supplemented**

**1.6. Inspection and Entry – Not Supplemented**

**1.7. Bypass – Not Supplemented**

**1.8. Upset – Not Supplemented**

**1.9. Other – Addition to Attachment D**

- 1.9.1. Neither the treatment nor the discharge of pollutants shall create pollution, contamination, or nuisance as defined by California Water Code section 13050.
- 1.9.2. Collection, treatment, storage, and disposal systems shall be operated in a manner that precludes public contact with wastewater. If public contact with wastewater could reasonably occur on public property, warning signs shall be posted.
- 1.9.3. If the Discharger submits a timely and complete Report of Waste Discharge for permit reissuance, this permit shall continue in force and effect until the permit is reissued or the Regional Water Board rescinds the permit.

**2. STANDARD PROVISIONS – PERMIT ACTION – NOT SUPPLEMENTED**

**3. STANDARD PROVISIONS – MONITORING**

**3.1. Sampling and Analyses – Supplement to Attachment D, Provisions 3.1 and 3.2**

- 3.1.1. **Certified Laboratories.** Water and waste analyses shall be performed by a laboratory certified for these analyses in accordance with California Water Code section 13176.
- 3.1.2. **Minimum Levels.** For the 126 priority pollutants, the Discharger should use the analytical methods listed in Table B unless the Monitoring and Reporting Program (MRP, Attachment E) requires a particular method or minimum level (ML). All monitoring instruments and equipment shall be properly calibrated and maintained to ensure accuracy of measurements.
- 3.1.3. **Monitoring Frequency.** The MRP specifies the minimum sampling and analysis schedule.
  - 3.1.3.1. **Sample Collection Timing**
    - 3.1.3.1.1. The Discharger shall collect influent samples on varying days selected at random and shall not include any plant recirculation or other sidestream wastes, unless otherwise stipulated in the MRP. The Executive Officer



may approve an alternative influent sampling plan if it is representative of plant influent and complies with all other permit requirements.

- 3.1.3.1.2. The Discharger shall collect effluent samples on days coincident with influent sampling, unless otherwise stipulated by the MRP. If influent sampling is not required, the Discharger shall collect effluent samples on varying days selected at random, unless otherwise stipulated in the MRP. The Executive Officer may approve an alternative effluent sampling plan if it is representative of plant discharge and in compliance with all other permit requirements.
- 3.1.3.1.3. The Discharger shall collect effluent grab samples during periods of daytime maximum peak flows (or peak flows through secondary treatment units for facilities that recycle effluent).
- 3.1.3.1.4. Effluent sampling for conventional pollutants shall occur on at least one day of any multiple-day bioassay the MRP requires. During the course of the bioassay, on at least one day, the Discharger shall collect and retain samples of the discharge. In the event that a bioassay result does not comply with effluent limitations, the Discharger shall analyze the retained samples for pollutants that could be toxic to aquatic life and for which it has effluent limitations.
- 3.1.3.1.4.1. The Discharger shall perform bioassays on final effluent samples; when chlorine is used for disinfection, bioassays shall be performed on effluent after chlorination and dechlorination; and
- 3.1.3.1.4.2. The Discharger shall analyze for total ammonia nitrogen and calculate the amount of un-ionized ammonia whenever test results fail to meet effluent limitations.

### 3.1.3.2. **Conditions Triggering Accelerated Monitoring**

- 3.1.3.2.1. **Average Monthly Effluent Limitation Exceedance.** If the results from two consecutive samples of a constituent monitored in a particular month exceed the average monthly effluent limitation for any parameter (or if the required sampling frequency is once per month or less and the monthly sample exceeds the average monthly effluent limitation), the Discharger shall, within 24 hours after the results are received, increase its sampling frequency to daily until the results from the additional sampling show that the parameter complies with the average monthly effluent limitation.
- 3.1.3.2.2. **Maximum Daily Effluent Limitation Exceedance.** If a sample result exceeds a maximum daily effluent limitation, the Discharger shall, within 24 hours after the result is received, increase its sampling frequency to daily until the results from two samples collected on consecutive days show compliance with the maximum daily effluent limitation.

- 3.1.3.2.3. **Acute Toxicity.** If final or intermediate results of an acute bioassay indicate a violation or threatened violation (e.g., the percentage of surviving test organisms of any single acute bioassay is less than 70 percent), the Discharger shall initiate a new test as soon as practical or as described in applicable State Water Board plan provisions that become effective after adoption of these Regional Standard Provisions. The Discharger shall investigate the cause of the mortalities and report its findings in the next self-monitoring report.
- 3.1.3.2.4. **Chlorine.** The Discharger shall calibrate chlorine residual analyzers against grab samples as frequently as necessary to maintain accurate control and reliable operation. If an effluent violation is detected, the Discharger shall collect grab samples at least every 30 minutes until compliance with the limitation is achieved, unless the Discharger monitors chlorine residual continuously. In such cases, the Discharger shall continue to conduct continuous monitoring.
- 3.1.3.2.5. **Bypass.** Except as indicated below, if a Discharger bypasses any portion of its treatment facility, it shall monitor flows and collect samples at affected discharge points and analyze samples for all constituents with effluent limitations on a daily basis for the duration of the bypass. The Discharger need not accelerate chronic toxicity monitoring. The Discharger also need not collect and analyze samples for mercury, dioxin-TEQ, and PCBs after the first day of the bypass. The Discharger may satisfy the accelerated acute toxicity monitoring requirement by conducting a flow-through test or static renewal test that captures the duration of the bypass (regardless of the method specified in the MRP). If bypassing disinfection units only, the Discharger shall only monitor bacteria indicators daily.
- 3.1.3.2.5.1. **Bypass for Essential Maintenance.** If a Discharger bypasses a treatment unit for essential maintenance pursuant to Attachment D section 1.7.2, the Executive Officer may reduce the accelerated monitoring requirements above if the Discharger (i) monitors effluent at affected discharge points on the first day of the bypass for all constituents with effluent limitations, except chronic toxicity; and (ii) identifies and implements measures to ensure that the bypass will continue to comply with effluent limitations.
- 3.1.3.2.5.2. **Approved Wet Weather Bypasses.** If a Discharger bypasses a treatment unit or permitted outfall during wet weather with Executive Officer approval pursuant to Attachment D section 1.7.4, the Discharger shall monitor flows and collect and retain samples for affected discharge points on a daily basis for the duration of the bypass. The Discharger shall analyze daily for TSS using 24 hour composites (or more frequent increments) and for bacteria indicators with effluent limitations using grab samples. If TSS exceeds 45 mg/L in any

composite sample, the Discharger shall also analyze daily the retained samples for all other constituents with effluent limitations, except oil and grease, mercury, PCBs, dioxin-TEQ, and acute and chronic toxicity. Additionally, at least once each year, the Discharger shall analyze the retained samples for one approved bypass for all other constituents with effluent limitations, except oil and grease, mercury, PCBs, dioxin-TEQ, and acute and chronic toxicity. This monitoring shall be in addition to the minimum monitoring specified in the MRP.

### **3.2. Standard Observations – Addition to Attachment D**

**3.2.1. Receiving Water Observations.** The following requirements only apply when the MRP requires standard observations of receiving waters. Standard observations shall include the following:

- 3.2.1.1. Floating and Suspended Materials (e.g., oil, grease, algae, and other macroscopic particulate matter) —** presence or absence, source, and size of affected area.
- 3.2.1.2. Discoloration and Turbidity —** color, source, and size of affected area.
- 3.2.1.3. Odor —** presence or absence, characterization, source, and distance of travel.
- 3.2.1.4. Beneficial Water Use —** estimated number of water-associated waterfowl or wildlife, fisherpeople, and other recreational activities.
- 3.2.1.5. Hydrographic Condition —** time and height of high and low tides (corrected to nearest National Oceanic and Atmospheric Administration location for the sampling date and time).
- 3.2.1.6. Weather Conditions —** wind direction, air temperature, and total precipitation during five days prior to observation.

**3.2.2. Wastewater Effluent Observations.** The following requirements only apply when the MRP requires standard observations of wastewater effluent. Standard observations shall include the following:

- 3.2.2.1. Floating and Suspended Material of Wastewater Origin (e.g., oil, grease, algae, and other macroscopic particulate matter) —** presence or absence.
- 3.2.2.2. Odor —** presence or absence, characterization, source, distance of travel, and wind direction.
- 3.2.3. Beach and Shoreline Observations.** The following requirements only apply when the MRP requires standard observations of beaches or shorelines. Standard observations shall include the following:

- 3.2.3.1. **Material of Wastewater Origin** — presence or absence, description of material, estimated size of affected area, and source.
- 3.2.3.2. **Beneficial Use** — estimate of number of people participating in recreational water contact, non-water contact, and fishing activities.
- 3.2.4. **Waste Treatment and/or Disposal Facility Periphery Observations.**  
The following requirements only apply when the MRP requires standard observations of the periphery of waste treatment or disposal facilities. Standard observations shall include the following:
  - 3.2.4.1. **Odor** — presence or absence, characterization, source, and distance of travel.
  - 3.2.4.2. **Weather Conditions** — wind direction and estimated velocity.

#### 4. STANDARD PROVISIONS – RECORDS

##### 4.1. Records to be Maintained – Supplement to Attachment D, Provision 4.1

The Discharger shall maintain records in a manner and at a location (e.g., the wastewater treatment plant or the Discharger's offices) such that the records are accessible to Regional Water Board staff. The minimum retention period specified in Attachment D, Provision IV, shall be extended during the course of any unresolved litigation regarding permit-related discharges, or when requested by Regional Water Board or U.S. EPA, Region IX, staff.

A copy of the permit shall be maintained at the discharge facility and be available at all times to operating personnel.

##### 4.2. Records of Monitoring – Supplement to Attachment D, Provision 4.2

Monitoring records shall include the following:

- 4.2.1. **Analytical Information.** Records shall include analytical method detection limits, minimum levels, reporting levels, and related quantification parameters.
- 4.2.2. **Disinfection Process.** For the disinfection process, records shall include the following:
  - 4.2.2.1. For bacteriological analyses:
    - 4.2.2.1.1. Wastewater flow rate at the time of sample collection; and
    - 4.2.2.1.2. Required statistical parameters for cumulative bacterial values (e.g., moving median or geometric mean for the number of samples or sampling period identified in the MRP).

- 4.2.2.2. For the chlorination process (when chlorine is used for disinfection), at least daily average values for the following:
  - 4.2.2.2.1. Chlorine residual of treated wastewater as it enters the chlorine contact basin (mg/L);
  - 4.2.2.2.2. Chlorine dosage (kg/day); and
  - 4.2.2.2.3. Dechlorination chemical dosage (kg/day).
- 4.2.3. **Wastewater Treatment Process Solids.** For each treatment unit process that involves solids removal from the wastewater stream, records shall include the following:
  - 4.2.3.1. Total volume or mass of solids removed from each collection unit (e.g., grit, skimmings, undigested biosolids, or combination) for each calendar month or other time period as appropriate, but not to exceed annually; and
  - 4.2.3.2. Final disposition of such solids (e.g., landfill, other subsequent treatment unit).
- 4.2.4. **Treatment Process Bypasses.** For all treatment process bypasses, including wet weather blending, records shall include the following:
  - 4.2.4.1. Chronological log of treatment process bypasses;
  - 4.2.4.2. Identification of treatment processes bypassed;
  - 4.2.4.3. Beginning and ending dates and times of bypasses;
  - 4.2.4.4. Bypass durations;
  - 4.2.4.5. Estimated bypass volumes; and
  - 4.2.4.6. Description of, or reference to other reports describing, the bypasses, their cause, the corrective actions taken (except for wet weather blending explicitly approved within the permit and in compliance with any related permit conditions), and any additional monitoring conducted.
- 4.2.5. **Treatment Plant Overflows.** The Discharger shall retain a chronological log of overflows at the treatment plant, including the headworks and all units and appurtenances downstream, and records supporting the information provided in accordance with Provision 5.5.2, below.

#### **4.3. Claims of Confidentiality – Not Supplemented**

### **5. STANDARD PROVISIONS – REPORTING**

#### **5.1. Duty to Provide Information – Not Supplemented**

#### **5.2. Signatory and Certification Requirements – Not Supplemented**

#### **5.3. Monitoring Reports – Supplement to Attachment D, Provision 5.3**

**5.3.1. Self-Monitoring Reports.** For each reporting period established in the MRP, the Discharger shall submit a self-monitoring report to the Regional Water Board in accordance with the requirements listed in the MRP and below:

**5.3.1.1. Transmittal Letter.** Each self-monitoring report shall be submitted with a transmittal letter that includes the following:

- 5.3.1.1.1.** Identification of all violations of effluent limitations or other waste discharge requirements found during the reporting period;
- 5.3.1.1.2.** Details regarding the violations, such as parameters, magnitude, test results, frequency, and dates;
- 5.3.1.1.3.** Causes of the violations;
- 5.3.1.1.4.** Corrective actions taken or planned to resolve violations and prevent recurrences, and dates or time schedules for implementation (the Discharger may refer to previously submitted reports that address the corrective actions);
- 5.3.1.1.5.** Explanation for any data invalidation. Data should not be submitted in a self-monitoring report if it does not meet quality assurance/quality control standards. However, if the Discharger wishes to invalidate a measurement after submitting it in a self-monitoring report, the Discharger shall identify the measurement suspected to be invalid and state the Discharger's intent to submit, within 60 days, a formal request to invalidate the measurement. The formal request shall include the original measurement in question, the reason for invalidating the measurement, all relevant documentation that supports invalidation (e.g., laboratory sheet, log entry, test results), and a discussion of the corrective actions taken or planned (with a time schedule for completion) to prevent recurrence of the sampling or measurement problem;
- 5.3.1.1.6.** Description of blending, if any. If the Discharger blends, it shall describe the duration of blending events and certify whether the blending complied with all conditions for blending;

- 5.3.1.1.7. Description of other bypasses, if any. If the Discharger bypasses any treatment units (other than blending), it shall describe the duration of the bypasses and effluent quality during those times; and
- 5.3.1.1.8. Signature. The transmittal letter shall be signed in accordance with Attachment D, Provision 5.2.
- 5.3.1.2. **Compliance Evaluation Summary.** Each self-monitoring report shall include a compliance evaluation summary that addresses each parameter for which the permit specifies effluent limitations, the number of samples taken during the monitoring period, and the number of samples that exceed the effluent limitations.
- 5.3.1.3. **More Frequent Monitoring.** If the Discharger monitors any pollutant more frequently than required by the MRP, the Discharger shall include the results of such monitoring in the calculation and reporting of the data submitted in the self-monitoring report.
- 5.3.1.4. **Analysis Results**
  - 5.3.1.4.1. **Tabulation.** Each self-monitoring report shall include tabulations of all required analyses and observations, including parameters, dates, times, sample stations, types of samples, test results, method detection limits, method minimum levels, and method reporting levels (if applicable), signed by the laboratory director or other responsible official.
  - 5.3.1.4.2. **Multiple Samples.** Unless the MRP specifies otherwise, when determining compliance with effluent limitations (other than instantaneous effluent limitations) and more than one sample result is available, the Discharger shall compute the arithmetic mean. If the data set contains one or more results that are “Detected, but Not Quantified (DNQ) or “Not Detected” (ND), the Discharger shall instead compute the median in accordance with the following procedure:
    - 5.3.1.4.2.1. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
    - 5.3.1.4.2.2. The median of the data set shall be determined. If the data set has an odd number of data points, the median is the middle value. If the data set has an even number of data points, the median is the average of the two values around the middle, unless one or both of these values is ND or DNQ, in which case the median shall be the lower of the two results (where DNQ is lower than a quantified value and ND is lower than DNQ).

**5.3.1.4.3. Duplicate Samples.** The Discharger shall report the average of duplicate sample analyses when reporting for a single sample result (or the median if one or more of the duplicates is DNQ or ND [see Provision 5.3.1.4.2, above]). For bacteria indicators, the Discharger shall report the geometric mean of the duplicate analyses.

**5.3.1.4.4. Dioxin-TEQ.** The Discharger shall report for each dioxin and furan congener the analytical results of effluent monitoring, including the reporting level, the method detection limit, and the measured concentration. The Discharger shall report all measured values of individual congeners, including data qualifiers. When calculating dioxin-TEQ, the Discharger shall set congener concentrations below the minimum levels (MLs) to zero. The Discharger shall calculate and report dioxin-TEQ using the following formula, where the MLs, toxicity equivalency factors (TEFs), and bioaccumulation equivalency factors (BEFs) are as provided in Table A:

$$\text{Dioxin-TEQ} = \sum (C_x \times \text{TEF}_x \times \text{BEF}_x)$$

where:  $C_x$  = measured or estimated concentration of congener x  
 $\text{TEF}_x$  = toxicity equivalency factor for congener x  
 $\text{BEF}_x$  = bioaccumulation equivalency factor for congener x

**Table A**  
**Minimum Levels, Toxicity Equivalency Factors,**  
**and Bioaccumulation Equivalency Factors**

| Dioxin or Furan Congener | Minimum Level (pg/L) | 2005 Toxicity Equivalency Factor (TEF) | Bioaccumulation Equivalency Factor (BEF) |
|--------------------------|----------------------|----------------------------------------|------------------------------------------|
| 2,3,7,8-TCDD             | 10                   | 1.0                                    | 1.0                                      |
| 1,2,3,7,8-PeCDD          | 50                   | 1.0                                    | 0.9                                      |
| 1,2,3,4,7,8-HxCDD        | 50                   | 0.1                                    | 0.3                                      |
| 1,2,3,6,7,8-HxCDD        | 50                   | 0.1                                    | 0.1                                      |
| 1,2,3,7,8,9-HxCDD        | 50                   | 0.1                                    | 0.1                                      |
| 1,2,3,4,6,7,8-HpCDD      | 50                   | 0.01                                   | 0.05                                     |
| OCDD                     | 100                  | 0.0003                                 | 0.01                                     |
| 2,3,7,8-TCDF             | 10                   | 0.1                                    | 0.8                                      |
| 1,2,3,7,8-PeCDF          | 50                   | 0.03                                   | 0.2                                      |
| 2,3,4,7,8-PeCDF          | 50                   | 0.3                                    | 1.6                                      |
| 1,2,3,4,7,8-HxCDF        | 50                   | 0.1                                    | 0.08                                     |
| 1,2,3,6,7,8-HxCDF        | 50                   | 0.1                                    | 0.2                                      |
| 1,2,3,7,8,9-HxCDF        | 50                   | 0.1                                    | 0.6                                      |
| 2,3,4,6,7,8-HxCDF        | 50                   | 0.1                                    | 0.7                                      |
| 1,2,3,4,6,7,8-HpCDF      | 50                   | 0.01                                   | 0.01                                     |
| 1,2,3,4,7,8,9-HpCDF      | 50                   | 0.01                                   | 0.4                                      |
| OCDF                     | 100                  | 0.0003                                 | 0.02                                     |



- 5.3.1.5. **Results Not Yet Available.** The Discharger shall make all reasonable efforts to obtain analytical data for required parameter sampling in a timely manner. Certain analyses may require additional time to complete analytical processes and report results. In these cases, the Discharger shall describe the circumstances in the self-monitoring report and include the data for these parameters and relevant discussions of any violations in the next self-monitoring report due after the results are available.
- 5.3.1.6. **Annual Self-Monitoring Reports.** By the date specified in the MRP, the Discharger shall submit an annual self-monitoring report covering the previous calendar year. The report shall contain the following:
- 5.3.1.6.1. Comprehensive discussion of treatment plant performance, including documentation of any blending or other bypass events, and compliance with the permit. This discussion shall include any corrective actions taken or planned, such as changes to facility equipment or operation practices that may be needed to achieve compliance, and any other actions taken or planned that are intended to improve the performance and reliability of wastewater collection, treatment, or disposal practices;
  - 5.3.1.6.2. List of approved analyses, including the following:
    - 5.3.1.6.2.1. List of analyses for which the Discharger is certified;
    - 5.3.1.6.2.2. List of analyses performed for the Discharger by a separate certified laboratory (copies of reports signed by the laboratory director of that laboratory need not be submitted but shall be retained onsite); and
    - 5.3.1.6.2.3. List of “waived” analyses, as approved;
  - 5.3.1.6.3. Plan view drawing or map showing the Discharger’s facility, flow routing, and sampling and observation station locations; and
  - 5.3.1.6.4. Results of facility report reviews. The Discharger shall regularly review, revise, and update, as necessary, the Operation and Maintenance Manual, Contingency Plan, Spill Prevention Plan, and Wastewater Facilities Status Report so these documents remain useful and relevant to current practices. At a minimum, reviews shall be conducted annually. The Discharger shall describe or summarize its review and evaluation procedures, recommended or planned actions, and estimated time schedule for implementing these actions. The Discharger shall complete changes to these documents to ensure that they remain up-to-date.

**5.4. Compliance Schedules – Not supplemented**

## **5.5. Twenty-Four Hour Reporting – Supplement to Attachment D, Provision 5.5**

### **5.5.1. Oil or Other Hazardous Material Spills**

- 5.5.1.1. Within 24 hours of becoming aware of a spill of oil or other hazardous material not contained onsite and completely cleaned up, the Discharger shall report as follows:
  - 5.5.1.1.1. If the spill exceeds reportable quantities for hazardous materials listed in 40 C.F.R. part 302. The Discharger shall call the California Office of Emergency Services (800 852-7550).
  - 5.5.1.1.2. If the spill does not exceed reportable quantities for hazardous materials listed in 40 C.F.R., part 302, the Discharger shall call the Regional Water Board (510-622-2369).
- 5.5.1.2. The Discharger shall submit a written report to the Regional Water Board within five working days following either of the above telephone notifications unless directed otherwise by Regional Water Board staff. A report submitted electronically is acceptable. The written report shall include the following:
  - 5.5.1.2.1. Date and time of spill, and duration if known;
  - 5.5.1.2.2. Location of spill (street address or description of location);
  - 5.5.1.2.3. Nature of material spilled;
  - 5.5.1.2.4. Quantity of material spilled;
  - 5.5.1.2.5. Receiving water body affected, if any;
  - 5.5.1.2.6. Cause of spill;
  - 5.5.1.2.7. Estimated size of affected area;
  - 5.5.1.2.8. Observed impacts to receiving waters (e.g., oil sheen, fish kill, water discoloration);
  - 5.5.1.2.9. Corrective actions taken to contain, minimize, or clean up the spill;
  - 5.5.1.2.10. Future corrective actions planned to prevent recurrence, and implementation schedule; and
  - 5.5.1.2.11. Persons or agencies notified.

### 5.5.2. **Unauthorized Municipal Wastewater Treatment Plant Discharges**<sup>1</sup>

5.5.2.1. **Two-Hour Notification.** For any unauthorized discharge that enters a drainage channel or surface water, the Discharger shall, as soon as possible, but not later than two hours after becoming aware of the discharge, notify the California Office of Emergency Services (800-852-7550) and the local health officer or director of environmental health with jurisdiction over the affected water body. Notification shall include the following:

- 5.5.2.1.1. Incident description and cause;
- 5.5.2.1.2. Location of threatened or involved waterways or storm drains;
- 5.5.2.1.3. Date and time that the unauthorized discharge started;
- 5.5.2.1.4. Estimated quantity and duration of the unauthorized discharge (to the extent known), and estimated amount recovered;
- 5.5.2.1.5. Level of treatment prior to discharge (e.g., raw wastewater, primary-treated wastewater, or undisinfected secondary-treated wastewater); and
- 5.5.2.1.6. Identity of person reporting the unauthorized discharge.

5.5.2.2. **Five-Day Written Report.** Within five business days following the two-hour notification, the Discharger shall submit a written report that includes, in addition to the information listed in Provision 5.5.2.1, above, the following:

- 5.5.2.2.1. Methods used to delineate the geographical extent of the unauthorized discharge within receiving waters;
- 5.5.2.2.2. Efforts implemented to minimize public exposure to the unauthorized discharge;
- 5.5.2.2.3. Visual observations of the impacts (if any) noted in the receiving waters (e.g., fish kill, discoloration of receiving water) and extent of sampling if conducted;
- 5.5.2.2.4. Corrective measures taken to minimize the impact of the unauthorized discharge;

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<sup>1</sup> California Code of Regulations, Title 23, section 2250(b), defines an unauthorized discharge to be a discharge, not regulated by waste discharge requirements, of treated, partially-treated, or untreated wastewater resulting from the intentional or unintentional diversion of wastewater from a collection, treatment, or disposal system.

- 5.5.2.2.5. Measures to be taken to minimize the potential for a similar unauthorized discharge in the future;
- 5.5.2.2.6. Summary of Spill Prevention Plan or Operation and Maintenance Manual modifications to be made, if necessary, to minimize the potential for future unauthorized discharges; and
- 5.5.2.2.7. Quantity and duration of the unauthorized discharge, and the amount recovered.

**5.6. Planned Changes** – Not supplemented

**5.7. Anticipated Noncompliance** – Not supplemented

**5.8. Other Noncompliance** – Not supplemented

**5.9. Other Information** – Not supplemented

## **6. STANDARD PROVISIONS – ENFORCEMENT – NOT SUPPLEMENTED**

## **7. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS – NOT SUPPLEMENTED**

## **8. DEFINITIONS – ADDITION TO ATTACHMENT D**

More definitions can be found in Attachment A of this NPDES Permit.

### **8.1. Arithmetic Calculations**

- 8.1.1. **Geometric Mean.** The antilog of the log mean or the back-transformed mean of the logarithmically transformed variables, which is equivalent to the multiplication of the antilogarithms. The geometric mean can be calculated with either of the following equations:

$$\text{Geometric Mean} = \text{Anti log} \left( \frac{1}{N} \sum \text{Log } C_i \right)$$

or

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_N)^{1/N}$$

Where “N” is the number of data points for the period analyzed and “C” is the concentration for each of the “N” data points.

- 8.1.2. **Mass Emission Rate.** The rate of discharge expressed in mass. The mass emission rate is obtained from the following calculation for any calendar day:

$$\text{Mass emission rate (lb/day)} = \frac{8.345}{N} \sum_{i=1}^N Q_i C_i$$

$$\text{Mass emission rate (kg/day)} = \frac{3.785}{N} \sum_{i=1}^N Q_i C_i$$

In which “N” is the number of samples analyzed in any calendar day and “Q” and “C<sub>i</sub>” are the flow rate (MGD) and the constituent concentration (mg/L) associated with each of the “N” grab samples that may be taken in any calendar day. If a composite sample is taken, “C<sub>i</sub>” is the concentration measured in the composite sample and “Q<sub>i</sub>” is the average flow rate occurring during the period over which the samples are composited. The daily concentration of a constituent measured over any calendar day shall be determined from the flow weighted average of the same constituent in the combined waste streams as follows:

$$C_d = \text{Average daily concentration} = \frac{1}{Q_t} \sum_{i=1}^N Q_i C_i$$

In which “N” is the number of component waste streams and “Q” and “C” are the flow rate (MGD) and the constituent concentration (mg/L) associated with each of the “N” waste streams. “Q<sub>t</sub>” is the total flow rate of the combined waste streams.

- 8.1.3. **Removal Efficiency.** The ratio of pollutants removed by the treatment facilities to pollutants entering the treatment facilities (expressed as a percentage). The Discharger shall determine removal efficiencies using monthly averages (by calendar month unless otherwise specified) of pollutant concentration of influent and effluent samples collected at about the same time and using the following equation (or its equivalent):

$$\text{Removal Efficiency (\%)} = 100 \times [1 - (\text{Effluent Concentration} / \text{Influent Concentration})]$$

- 8.2. **Blending** – the practice of bypassing biological treatment units and recombining the bypass wastewater with biologically-treated wastewater.
- 8.3. **Composite Sample** – a sample composed of individual grab samples collected manually or by an automatic sampling device on the basis of time or flow as specified in the MRP. For flow-based composites, the proportion of each grab sample included in the composite sample shall be within plus or minus five percent (+/-5%) of the representative flow of the waste stream being measured at the time of grab sample collection. Alternatively, equal volume grab samples may be individually analyzed with the flow-weighted average calculated by averaging flow-weighted ratios of each grab sample analytical result. Grab samples comprising time-based composite samples shall be collected at intervals not greater than those specified in the MRP. The quantity of each grab sample comprising a time-based composite sample shall be a set of flow proportional volumes as specified in

the MRP. If a particular time-based or flow-based composite sampling protocol is not specified in the MRP, the Discharger shall determine and implement the most representative protocol.

- 8.4. Duplicate Sample** – a second sample taken from the same source and at the same time as an initial sample (such samples are typically analyzed identically to measure analytical variability).
- 8.5. Grab Sample** – an individual sample collected during a short period not exceeding 15 minutes. Grab samples represent only the condition that exists at the time the sample is collected.
- 8.6. Overflow** – the intentional or unintentional spilling or forcing out of untreated or partially-treated waste from a transport system (e.g., through manholes, at pump stations, or at collection points) upstream of the treatment plant headworks or from any part of a treatment plant.
- 8.7. Priority Pollutants** – those constituents referred to in 40 C.F.R. part 122 as promulgated in the Federal Register, Vol. 65, No. 97, Thursday, May 18, 2000, also known as the California Toxics Rule.
- 8.8. Untreated waste** – raw wastewater.

**Table B**  
**List of Monitoring Parameters, Analytical Methods, and Minimum Levels (µg/L)<sup>[1]</sup>**

| CTR No. | Pollutant / Parameter                                                 | Analytical Method <sup>[2]</sup> | GC  | GC MS | LC | Color | FAA | GFAA | ICP | ICP MS | SPGF AA | HYD RIDE | CVAA | DCP    |
|---------|-----------------------------------------------------------------------|----------------------------------|-----|-------|----|-------|-----|------|-----|--------|---------|----------|------|--------|
| 1       | Antimony                                                              | 204.2                            | -   | -     | -  | -     | 10  | 5    | 50  | 0.5    | 5       | 0.5      | -    | 1000   |
| 2       | Arsenic                                                               | 206.3                            | -   | -     | -  | 20    | -   | 2    | 10  | 2      | 2       | 1        | -    | 1000   |
| 3       | Beryllium                                                             | -                                | -   | -     | -  | -     | 20  | 0.5  | 2   | 0.5    | 1       | -        | -    | 1000   |
| 4       | Cadmium                                                               | 200 or 213                       | -   | -     | -  | -     | 10  | 0.5  | 10  | 0.25   | 0.5     | -        | -    | 1000   |
| 5a      | Chromium (III)                                                        | SM 3500                          | -   | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 5b      | Chromium (VI)                                                         | SM 3500                          | -   | -     | -  | 10    | 5   | -    | -   | -      | -       | -        | -    | 1000   |
|         | Chromium (total) <sup>[3]</sup>                                       | SM 3500                          | -   | -     | -  | -     | 50  | 2    | 10  | 0.5    | 1       | -        | -    | 1000   |
| 6       | Copper                                                                | 200.9                            | -   | -     | -  | -     | 25  | 5    | 10  | 0.5    | 2       | -        | -    | 1000   |
| 7       | Lead                                                                  | 200.9                            | -   | -     | -  | -     | 20  | 5    | 5   | 0.5    | 2       | -        | -    | 10,000 |
| 8       | Mercury                                                               | 1631 <sup>[4]</sup>              | -   | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 9       | Nickel                                                                | 249.2                            | -   | -     | -  | -     | 50  | 5    | 20  | 1      | 5       | -        | -    | 1000   |
| 10      | Selenium                                                              | 200.8 or SM 3114B or C           | -   | -     | -  | -     | -   | 5    | 10  | 2      | 5       | 1        | -    | 1000   |
| 11      | Silver                                                                | 272.2                            | -   | -     | -  | -     | 10  | 1    | 10  | 0.25   | 2       | -        | -    | 1000   |
| 12      | Thallium                                                              | 279.2                            | -   | -     | -  | -     | 10  | 2    | 10  | 1      | 5       | -        | -    | 1000   |
| 13      | Zinc                                                                  | 200 or 289                       | -   | -     | -  | -     | 20  | -    | 20  | 1      | 10      | -        | -    | -      |
| 14      | Cyanide                                                               | SM 4500 CN <sup>-</sup> C or I   | -   | -     | -  | 5     | -   | -    | -   | -      | -       | -        | -    | -      |
| 15      | Asbestos (only required for dischargers to MUN waters) <sup>[5]</sup> | 0100.2 <sup>[6]</sup>            | -   | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 16      | 2,3,7,8-TCDD and 17 congeners (Dioxin)                                | 1613                             | -   | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 17      | Acrolein                                                              | 603                              | 2.0 | 5     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 18      | Acrylonitrile                                                         | 603                              | 2.0 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 19      | Benzene                                                               | 602                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 33      | Ethylbenzene                                                          | 602                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 39      | Toluene                                                               | 602                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 20      | Bromoform                                                             | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |
| 21      | Carbon Tetrachloride                                                  | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -      |

| CTR No. | Pollutant / Parameter                        | Analytical Method <sup>[2]</sup> | GC  | GC MS | LC | Color | FAA | GFAA | ICP | ICP MS | SPGF AA | HYD RIDE | CVAA | DCP |
|---------|----------------------------------------------|----------------------------------|-----|-------|----|-------|-----|------|-----|--------|---------|----------|------|-----|
| 22      | Chlorobenzene                                | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 23      | Chlorodibromomethane                         | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 24      | Chloroethane                                 | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 25      | 2-Chloroethylvinyl Ether                     | 601                              | 1   | 1     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 26      | Chloroform                                   | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 75      | 1,2-Dichlorobenzene                          | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 76      | 1,3-Dichlorobenzene                          | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 77      | 1,4-Dichlorobenzene                          | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 27      | Dichlorobromomethane                         | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 28      | 1,1-Dichloroethane                           | 601                              | 0.5 | 1     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 29      | 1,2-Dichloroethane                           | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 30      | 1,1-Dichloroethylene or 1,1-Dichloroethene   | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 31      | 1,2-Dichloropropane                          | 601                              | 0.5 | 1     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 32      | 1,3-Dichloropropylene or 1,3-Dichloropropene | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 34      | Methyl Bromide or Bromomethane               | 601                              | 1.0 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 35      | Methyl Chloride or Chloromethane             | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 36      | Methylene Chloride or Dichloromethane        | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 37      | 1,1,2,2-Tetrachloroethane                    | 601                              | 0.5 | 1     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 38      | Tetrachloroethylene                          | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 40      | 1,2-Trans-Dichloroethylene                   | 601                              | 0.5 | 1     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 41      | 1,1,1-Trichloroethane                        | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 42      | 1,1,2-Trichloroethane                        | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 43      | Trichloroethene                              | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 44      | Vinyl Chloride                               | 601                              | 0.5 | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 45      | 2-Chlorophenol                               | 604                              | 2   | 5     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 46      | 2,4-Dichlorophenol                           | 604                              | 1   | 5     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 47      | 2,4-Dimethylphenol                           | 604                              | 1   | 2     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |



| CTR No. | Pollutant / Parameter                                | Analytical Method <sup>[2]</sup> | GC | GC MS | LC   | Color | FAA | GFAA | ICP | ICP MS | SPGF AA | HYD RIDE | CVAA | DCP |
|---------|------------------------------------------------------|----------------------------------|----|-------|------|-------|-----|------|-----|--------|---------|----------|------|-----|
| 48      | 2-Methyl-4,6-Dinitrophenol or Dinitro-2-methylphenol | 604                              | 10 | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 49      | 2,4-Dinitrophenol                                    | 604                              | 5  | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 50      | 2-Nitrophenol                                        | 604                              | -  | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 51      | 4-Nitrophenol                                        | 604                              | 5  | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 52      | 3-Methyl-4-Chlorophenol                              | 604                              | 5  | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 53      | Pentachlorophenol                                    | 604                              | 1  | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 54      | Phenol                                               | 604                              | 1  | 1     | -    | 50    | -   | -    | -   | -      | -       | -        | -    | -   |
| 55      | 2,4,6-Trichlorophenol                                | 604                              | 10 | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 56      | Acenaphthene                                         | 610 HPLC                         | 1  | 1     | 0.5  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 57      | Acenaphthylene                                       | 610 HPLC                         | -  | 10    | 0.2  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 58      | Anthracene                                           | 610 HPLC                         | -  | 10    | 2    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 60      | Benzo(a)Anthracene or 1,2 Benzanthracene             | 610 HPLC                         | 10 | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 61      | Benzo(a)Pyrene                                       | 610 HPLC                         | -  | 10    | 2    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 62      | Benzo(b) Fluoranthene or 3,4 Benzofluoranthene       | 610 HPLC                         | -  | 10    | 10   | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 63      | Benzo(ghi)Perylene                                   | 610 HPLC                         | -  | 5     | 0.1  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 64      | Benzo(k)Fluoranthene                                 | 610 HPLC                         | -  | 10    | 2    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 74      | Dibenzo(a,h)Anthracene                               | 610 HPLC                         | -  | 10    | 0.1  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 86      | Fluoranthene                                         | 610 HPLC                         | 10 | 1     | 0.05 | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 87      | Fluorene                                             | 610 HPLC                         | -  | 10    | 0.1  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 92      | Indeno(1,2,3-cd)Pyrene                               | 610 HPLC                         | -  | 10    | 0.05 | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 100     | Pyrene                                               | 610 HPLC                         | -  | 10    | 0.05 | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 68      | Bis(2-Ethylhexyl)Phthalate                           | 606 or 625                       | 10 | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 70      | Butylbenzyl Phthalate                                | 606 or 625                       | 10 | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 79      | Diethyl Phthalate                                    | 606 or 625                       | 10 | 2     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 80      | Dimethyl Phthalate                                   | 606 or 625                       | 10 | 2     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 81      | Di-n-Butyl Phthalate                                 | 606 or 625                       | -  | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 84      | Di-n-Octyl Phthalate                                 | 606 or 625                       | -  | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 59      | Benzidine                                            | 625                              | -  | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 65      | Bis(2-Chloroethoxy)Methane                           | 625                              | -  | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |

| CTR No. | Pollutant / Parameter                | Analytical Method <sup>[2]</sup> | GC    | GC MS | LC   | Color | FAA | GFAA | ICP | ICP MS | SPGF AA | HYD RIDE | CVAA | DCP |
|---------|--------------------------------------|----------------------------------|-------|-------|------|-------|-----|------|-----|--------|---------|----------|------|-----|
| 66      | Bis(2-Chloroethyl)Ether              | 625                              | 10    | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 67      | Bis(2-Chloroisopropyl) Ether         | 625                              | 10    | 2     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 69      | 4-Bromophenyl Phenyl Ether           | 625                              | 10    | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 71      | 2-Chloronaphthalene                  | 625                              | -     | 10    | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 72      | 4-Chlorophenyl Phenyl Ether          | 625                              | -     | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 73      | Chrysene                             | 625                              | -     | 10    | 5    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 78      | 3,3'-Dichlorobenzidine               | 625                              | -     | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 82      | 2,4-Dinitrotoluene                   | 625                              | 10    | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 83      | 2,6-Dinitrotoluene                   | 625                              | -     | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 85      | 1,2-Diphenylhydrazine <sup>[7]</sup> | 625                              | -     | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 88      | Hexachlorobenzene                    | 625                              | 5     | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 89      | Hexachlorobutadiene                  | 625                              | 5     | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 90      | Hexachlorocyclopentadiene            | 625                              | 5     | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 91      | Hexachloroethane                     | 625                              | 5     | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 93      | Isophorone                           | 625                              | 10    | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 94      | Naphthalene                          | 625                              | 10    | 1     | 0.2  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 95      | Nitrobenzene                         | 625                              | 10    | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 96      | N-Nitrosodimethylamine               | 625                              | 10    | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 97      | N-Nitrosodi-n-Propylamine            | 625                              | 10    | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 98      | N-Nitrosodiphenylamine               | 625                              | 10    | 1     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 99      | Phenanthrene                         | 625                              | -     | 5     | 0.05 | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 101     | 1,2,4-Trichlorobenzene               | 625                              | 1     | 5     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 102     | Aldrin                               | 608                              | 0.005 | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 103     | α-BHC                                | 608                              | 0.01  | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 104     | β-BHC                                | 608                              | 0.005 | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 105     | γ-BHC (Lindane)                      | 608                              | 0.02  | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 106     | δ-BHC                                | 608                              | 0.005 | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 107     | Chlordane                            | 608                              | 0.1   | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 108     | 4,4'-DDT                             | 608                              | 0.01  | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 109     | 4,4'-DDE                             | 608                              | 0.05  | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 110     | 4,4'-DDD                             | 608                              | 0.05  | -     | -    | -     | -   | -    | -   | -      | -       | -        | -    | -   |

| CTR No. | Pollutant / Parameter                                   | Analytical Method <sup>[2]</sup> | GC   | GC MS | LC | Color | FAA | GFAA | ICP | ICP MS | SPGF AA | HYD RIDE | CVAA | DCP |
|---------|---------------------------------------------------------|----------------------------------|------|-------|----|-------|-----|------|-----|--------|---------|----------|------|-----|
| 111     | Dieldrin                                                | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 112     | Endosulfan (alpha)                                      | 608                              | 0.02 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 113     | Endosulfan (beta)                                       | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 114     | Endosulfan Sulfate                                      | 608                              | 0.05 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 115     | Endrin                                                  | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 116     | Endrin Aldehyde                                         | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 117     | Heptachlor                                              | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 118     | Heptachlor Epoxide                                      | 608                              | 0.01 | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 119-125 | PCBs: Aroclors 1016, 1221, 1232, 1242, 1248, 1254, 1260 | 608                              | 0.5  | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |
| 126     | Toxaphene                                               | 608                              | 0.5  | -     | -  | -     | -   | -    | -   | -      | -       | -        | -    | -   |

Footnotes:

- <sup>[1]</sup> Minimum levels are from the *State Implementation Policy*. They are the concentration of the lowest calibration standard for that technique based on a survey of contract laboratories. Laboratory techniques are defined as follows: GC = Gas Chromatography; GCMS = Gas Chromatography/Mass Spectrometry; LC = High Pressure Liquid Chromatography; Color = Colorimetric; FAA = Flame Atomic Absorption; GFAA = Graphite Furnace Atomic Absorption; ICP = Inductively Coupled Plasma; ICPMS = Inductively Coupled Plasma/Mass Spectrometry; SPGFAA = Stabilized Platform Graphite Furnace Atomic Absorption (i.e., U.S. EPA 200.9); Hydride = Gaseous Hydride Atomic Absorption; CVAA = Cold Vapor Atomic Absorption; DCP = Direct Current Plasma.
- <sup>[2]</sup> The suggested method is the U.S. EPA Method unless otherwise specified (SM = Standard Methods). The Discharger may use another U.S. EPA-approved or recognized method if that method has a level of quantification below the applicable water quality objective. Where no method is suggested, the Discharger has the discretion to use any standard method.
- <sup>[3]</sup> Analysis for total chromium may be substituted for analysis of chromium (III) and chromium (VI) if the concentration measured is below the lowest hexavalent chromium criterion (11 ug/l).
- <sup>[4]</sup> The Discharger shall use ultra-clean sampling (U.S. EPA Method 1669) and ultra-clean analytical methods (U.S. EPA Method 1631) for mercury monitoring. The minimum level for mercury is 2 ng/l (or 0.002 ug/l).
- <sup>[5]</sup> MUN = Municipal and Domestic Supply. This designation, if applicable, is in the Findings of the permit.
- <sup>[6]</sup> Determination of Asbestos Structures over 10 [micrometers] in Length in Drinking Water Using MCE Filters, U.S. EPA 600/R-94-134, June 1994.
- <sup>[7]</sup> Detected as azobenzene.